



FOREWORD



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It is with great pleasure that I present the “ICAR Kharif Agro-Advisories for Farmers 2025”- a timely and comprehensive guide developed to empower India’s farming community with science-based, region-specific advisories for the upcoming Kharif season. This publication marks a significant step toward strengthening climate-resilient agriculture and enhancing farm productivity across diverse agro-climatic zones.

Agriculture in India stands at a critical juncture. While it continues to serve as the backbone of rural livelihoods, it faces unprecedented challenges due to climate variability, resource degradation, pest dynamics, and market volatility. In this context, the role of timely and location-specific agro-advisories has become more crucial than ever. These advisories act as a bridge between cutting-edge agricultural research and its practical application in farmers’ fields.

This publication offers practical guidance on region-based suitable varieties, nutrient management, pest and disease control, and harvesting practices tailored for major Kharif crops such as rice, maize, pulses, oilseeds, and cotton. It is the result of collaborative efforts by ICAR Institutes, Agricultural Universities, Krishi Vigyan Kendras (KVKs), and Subject Matter Experts working at the grassroots level.

At ICAR, we remain committed to the vision of a knowledge-driven, inclusive, and sustainable agricultural system. The advisories compiled herein reflect our continuous efforts to translate scientific knowledge from laboratories into actionable insights for every farmer. I urge extension officers, KVK teams, farmer producer organizations (FPOs), and progressive farmers to widely disseminate and implement the recommendations provided in this book.

I extend my heartfelt appreciation to all scientists, experts, and field professionals who have contributed to the preparation of this publication. I am confident that this agro-advisory compilation will serve as a valuable resource for farmers during upcoming Kharif season, enabling them to make informed decisions that lead to improved productivity, profitability, and sustainability.

Let us move forward together to build a resilient, prosperous, and empowered farming community #Viksit Bharat 2047.

Jai Kisan, Jai Vigyan!

(M.L. Jat)

INTRODUCTION

Agriculture remains the cornerstone of India's economy, supporting nearly half of the population and providing employment, nutrition, and rural development. The Kharif season, spanning from June to October, is particularly vital due to its reliance on the south-west monsoon. Approximately 55% of India's net sown area is rainfed, making the timely and equitable distribution of monsoon rains crucial for agricultural success.

In 2025, the India Meteorological Department (IMD) forecasts an above-normal monsoon season, with cumulative rainfall expected at 105% of the Long Period Average (LPA) of 87 cm. The monsoon's onset is anticipated to be slightly delayed in southern India but should reach Maharashtra by May 27, facilitating timely sowing preparations. The IMD's forecast of an above-normal monsoon offers a promising outlook but still carries risks of intra-seasonal dry spells and localized flooding. The Indian government's initiatives, such as the "Panchayat Mausam Seva," aim to deliver hyper-local weather forecasts at the panchayat level, ensuring that farmers receive accurate and timely information.

The Kharif season witnesses the cultivation of a diverse array of crops across India's agro-climatic zones, including rice, maize, cotton, soybean, groundnut, bajra, jowar, tur, urad, and moong. Despite advancements in agricultural technology, the sector remains vulnerable to climatic fluctuations such as delayed rains, uneven distribution, and extreme weather events. These challenges can adversely affect crop yields, farm incomes, and rural livelihoods.

To mitigate these risks, it is imperative to provide farmers with location-specific, research-backed guidance. Agro-advisories play a pivotal role by offering timely, region-specific, science-based information on weather patterns, crop management, pest control, harvesting, and storage. These advisories integrate seasonal outlooks with crop calendars and state-wise sowing windows, enabling farmers to select appropriate crop varieties and sowing dates, optimize resource use, and prevent yield losses.

This advisory serves as a crucial tool for agricultural officers, Farmer Producer Organizations (FPOs), and NGOs involved in rural development. By bridging the information gap between research and practice, it empowers farmers with the knowledge needed to make informed decisions, thereby contributing to the vision of a self-reliant, prosperous, and inclusive India @ VIKSIT BHARAT 2047.



Andaman & Nicobar Islands

The Andaman and Nicobar Islands have a warm, humid tropical climate with high annual rainfall and fertile loamy soils, making them well-suited for diverse agricultural activities. The climatic conditions are ideal for cultivating paddy along with plantation crops such as coconut and arecanut. Major fruit crops include banana, mango, sapota, papaya, pineapple, and citrus, while vegetables like brinjal, okra, chilli, cowpea and cucurbits, tuber crops- tapioca, sweet potato, dioscorea and elephant foot yam as well as spices-black pepper, cinnamon, turmeric and ginger are also widely grown.

CEREAL CROPS

Rice

Recommended varieties of medium duration are Dweep Dhan 1,2,3,6 and 7, Jaya, ANR-47, and Gayatri whereas long duration varieties are Dweep Dhan 4, 5 and ANR 40. Soak paddy seeds in normal water for 12-24 hours, discard the floating seeds, and use only the fully submerged seeds for sowing. It is recommended to treat seeds with Bijamrit @ 300 ml/kg seed. About 1000 m² nursery area is adequate for one hectare of field. Apply FYM @ 10q or vermicompost @ 5 q along with along Azospirillum @ 2 kg/1000 m² at 2-3 days prior to sowing in nursery. For nursery, prepare beds of 10 m × 1.25 m; apply 20-30 kg FYM per bed. Maintain a 30 cm drainage channel between beds. Sow 650-1000 g of sprouted seed per bed. Maintain saturated condition of 2-3 cm standing water 2-3 days before uprooting seedlings. Before transplanting, root dipping of seedlings with Jeevamrit @ 50 ml/L may be followed for better establishment. In main field, spacing should be maintained by 20 cm x 20 cm in ICM and 25 cm x 25 cm for SRI methods. Apply either FYM@ 2.6 t/ha or vermicompost @2.5 t/ha in the main field. Cono-weeder can be used for weeding purpose. Ensure bunds are repaired and 20-30 cm high to retain water. Use neem-based formulations (@ 0.5% or 5 ml/L) 2-3 times for pest control.

Cinnamon

Transfer six months old seedlings to polythene bags when they attain a height of 15 cm. Employ air layering techniques during the first week of July, using coir pith compost as a rooting medium. Apply 10 kg of FYM/plant/year and mulch with leaves to retain soil moisture. Utilize neem-based products and *Trichoderma* formulations to manage common pests and diseases.

Ginger

Prefer well-drained, loamy soils rich in organic matter for planting. Prepare raised beds of 15 cm height and 1 m width, ensuring proper drainage. Incorporate organic manures like FYM at 25-30 t/ha during bed preparation. Use healthy and disease-free rhizome for planting. Treat seed rhizomes with *Trichoderma harzianum* before planting to prevent fungal diseases. Apply neem cake at 2 t/ha and vermicompost at 5 t/ha. Also incorporate biofertilizers like *Azospirillum* and phosphate-solubilizing bacteria to enhance nutrient availability. Adopt crop rotation with legumes to break pest cycles. Use neem-based products and *Trichoderma* formulations to manage common pests and diseases.

Turmeric

Use healthy rhizomes (2000-2500 kg/ha) for planting and maintain spacing at 40 cm × 40 cm. Use *Trichoderma* for seed treatment and to control rhizome rot. Apply 20-25 t/ha of compost. Also mulch with locally available dry leaves to retain moisture and suppress weeds. Irrigate during dry spells but avoid waterlogging.

Root and Tuber Crops

Use raised beds or ridges for better drainage. Treat planting materials with *Trichoderma* to prevent rots. Use high-yielding varieties for sweet potato- CIARI Swarna, CIARI Aparna and Bhu Sona; for Cassava- Sree Vijaya and Sree Jaya; for elephant foot yam- Gajendra; for Taro- Mega Taro -2, Sree Kiran, Sree Pallavi and Sree Rashmi; and for dioscorea- Nicobari Alu (Traditional variety). Apply compost (10-15 t/ha) and biofertilizers like *Azospirillum* and PSB. Install pheromone traps- one trap/100 m² area to control sweet potato weevil.

LIVESTOCK

Follow strict bio-security measures-i) sprinkling of disinfectant solution (phenyl solution, bleaching powder and fumigation with KMnO₄ and formalin), ii) use of foot bath/tray at the entry of livestock/poultry shed with KMnO₄/phenyl/savlon at the farm. Deworm livestock/poultry and practice dipping with proper medicines suggested by the Veterinarians to prevent them from internal and external parasites infestations. Vaccinate livestock/ poultry to prevent them from diseases. Protect livestock/poultry from heavy rain and provide well ventilated shelter. Artificial heat along with dry floor space should be ensured for neonates to avoid neonatal mortality. Wound should be treated immediately with proper medication including topical application of fly repellent ointment or spray to avoid development of myiasis (maggot wound). Trimming of hooves should be done in cattle to prevent foot-rots and other complications. Provide balanced feed for livestock/poultry preferably with home-made low-cost locally available feed ingredients viz. broken rice, rice

bran, coconut cake, chunni (pulse), till cake, azolla, noni fruits, dry fish, hatchery waste etc. Store feed in a dry and cool place in order to prevent from fungus. Supplementation of Mineral mixture (@ 50g cattle) should be provided for better productivity and reproductive performance in livestock. Cultivate fodder viz. para grass, guinea grass, hybrid napier (CO-5), supper napier, sorghum etc. for ensuring proper nutrition around the year. Fodder trees like jackfruits, noni, gliricidia, moringa, sesbania etc. may be planted for higher survivability during monsoon. For poultry, litter materials should be frequently turned and mixed with lime powder and fresh saw-dust to prevent dampening and cake formation leading fungal growth. Artificial brooding should be provided to the chick's up to 4-6 weeks for healthy growth and to reduce mortality.

FISHERIES

Pre-stocking management (Pond preparation)

Prior to the onset of monsoon rains (April-May), it is advisable to desilt the fish ponds and apply agricultural lime (CaCO_3) to maintain water pH between 7.0 and 8.5. Adequate drainage systems should be in place to prevent overflow or contamination from runoff, and inlet-outlet structures must be fitted with screens to prevent the entry of predators or wild fish during rains.

Species selection and stocking of fingerlings

Stocking of fish fingerlings should be planned early in the kharif season (June-July) to take advantage of rising water levels. For composite fish culture, IMC (Katla, Rohu, and Mrigal) is recommended. Side by side, consider species like GIFT Tilapia, Pangasius or Magur based on the market demand. A stocking density of 4000-6000 fingerlings per ha is recommended, depending on the level of management and availability of inputs.

Post stocking management

Feeding during the monsoon should be balanced and regular. Use either commercial feed with 24-28%CP or a mixture of rice bran and oilcake. Organic manures like cow dung (500 kg/ha/month) and poultry dropping may be applied to enhance pond fertility and plankton production. Water colour should be monitored regularly, as deep green or blackish water may indicate harmful algal blooms, which can be fatal to fish. Be vigilant for symptoms of common fish diseases like bacterial gill rot, EUS, and fungal infections. Diseased fish should be isolated immediately, and disinfectants like KMnO_4 and salt treatment may be used under expert guidance. Harvest fish during last week of September.

Seed production

For the production of quality fish seed, healthy and disease-free broodstock aged 2-3 years should be selected and provided with a protein-rich diet (25-30% CP). Induce breeding between June to August using synthetic hormones like Ovaprim or WOVA-FH. The recommended male-to-female ratio for breeding in hapas is between 1:1 to 2:1.

CEREAL CROPS

Paddy

Varieties

North Coastal Zone: Pushkala, Srisatya, Cotton Dora Sannalu, Tarangini, Nellore Masoori, Nellore Dhanyarashi, NLR Rice 3238.

Southern Zone: Sweta, Bharani, Somasila, Nellore Masoori, Nellore Dhanyarashi, Nellore Sugandha, NLR 3238, Cotton Dora Sannalu, Tarangini, Chandra.

HAT Zone: Cotton Dora Sannalu, Tarangini, Chandra, Pushkala, Srisatya, NLR 3238.

Krishna Zone: Swarna, Indra, Amara, Krishnaveni, Varam, MTU Rice 1318, MTU 1232, Panduranga, MCM Rice 103, Samba Masoori, Akshaya, Bhavapuri Sannalu, Bapatla Masoori, Nellore Siri, Teja, Bhavathi, Sasya, BPT Rice 2841 & 2846, Maruteru Samba, Maruteru Masoori, MTU Rice 1271 & 1278, Nellore Sona, Nellore Siri, NLR 3238.

Late Nursery Transplanting: Swarna, Indra, Amara, Maruteru Masoori, MTU Rice 1318, Bapatla Masoori, Bhavapuri Sannalu, Teja, Panduranga, MCM Rice 103. BPH (Brown Plant Hopper) Infestation-Prone Areas: Krishnaveni, Indra, Amara, Maruteru Samba & Masoori, MTU 1271, Bhavapuri Sannalu, Teja, Sasya, BPT Rice 2846, Nellore Siri.

Flood-Prone Areas: Amara, Indra, Bhima, MTU Rice 1232 & 1318. *Saline Soils:* Indra, Panduranga, MCM Rice 103.

Late Planting: Cotton Dora Sannalu, Tarangini, MTU 1293, Nellore Sona, Nellore Masoori, Nellore Siri, NLR 3238.

Drought-Prone Areas: Cotton Dora Sannalu, Vijetha, Tarangini, Chandar, MTU 1293, NLR 3238.

Godavari Zone: Swarna, Indra, Amara, Bhima, Pushyami, Ksheera, Sravani, Maruteru Samba & Masoori, MTU 1212, 1280, 1281, 1293, 1310, 1321, 1318, 1232,

1271, 1275, Samba Masoori, Teja, Bhavathi, Bapatla Masoori, BPT 2841, MCM Rice 103.

Late Nursery Transplanting: Swarna, Indra, Amara, MTU 1318, Maruteru Masoori, Srikakulam Sannalu.

BPH-Prone: Krishnaveni, Indra, Amara, Pushyami, Maruteru Samba, MTU 1271, BPT 2846.

Flood-Prone Areas: Amara, Indra, Bhima, MTU 1232, 1318.

Saline Soils: Indra, Panduranga, MCM Rice 103.

Late Planting: Cotton Dora Sannalu, Tarangini, MTU 1293, Nellore Siri, NLR 3238.

Drought-Prone Areas: Cotton Dora Sannalu, Vijetha, Tarangini, Chandra, Maruteru Samba, MTU 1293, NLR 3238.

North Coastal Region: Swarna, Indra, Amara, Bhima, Pushyami, Ksheera, Sravani, Maruteru Samba, Masoori, MTU 1310, 1318, 1232, 1271, 1275, Samba Masoori, Teja, Bhavathi, Bapatla Masoori, BPT 2841, 2846, MCM Rice 103.

Late Nursery Planting: Swarna, MTU 1318, Maruteru Masoori, Srikakulam Sannalu.

BPH-Prone Areas: Vijetha, Amara, Indra, Sujatha, Pushyami, Maruteru Samba & Masoori, MTU 1271, Bapatla Masoori, BPT 2646, Nellore Siri.

Flood-Prone Areas: Srikakulam Sannalu, Indra, Bapatla Masoori, Bhima, Amara, Ksheera, MTU 1318, 1232.

Saline Soils: Indra, Panduranga, MCM Rice 103. *Late Planting:* Vasundhara, Cotton Dora Sannalu, Tarangini, Sridhruti, Chandra, MTU 1293, NLR 3238.

Drought-Prone Areas: Sona Masoori, Swarna, Cotton Dora Sannalu, Vijetha, Chandra, Tarangini, Sujatha, Maruteru Samba, MTU 1293.

Southern Region: Nellore Sona, Nellore Masoori, Nellore Dhanyarashi, Nellore Siri, Somasila, Bharani, Sweta, NLR 3238, Samba Masoori, Bapatla Masoori, Teja, Bhavathi, Sasya, BPT Rice-2841, BPT Rice-2846, Varam, Maruteru Samba & Maruteru Masoori, MTU Rice-1232, MTU Rice-1271, MCM Rice-103, Nandyala Sona.

Late Nursery Planting: Parthiva, Bapatla Masoori, Teja, Bhavapuri Sannalu, Maruteru Masoori.

BPH-Prone Areas: Maruteru Samba & Maruteru Masoori, MTU 1271, Bapatla Masoori, BPT 2846, Parthiva.

Saline Soils: Nellore Masoori, Panduranga, MCM Rice 103.

Late Planting: MTU 1293, NLR 3238, Nellore Masoori, Cotton Dora Sannalu, Chandara.

Scarce Rainfall Zone: Nellore Sona, Masoori, Siri, NLR 3238, Samba Masoori, BPT 2841, 2846, Varam, Maruteru Samba & Maruteru Masoori, MTU 1271, MCM Rice 103, Nandyala Sona.

Late Nursery Planting: Bapatla Masoori, Bhavapuri Sannalu, Maruteru Masoori.

BPH-Prone Areas: Maruteru Samba & Masoori, MTU Rice -1271, Bapatla Masoori, BPT Rice- 2846, Parthiva, Nandyala Sona.

Late Planting: Samba Masoori, Nandyala Sannalu, MTU Rice-1293, NLR Rice -3238, Nellore Masoori, Cotton Dora Sannalu, Chandra.

Fertilizer Application: Grow green manure crops (Dhaincha, Sunhemp, Pillipesara and Cowpea) as preceding crop to Paddy and apply soil test-based fertilizers. The entire dose of phosphorus fertilizer should be applied during the final puddling stage. In heavy soils, the full dose of potash fertilizer should be applied at the final puddling. In light soils, half the dose should be applied during puddling and the remaining half at the panicle initiation stage.

Weed Control: In wet direct-seeded rice fields, within 3-5 days of sowing, apply Pyrazosulfuron-ethyl 10% WP @ 250 g/ha or use Pretilachlor + Safener 37% EW @ 1500 ml/ha mixed with 50 kg of dry sand and broadcast evenly over the field. In transplanted paddy fields, within 3-5 days of planting, when the field has shallow standing water, apply either Pyrazosulfuron-ethyl 10% WP @ 200 g/ha mixed with 50 kg sand, or Butachlor 50% EC @ 2.5-3.12 litres/ha, or Pretilachlor 50% EC 1000 ml/ha. If the field has a heavy infestation of grassy, sedge and broadleaf weeds, use Bensulfuron Methyl 0.6% + Pretilachlor 6.0% Granules @ 10 kg/ha mixed with 50 kg sand and broadcast evenly. At 20 days after transplanting, for grassy weed control, apply Cyhalofop-P-butyl 10% EC @ 1000 ml/ha. If weed infestation is high, apply Bispyribac Sodium 10% SL @ 250 ml/ha or Triafamone + Ethoxysulfuron 30% WG at 225 g/ha in 500 liters of water.

False smut: Spraying of propiconazole @ 200ml/acre two times at booting stage and at 50% panicle emergence stage. At severe incidence of diseases spray tebuconazole + trifloxystrobin 75WG@80g/acre.

Bacterial leaf blight: Affected stubbles are to be destroyed by burning or through ploughing, judicious use of nitrogenous fertilizers, avoid clipping of tip of seedling at the time of transplanting. Avoid flooded conditions or drying of the field (not at the time of flowering). Application of potassium two time @ 15 kg/acre at planting time and 20 kg/acre at panicle initiation stage.

Trap Barrier System (TBS): Erect 2 ft height polythene sheet barrier around the crop at 15 days after transplanting. The polythene barrier is provided with 3 inch radius holes above the water level at 15-30 m distance and wonder traps are arranged firmly inside the crop, their entry mouths matching the holes made in the polythene sheet. Arrange a soil moat before the entry hole to facilitate the animal to enter easily inside the trap. Maintain water around the TBS and collect the successful traps every day and replace with new / emptied traps.

Precautions during natural calamities: If the crop is submerged in water for more than 1 to 2 days in the boot leaf or flowering stage, the panicle exertion may be incomplete. Water accumulation in the spikelets, would also result in reduced fertilization and formation of chaffy seeds. Spraying with Carbendiazim @ 1g/L or Carbendiazim + Mancozeb @ 2g/L or Propiconazole @1ml/L is recommended to avoid discoloration of grains. If the harvested sheaves in the field are drenched, 5% salt solution should be sprayed on the sheaves to prevent seed germination. The sheaves should be turned over, dried and threshed after cessation of the rains. In

case of stormy weather conditions, while heaping of incompletely dried paddy sheaves, 25 kg of salt per acre of paddy sheaves may be sprinkled and heaped to avoid damage.

Maize

In normal conditions, rainfed crop should be sown between June 15 and July 15. In the Kharif season, under rainfed conditions, maize should be sown only after a substantial rainfall (more than 50 mm). Seed rate is 17.5-20 kg/ha for normal hybrids, 7.5-10 kg/ha for sweetcorn, 12.5 kg/ha for popcorn and 25 kg/ha for baby corn. Within two to three days after sowing the crop, spray Atrazine herbicide @ 2 kg/ha in light soils and 3 kg/ha in heavy soils mixed with 500 liters of water when there is sufficient moisture which helps in controlling broadleaved and some grassy weeds for about 30 DAS. To control broadleaved and grassy weeds in normal and zero tillage maize crop at 15-18 DAS or 4 leaf weed stage, Tembotrione 34.4% SC @ 287.5 ml/ha + Atrazine 50% WP @ 1000 g/ha or Topramezone 33.6% SC 75 ml + Atrazine 50% WP @ 1000 g/ha should be sprayed. If the problem of *Cyperus spp.* is more, spray Halosulfuron methyl @ 100 g/ha mixed with 500 L of water.

Fall army worm: Summer ploughing, collection & destruction of egg masses, installation of pheromone traps (*S. frugiperda*) @ 4/acre sowing of border crop with grain sorghum & inter crop with cowpea (few rows) spraying of Azadirachtin 10000 ppm @ 2ml/l (10 to 15 DAS), EPN or Bt spray @ 2 ml/l (15 to 21 DAS). First insecticide spray - Emamectin Benzoate 5SG @ 0.4g/l (or) Spinosad 480SC @ 0.5 ml/l (21-28 DAS). Metarhizium anisopliae spray (1x107) @ 2 ml/l (30 -35DAS). Second Insecticide spray - Flubendiamide 480SC @ 0.3 ml/l (or) Chlorontriliniprole 18.5SC @ 0.3ml/l or Spinetoram 11.7 SC @ 0.3 ml/l (36 - 42 DAS). Poison Baiting - 45-65 DAS using Thiodicarb 75WP (250 g thiodicarb + 25kg rice bran + 5 kg jaggery / ha).

OILSEED CROPS

Groundnut

Farmers are advised to take up summer ploughing and also suggested to select high yielding varieties like TCGS 1694, K-Chitravathi, K-Lepakshi. The suitable time for sowing of rainfed groundnut is July to first week of August. Farmers are advised to adopt intercropping with redgram at 7:1/11:1/15:1 ratio as per their convenience and availability of sowing equipment. Use fertilizer cum seed drill instead of manual labour for sowing behind the country plough. Formation of conservation furrows every 3.6m interval at sowing/20-30 days after sowing will help in soil moisture conservation.

Farmers are also suggested to follow seed treatment with imidacloprid 600 FS @ 1ml per kg seed and followed by tebuconazole 2 DS @ 1g or mancozeb @ 3 g/kg or *Trichoderma viride* @ 10g/kg seed. Take up sowing of bajra/jowar (4-8 rows) as border crops all around the field to avoid transmission of vectors like thrips which spread PSND/PBND diseases. Spray KNO₃ @ 0.5% or 19:19:19 immediately after dry spell. Provide supplemental irrigation 20mm at flowering and peg penetration stage. Install sticky traps (10/acre) and pheromone traps (4/acre) in field to monitor

enrich the soil. Tractor operated FYM Pulverizer cum spreader is useful for FYM application in the field without any labour. It covers up to 10 to 12 acres per day. Cost of the implement is Rs. 76000/-.

DSR with Seed cum fertilizer drill increases 14% higher yields compared to traditional transplanting. Crops matured 7-10 days earlier. Farmers saved on 12 men labour and 25 women labour per hectare leading to a 17 % reduction in cost of cultivation. Tractor drawn Ananta groundnut planter (8 rows) is used for groundnut sowing. The intercrops of redgram or castor can also be sown with this implement. It covers 6 to 7 ha per day. For small and marginal farmers, use Handpush seed drill in light soils for Maize, Castor, Redgram, Cowpea, field bean and other bold seeded crops and in heavy soils on ridges sowing should be done for better crop stand establishment. Motor-Operated Groundnut Sheller is capable of shelling 25 to 250 kg of pods per hour. The distance between the seed drum and the concave can be adjusted according to pod size for efficient shelling. Powered by a 1 or 2 HP motor/engine depending on size. Use cono weeders for Drum seeder/transplanted rice in wet conditions. Power weeders help in upland rice systems or DSR & Early-stage weeding (15 to 30 DAS) is crucial for effective control. Self-propelled Weeder in paddy fields is operated by a 3 to 5 HP diesel engine, it can cover 0.15-0.25 acres per hour at low speed. & with a 5 to 6.5 HP engine, it can cover up to 1 acre per hour. Cost of cultivation is Rs.52,000 to Rs.56,000/ha.

Rain pipe irrigation system is effortless to install, reallocate and reduce the consumption of energy and water by more than 50 percent. These pipes are light weight, portable and flexible. Works on low water pressure. Expected life for 3 to 5 Years. Solar Fence system is used to control Wild boars a major problem in groundnut, paddy, maize and other crops. The solar fence cost varies from Rs. 15,000 to 25,000 by different companies like Pavani Solar systems, Nellore (Mobile numbers: 9848582332 & 9440171938) and Krupa Solar fence, Hyderabad. This unit protects 5 acres field. Use Acoustics (Sound emitting machinery) to repel the Birds, Wild boars, Monkeys etc., in crops like Groundnut, Sunflower, Maize, jowar etc. Use Bird scarer to repel the birds in the fields. The implement will create sound by rotating fan when natural air occurs. The cost is Rs.800/-. Straw Baling Machine is used to bundle paddy straw into either round or square/rectangular shapes for easy handling and transport. ANGRAU blade guntaka (Groundnut Digger) separates groundnut pods from the soil efficiently without wasting time. Built to dig up to 30 cm depth and can harvest up to 1 acre per hour.

Groundnut Digger, shaker cum Windower digs plants up to about 25 cm depth, shakes off the soil, and arranges the plants into windrows behind the machine. Cost: Rs.1,68,000/-. Groundnut Stripper separates pods from the plant by using a 1 to 3 HP motor. Requires two labour. Capacity: 100-125 kg per hour. Tractor operated pit digger for sugarcane planting will dig pits 250 to 300 holes per hour with a diameter of 90 cm. Cost is Rs.1,20,000. Cutting sugarcane sett with single bud by Sugarcane sett cutter. Capacity about 1500 setts per hour and Cost of the implement is Rs.5000/-. Damage is less when compared to manual cutting. Sugarcane detrasher can be used for detrashing. Easy for handling and detrashing, reduces the labour

requirement and Cost of the implement is Rs.1200/-. Cotton Picker mechanically plucks open cotton bolls from the plant without damaging the fiber. Used in large-scale operations.

Fruit Crops

Secateurs are used during June to August for pruning mango branches lesser than 1 inch thickness and for removing dry twigs. Cost of the falcon company implement is Rs. 450. Loppers are used during June to August for pruning mango branches and dry twigs less than 3 inches thickness. Cost of the falcon company implement is Rs. 1500. Mango harvester is used to harvest mangoes without damaging the fruits. Cost of the implement is Rs. 3150. Banana bunch cutter is used to harvest banana bunches. Coconut climbing cycles are used to easily climb coconut trees for harvesting coconuts and fronds and for pest and disease management. Cost of the implement is Rs. 3500/-. Coconut dehusker is used to remove the husk from coconut shells. Cost of the implement is Rs. 950. Oil palm harvester is used to harvest oil palm bunches, bring them down, and load them into a tractor.

Livestock

A milking machine is beneficial for farmers with more than 4-5 dairy cows as it extracts milk 3-4 times faster than manual milking and reduces physical labour. Farmers rearing more than 30-50 backyard poultry birds can use a vaccinator to administer the specified dose of vaccine in a very short time. When raising backyard poultry birds in large numbers, de-beaking with the help of a debeaker is a good management practice to reduce pecking. With the help of a duplex poultry shed, 10 - 15 chickens can be kept in a protective shelter providing with integrated feed and protection, and manage very easily at very low cost.

Drone Technology

Spraying of pesticides through Drones reduces time and cost of application and it is the Measure against labour scarcity. In Paddy, Maize, Jowar, Groundnut, Bengalgram, Chilli crops, this technology is already proved effective. An Acre can be sprayed with 10 litres capacity of Drone in 6-10 minutes.

ENTERPRISES

Adopt secondary agriculture through value addition to primary agriculture for enhancing farm income and sustainable rural livelihoods. Agripreneurship promotion through farm mechanization, crop diversification, processing, value addition, organic farming. Take up Agro based enterprises such as commercial horticulture, cultivation of high value crops, animal husbandry, fisheries, poultry, and Mushroom cultivation for doubling farm income. Setting up of Agro produce processing units like Mini dhal mills, Rice Mills and decorticating mills for promotion of agripreneurship among rural youth & women. Value addition of agriculture producing through Agro produce manufacturing units such as preparation of millet value added products, bakery products, ready to cook products, ready to eat products and nutraceuticals. Entrepreneurship by agro- input manufacturing units such as production of organic manures, fertilizers, agricultural inputs, vermin compost units, Bio fertilizers and agricultural implements. Establishments and

sunning of custom hiring centers, agro service centers, vegetable and fruit retail outlets. Organic certification for organic products and FSSAI Registration for value added products. Grading, packing, labeling for enhancing shelf life and marketing of value-added products. Marketing of products through FPOs, SHGs, and commodity interest Groups.

ICT TOOLS

ANGRAU ATARI CFLD: Mobile application provides complete information on Crop Production, Varietal, Crop Protection and Marketing information on Oilseed's and Pulse crops on Smartphone. In addition, it provides other useful information and useful apps in agriculture and allied sectors and connecting with experts.

ANGRAU Pashu Poshan: The ANGRAU Pashu Poshan application serves as a comprehensive resource for all aspects related to animal husbandry. This app is designed to provide farmers and livestock enthusiasts with valuable insights into various practices and techniques essential for the successful management and breeding of animals.

Fertilizer Calculator: This app covers complete nutrient management of major crops in agriculture and vegetable crops like chilli and onion. It totally covers 21 crops.

Meghdoot IMD: The app seamlessly aggregates contextualised district and crop wise advisories issued by Agro Met Field Units every Tuesday and Friday with the forecast and historic weather information to farmers in local vernacular.

Vyavasaya Vatavaranam: App designed to provide farmers with accurate and timely weather information to help them make informed decisions about their agricultural activities.

eNAM: The eNAM app is a mobile application that facilitates online trading of agricultural commodities, primarily for farmers and traders. It allows farmers to register and trade their produce directly through the app or via registered commission agents, connecting them with buyers on the eNAM platform. The application provides features like viewing lot information, confirming sale prices, tracking lot progress, and accessing various other APMC-related services.

Agri Central: Agri Central is the best Agriculture application for Indian farmers which helps them make critical decisions in their farming business to increase profitability.

IFFCO KISAN UDAY: It facilitates the booking, viewing, and management of drone-based sprayings of IFFCO nano fertilizers. The app also allows farmers to create profiles, save farm information, access product details, and scan QR codes on IFFCO nano products.

ITC Krishi Mitra/ITCMAARS: The app supports various aspects of farming, such as crop management, pest and disease control, soil health, water conservation, weather forecasting, market linkages, and government programmes.

DeHaat: Connects farmers to suppliers and buyers on a single platform. Online marketplace providing all the agricultural products and services to farmers.



Arunachal Pradesh

CEREAL CROPS

Rice

Selection of Varieties: RCM-13, RCM-15, RCM-16, Ranjit Sub 1, Gitesh, Luit, Disang, Numoli

Nursery Management: During nursery seed bed preparation, well decomposed FYM should be applied 2-3 days before sowing @ 10 q/ 1000 m² or vermicompost can be used @ 500kg/1000 sq.m and *Azospirillum* may be applied @ 2 kg /1000 sq.m area. 1000 sq.m nursery area is sufficient for transplanting of one hectare land.

Transplanting: First ploughing should be given at least 21 days before transplanting. 2nd ploughing at 10-12 days before transplanting and final puddling should be done 4-5 days before transplanting. The rice transplanting for mid and high altitude should be completed by month of June and in low altitude it should be completed by first fortnight of July. Transplanting should be done with 3 seedlings per hill. Spacings should be 20 × 15 cm (33 hills /sq.m) for semi-dwarf and 20 × 20 cm (25 hills/sq.m) for tall traditional varieties.

Fertilizers application and interculture operation: Well-rotted FYM or compost has to be applied @ 10 t/ha. Fertilizers @40:20:20 kg N, P₂O₅ & K₂O per hectare respectively. Green manuring crop viz. dhaincha can be sown in the month of May and crop should be incorporated through ploughing (Trampling) 10-15 days prior to transplanting. Preferably, two weeding's at 20 and 40 days after transplanting should be done. Critical stages for weeding are 15-45 DAS. Weeding can be done by Cono weeder or manual weeding depending upon stage of crop.

Plant protection measure: Clipping of leaf tips during transplanting to remove egg masses laid by stem borer; Release of *Trichogramma* @50000/ha for 4 times to control stem borer; Formation of alleyways during transplanting (20cm); and application of *Pseudomonas fluorescens* @ 20 g/m² at the time of sowing are the common plant protection measure to be followed.



CEREAL CROPS

Sali rice

Selection of varieties: The Semi dwarf varieties are: AAU-TTB-Dhan 44 (Prachur), AAU-TTB-Dhan 43(Shatabdi), AAU-TTB-Dhan 42 (Patkai). Farmers interested in multiple cropping situation can select the medium duration varieties like Numoli, Dholi, Shraboni. For staggered planting in sali season suitable varieties are Prafulla and Gitesh. Farmers of flood prone district can take submergence tolerant rice varieties for flash-flood situation: Ranjit Sub1, Bahadur Sub1, Jalashree, Jalkunwari, Dholi. In case of late planting in flood-affected area/direct seeded late Sali are Luit, Kapilee, Dishang.

Seed treatment: Put the seeds in normal water, stirred well, sunken seeds are selected and the floated ones are rejected. For wet method, selected seeds should be soaked directly in fungicidal suspensions of carboxin @ 2.0 g/l of water for 24 hours. One liter of fungicide solution is required to treat one kg of seed. In case of dry method, put the seeds in a container and add carboxin @ 2.0 g/kg seed and mixed thoroughly by agitating them for five minutes.

Nursery management: Land is thoroughly puddled and seedbeds of 10 m length and 1.25 m breadth are prepared with 30 cm gap in between the beds. The length of the bed may vary according to convenience. In each seedbed 20-30 kg cow dung/compost, 80 g urea, 80g SSP and 40g MOP are to be applied and mixed well with the soil. Well germinated seeds are to be sown @ 650 g to 1 kg per bed depending on grain size. Requirement of seed for transplanting one hectare of main field is 40-45 kg. Irrigation water should be applied in furrows to maintain saturated condition in the surface soil of the nursery bed. However, standing water to a depth of 2-3 cm should be maintained at least 2-3 days before uprooting. As soon as one or two blast spots are seen, ediphenphos @ 1 ml/l of water is to be sprayed. For management of root knot nematode, apply, *Pseudomonas fluorescens* @ 20 g /sq. m at the time of sowing.

Transplanting: One ploughing should be given at least 21 days prior to transplanting. An irrigation for land soaking should be applied before preparatory tillage. The final puddling should be done 4-5 days prior to transplanting. One irrigation should be applied before final puddling. 25 days old seedlings of short and medium duration varieties should be transplanted by maintaining a row-to-row and plant-to-plant spacing of 20 cm × 15 cm. In case of long duration varieties, 35-40 days old seedlings are to be transplanted at 25 cm × 15 cm spacing. It is advised to plant 2-3 seedlings per hill, at a depth of 4-5 cm, for all varieties.

Nutrient management: Well rotten FYM or compost @ 10 t/ha has to be applied during field preparation. In addition, for semidwarf varieties, 60 kg N/ha, 20 kg P₂O₅/ha and 40 kg K₂O/ha and for tall varieties, 20 kg N/ha, 10 kg P₂O₅/ha and 10 kg K₂O/ are to be applied in areas with moderate fertility level. In case of poor soil, the rates of fertilizers may be required to increase to the extent of 60:30:30 kg/ha N, P₂O₅ and K₂O respectively. Half of urea and whole of super phosphate and muriate of potash should be applied at the time of final puddling. Of the remaining part of urea, half at tillering stage i.e. 20-30 days after transplanting and other half at panicle initiation stage should be applied. Two weedings should be given with paddy weeder or hoe at 20 and 40 days after transplanting. For weed control, pretilachlor @ 0.75 kg/ha or anilofos is to be applied @ 0.4 kg/ha at 3 days after transplanting. In sali rice, application of 5 cm irrigation water 3 days after disappearance of ponding water is recommended in medium and heavy soils. In rainfed kharif rice, height of bunds should be 30 cm to retain rainwater.

Plant protection measures: To control the rice stem borer, whorl maggot, gall midge and leaf folder, spray fipronil 5SC @ 1.5-2 ml/l of water. To control thrips and plant hopper, spray imidacloprid 70 WG @ 0.3 g/l of water or thiamethoxam 25 WG @ 0.03 g/l of water. Against Rice Hispa, spray lamda-cyhalothrin 5 EC @ 12.5g /per ha. To control rice pests, erect 50 'T'-perches per ha 2 ft (60 cm) above crop canopy as roosting site for insectivorous birds, which are to be removed before flowering in order to prevent activity of granivorous birds. To prevent rice blast, spray hexaconazole 5EC @ 2g/l of water at tillering stage (40-55 days after sowing) and subsequently give two more sprays of ediphenphos @ 1ml/l of water, one at panicle initiation stage and the other when the tip of the panicle just comes out.

Management of direct seeded late Sali: Field should be prepared just after recession of flood by ploughing, cross ploughing and laddering to bring it to a puddle



Seedlings of Flood Tolerant Rice variety-
Ranjit sub-1 during flood



Installation of T Perch in Rice Field

with short-duration pulses (e.g., moong, urd) is recommended during the early monsoon to improve soil fertility and farmer income.

Timely weeding and hoeing should be completed by July to avoid competition during the peak growth phase. For zinc-deficient fields, apply 25 kg zinc sulphate per hectare or spray 0.5% $ZnSO_4$ solution on standing crop. Pest and disease surveillance is essential from June onwards. Top borer infestation is common during this period; apply Chlorantraniliprole 0.4% GR @10 kg/ha in leaf whorls during early July. For pyrrilla, release of *Epiricania melanoleuca* parasitoids or spraying Acephate 75 SP @1g/litre can be effective. Monitoring for red rot and wilt, especially in susceptible varieties; promptly remove and destroy infected clumps. Maintain proper drainage to prevent water stagnation during heavy rains. With good management, the crop can achieve 60-80 t/ha yield by harvesting in January-February.

OTHER CROPS

Farmer are advised to use varieties such as A-404, BM-2, BM-3, JPU-28, JPU-67, VL-379 for Finger Millet cultivation. In case of Turmeric and Ginger, prepare the field for sowing of. Apply 25-30 t FYM/ha at final ploughing of field and complete sowing by middle of May. Leguminous fodder crops like Cowpea & Guar bean can be cultivated during Kharif season. Farmers are encouraged to practice cultivation of milky, oyster and paddy straw mushrooms for additional income. Seed sowing of Mahogni and Gamhar is to be started in prepared bed or in polytube.



Mulching in crop field

LIVESTOCK

The onset of the monsoon in June increases humidity and causes a spike in vector-borne and waterborne diseases in livestock. Farmers are advised to ensure clean housing and proper drainage to avoid water stagnation around sheds. Ensure adequate drinking water and shade (thatched or tree-based) in animal sheds. Clean water should be provided to the animals three times a day. Milk utensils and other equipment should be frequently cleaned. Maintain strict biosecurity and avoid overcrowding in livestock and poultry farm. Sufficient space should be maintained to avoid overcrowding. The floor of the animal shed should be cleaned regularly to avoid disease infestations. Sanitize the vehicle/cart used by infected animals, premises, shed and other equipment properly by solution of 2% red potash. Dusting of bleaching powder or lime in the vicinity.

For poultry, ensure dry bedding with rice husk or sawdust, and use potassium permanganate or lime powder around sheds to reduce microbial load. Avoid overgrazing during monsoon as it increases the risk of parasitic infections. Regular deworming should be done before the rainy season especially in young calves and kids using albendazole or fenbendazole under veterinary guidance. Deworming of birds with Piperazin hydrochloride @5ml per 10 birds is recommended. Proper

vaccination schedule should be followed before the onset of rainy season for prevention of Foot and Mouth Disease (FMD), Haemorrhagic Septicaemia (HS), Black Quarter (BQ) in cattle and buffaloes, and Peste des Petits Ruminants (PPR) in goats. Poultry must be protected from Ranikhet and fowl pox.

To increase the feed intake in high yielding dairy animal during heat stress, farmers should practice urea treatment of dry straw (4 kg urea in 40 litres water per 100 kg straw) to enhance its digestibility. Supplement animal diets with mineral mixture and common salt, and in lactating animals, give bypass protein and calcium supplements to maintain milk production. Ensure availability of green fodder during this season. Cultivate fodder crops like Maize, Sorghum, Cowpea and Napier to provide fresh and nutritious feed. In goats and sheep, provide bamboo slat flooring or dry raised platforms to prevent foot rot during wet months. Clean water should be provided to the animals three times a day. Farmers are advised to administer Electrol Powder 2g and Vimeral 5 ml per 5 litre water for adults. In case of disease outbreak in animals, wash the infected animal tongue, mouth, feet and nose by 1% red potash or 2% baking soda 2-3 times in a day. For adult birds, mix 2 g of Electrol powder and 5 ml of Vimeral in 5 litres of water. If the chick shows symptoms like drowsiness, cough and distorted feet appearing or change in colour of excreta then immediately consult the nearest Veterinarian.

Fisheries

Fisherman are advised to prepare their own feed from locally available materials like oil cake, undecomposed grains, rice bran, mineral salt etc. (To prepare 100 kg fish feed there is need of 40 kg oil cake, 19 kg undecomposed grain (floor), 40 kg rice bran and 1 kg mineral mixture). In order to maintain the water colour light green, fill the pond with fresh water as and when needed. Also to maintain the proper water quality, apply lime @250 kg/ha and Potassium permagnate 500g/ha. Harvest only that much quantity which can be sold locally. Beware of gathering of fish while catching and selling the fish and at any cost maintain the social distancing.



Diagnostic visit to pond



CEREAL CROPS

Rice

Field preparation: Cleaning of the field and bund management should be done before the monsoon, and summer ploughing to done which helps to destroy pathogens and weed seeds.

Seed rate: Paddy seeds should be treated with a 17% salt solution before raising the nursery or direct seeding. Use the recommended seed rate: for line sowing, 80-90 kg/ha; for the Biasi and broadcast methods, 100-120 kg/ha. For the transplanting method, prepare the nursery in approximately 1/10th of the total field area. Use 50 kg/ha of seed for thick-grained varieties and 40 kg/ha for thin-grained varieties.

Varieties: Farmers are advised to grow high-yielding varieties rice varieties for sowing in the Kharif season, such as Vikram-TCR, Durgeshwari, Chhattisgarh Sugandhit Bhog, Indira Aerobic-1, CG-1919, MTU-1318, CG Devbhog. Paddy Seeds should be treated with *Azospirillum* and *PSB* @ 5-10 g/kg of seed before sowing. Line sowing method should be adopted in paddy cultivation and a pre-emergence herbicide to be sprayed three days after sowing. Line sowing can be done using a seed-cum-fertilizer drill in a well-pulverized field. In leveled fields, sowing should be carried out at a row spacing of 20 cm using implements such as a tractor-drawn seed drill, Indira seed drill, Nari plough, or Borhamdev seed drill. The seeding depth should not exceed 3-4 cm. Proper calibration of the seed drill is essential to maintain the recommended seed rate accurately.

Soil testing: Farmers are advised to get their soil tested. For this, they should contact the local agricultural officer or nearby Krishi Vigyan Kendra. Based on the soil test results, balanced fertilizers should be applied accordingly.

Weed management: Paddy should be sown in line and pre-emergence herbicide like Pretilachlor 50% EC @1500 ml/ha or Pyrazosulfuron 10% WP@ 200 ml/ha

should be applied within 3 days after sowing. Followed by post emergence application of Bispyribac sodium (10%) @ 250 ml/ha.

Pest and disease management: If symptoms of sheath blight disease are observed in paddy field, spray Hexaconazole @ 1000 ml/ha. Bacterial Leaf Blight in rice fields can be controlled by draining excess water and spraying a mixture of Validamycin @ 1250 ml, Streptomycin @ 30 g, and Potash @ 37.5 kg per ha. It is recommended not to apply/spray urea during this period. For the management of panicle mite, spray Diafethiuron 50% WP @ 1.2 g/L water along with Propiconazole @ 1ml/L at late tillering stage. If adult insects of Yellow Stem Borer are observed in the rice crop, destroy the egg masses immediately. Remove 'dead hearts' (dried central shoots) from the plants. Install pheromone traps @ 5 traps/ha to monitor stem borer activity in the rice field. For control, spray Fipronil 5% SC @ 1000 ml/ha. For control of leaf folder and army worm caterpillars, observe the population of larvae if it >1 per leaf, take up a spray of Fipronil 5% SC 1000 ml/ha.

Utera

When the rice crop is approaching maturity and the field retains some moisture, farmers can opt for relay cropping (utera) with linseed or lathyrus. Before sowing Kharif crops, farmers should arrange for necessary agricultural tools and equipment, carry out any required repairs, and perform deep ploughing using a MB plough. Cover the fields with polythene can help destroy weeds, soil-borne diseases, and insect eggs. If green algae appear in paddy fields, drain the water. Place a pouch of copper sulfate (blue vitriol) where water enters the field.

Maize

Use of high-yielding varieties such as Vivek Hy-17, Vivek Hy-43, Pusa Vivek QPM-9, VLQPM-45, Hybrid Maize-1, HQPM-1, Vivek Makka Hy-51, Pusa HM-4, DMRH-1308, IMH 230, and IMH 231. A germination test is essential to ensure an optimum number of healthy plants in the field. This should be followed by bio-priming or seed treatment using Carbendazim 50% WP @ 2 g/kg seed, Imidacloprid 600 FS @ 2 g/kg seed, *Trichoderma* @ 10 g/kg seed, and *Azotobacter* @ 10 g/kg seed to enhance productivity. The optimum sowing time is from June 15 to June 30, with a spacing of 60 x 25 cm (row x plant). Apply balanced fertilizers, including zinc-based fertilizers, for better crop growth.

For weed control, apply Tropicamezone 33.6% SC @ 30 ml per acre as a post-emergence spray at 20-25 days after sowing. The pupae of Fall Armyworm and stem borer, as well as the spores of *Rhizoctonia solani* (the causal agent of sheath blight), remain in the soil. Therefore, deep summer ploughing is crucial to reduce the inoculum potential. The infestation of Fall Armyworm can be significantly minimized by dry sowing or sowing at the onset of the monsoon, practicing crop rotation, and alternating maize with other crops that are not affected by Fall Armyworm. Additionally, apply a basal dose of nutrients and use the ridge and furrow method of bed preparation. These practices should be completed by the last week of May or the first week of June. Monitor regularly to prevent Fall Armyworm infestations. If necessary, spray Spinetoram 11.7 SC @ 350 ml/ha or Emamectin Benzoate 5 SG @ 200 g/ha. To control leaf blight disease, take up spray of Mancozeb 75 WP @ 1000 g/ha.



Goa's tropical monsoon climate, with high humidity and heavy rainfall, supports Kharif crops like paddy (the staple), pulses, groundnut, and vegetables. The region's lateritic and alluvial soils, ample rain, and moderate temperatures favor crop growth. To optimize yields and manage risks such as pest outbreaks and waterlogging, timely and location-specific practices are essential. This advisory provides key guidelines on land preparation, crop choice, pest and nutrient management, and sustainable farming for the Kharif season in Goa.

Prepare land for Kharif paddy and new orchards. Begin nursery sowing of Kharif rice
Crop-wise Advisory.

CEREAL CROPS

Paddy

Sowing of Kharif rice in nursery can be done. Procure good quality seeds, fertilizers and other inputs. As the Rainy season is approaching, all the farmers are advised to repair the bunds of rice fields and keep bunds weed free. While puddling the field, green leaf manures or biomass could be added and incorporated @5t/ha. When the intensity of rainfall is low, apply 1 kg urea per 100 m² to the rice nursery. Control the crabs in the rice field by collecting during night time.

FRUIT CROPS

Banana

Clear drainage; avoid waterlogging. Provide staking support to young plants

Mango

Harvest stage: Remove fallen fruits; orchard hygiene. Fruit rot risk: Hot water treat fruits (48°C, 1 hr). Anthracnose: Spray Carbendazim @2g/L on seedlings. New planting: Space 5×5m; fill pits with FYM, neem cake, rock phosphate

season. Weed management should start with a pre-emergence spray of Pendimethalin at 1 kg a.i./ha immediately after sowing, followed by manual weeding at 20 and 40 days after sowing (DAS).

Intercropping with green gram, black gram, or cowpea during the early growth stages can provide additional income and improve soil health. For plant protection, monitor for sucking pests like aphids, jassids, and whiteflies, and spray Thiamethoxam at 0.3 g/l or Imidacloprid at 0.3 ml/l if necessary. Bollworm control can be managed using pheromone traps, and sprays of Spinosad, Emamectin benzoate, or Flubendiamide should be applied when pests reach threshold levels. Biocontrol agents like *Trichogramma* can be used where feasible. To prevent fungal diseases such as Alternaria and anthracnose, spray a combination of Carbendazim and Mancozeb. Ensure clean cultivation practices and remove infected plants to prevent the spread of disease. Avoid waterlogging during the monsoon, and provide protective irrigation during dry spells, especially during boll formation. Growth regulation can be enhanced by spraying 2% DAP or 1% KNO₃ at flowering to improve boll retention.

Sugarcane

For sugarcane cultivation, it is important to select region-specific, high-yielding, and disease-resistant varieties such as Co 86032, CoN 05071, CoN 09072, and Co 99004. The ideal planting time is from mid-June to July, especially under rainfed conditions. Land preparation involves deep ploughing followed by harrowing, levelling the fields, and ensuring good drainage to prevent water stagnation. Use 3-budded setts from healthy, disease-free canes, and treat them with fungicides like Carbendazim 0.1% and insecticides such as Chlorpyrifos 0.1% before planting. The recommended row-to-row spacing is 120 cm, with a sett-to-sett spacing of 30 cm to ensure proper aeration and sunlight. Use 300 kg of nitrogen, 75 kg of phosphorus, and 75 kg of potassium (K₂O) per hectare. Apply the full dose of phosphorus and potassium, along with one-third of the nitrogen, at planting. The remaining nitrogen should be applied in two equal splits, one at 45 days and the other at 90 days after planting.

Weed management can be achieved using pre-emergence herbicides such as Atrazine at 1 kg/ha or Metribuzin at 1.25 kg/ha. During dry spells, provide light irrigation, particularly at the formative and grand growth stages, to maintain adequate moisture for proper tillering. Earthing up should be carried out at 45-60 days after planting (DAP) for partial earthing and at 90-120 DAP for full earthing to promote root growth and prevent lodging. Release *Trichogramma chilonis* at 50,000/ha/week to control early shoot borers, and apply Chlorpyrifos at 2.5 L/ha with irrigation to manage white grubs. In case of top shoot borers, spray Chlorantraniliprole or Spinosad when pests reach threshold levels. To prevent red rot and smut, use disease-free setts and rotate with non-host crops such as rice or legumes, and promptly remove and destroy infected clumps. Intercropping with crops like green gram, black gram, cowpea, or groundnut during the early growth stages can enhance income and improve soil fertility. Additionally, retaining sugarcane trash in the inter-row spaces helps conserve moisture and suppress weeds.

To ensure the well-being of livestock, provide adequate shade, cool water, and a mineral mix for proper nutrition. Regularly shower animals to help keep them cool. Vaccinate for Hemorrhagic Septicemia (HS) and Black Quarter (BQ), and deworm young animals to maintain their health. For tick control, use Deltamethrin. Additionally, use sanitizers in the sheds to reduce the presence of flies and mosquitoes, promoting a clean and hygienic environment for the animals.

CENTRAL GUJARAT

This region of Gujarat includes Ahmedabad, Anand, Kheda, Vadodara, Panchmahal, Mahisagar, and Dahod. They receive 600-900 mm of rainfall annually. The region has shallow to medium-depth black and red soils, which are typically low in fertility and organic carbon.

Maize

For sowing in Central Gujarat, use improved varieties like Narmda Moti, GM-2, GM-4, and GM-6, and hybrids such as GAYMH-1 and GAWMH-2. The ideal sowing time is from June 15th to July 15th. Prepare the land with 1-2 harrowings and 1 planking, applying manure before sowing. For spacing, use 60x20 cm for improved varieties and 75x20 cm for hybrids. The seed rate is 20-25 kg/ha for improved varieties and 25-30 kg/ha for hybrids. Fertilizer application should be 100:50:00 (N:P:K) kg/ha for improved varieties and 120:60:00 (N:P:K) kg/ha for hybrids. For plant protection against Fall Armyworm, use light traps, pheromone traps, and Neem seed powder/oil spray.

Rice

For transplanted rice cultivation, use 20-25 days old seedlings, with 2-3 plants per hill. Treat seedlings with *Azotobacter* and PSB liquid before transplanting. Irrigate with 5-7 cm of stagnant water. For top dressing, apply 40-48 kg N/ha at the tillering stage. For plant protection, control stemborer with light traps and spray Flubendiamide or Indoxacarb. Manage Leaf Blight with Streptocycline + Copper Oxychloride, and control Blast with Tricyclazole spray.

Soybean

Use recommended varieties: NRC-37, JS-335, KDS 726, Gujarat Soybean-2, and JS-9305. The ideal sowing time is from June to mid-July, with early sowing if conditions permit. Land preparation involves applying 10-15 tonnes of manure, followed by ploughing and harrowing. For spacing, use 45x7.5 cm or 30x5 cm, with a seed rate of 70-80 kg/ha. Fertilizer application should be 45:60:00 (N:P:K) kg/ha before sowing. Weeding can be done manually or with a Propaquizafop + Imazethapyr spray. For plant protection, control Bihar Hairy Caterpillar with Neem powder or pesticide spray, and manage Girdle Beetle with Novaluron or Quinalphos spray.

Pigeon Pea

For pigeon pea cultivation, early varieties include GT-1 and Vaishali, while late varieties are BDN-2 and IPCL-87119. The ideal sowing time is June-July, after rainfall. Land preparation involves applying 8-10 tonnes of manure, followed by ploughing and harrowing. For spacing, use 45-60 x 15-20 cm for early varieties and 90-120 x 25-30 cm for late varieties. Fertilizer application should be 25:50:00 (N:P:K) kg/ha.

CEREAL CROPS

Rice

Improved Varieties: Medium duration: HKR-127, HKR-126, HKR-120, HSD-1, Jaya, PR-106. Mid to early duration: HKR-47, IR-64, HKR 46, HKR 128, Pusa-1509, Pusa-1121. Early duration: HKR-48, Govind.

Nursery sowing: For early duration varieties: 15 May to 31 June. Medium, mid-early and hybrid rice: 15 May to 31 May.

Seed selection: Select bold and disease-free seed for better results. Treatment of seed with 10 % salt solution should be done to get bold seeds.

Seed rate & seed treatment: 20-25 kg/ha seed is sufficient for transplanting. Soaking of 1 ha seed (20-30 kg seed) in 25 litre water solution with 25 g bavistin and 2.5 g streptocycline is done for 24 hrs. Treated seed should be covered with wet gunny bags for 24-36 hrs for getting germination of seeds.

Nursery raising: Apply 15 to 20 tonnes of FYM per hectare in the nursery field before sowing of the nursery. Apply 25 kg N, 25 kg P and 25 kg ZnSO_4 per hectare for raising the nursery. Apply second doze of 25 kg N/ha at two weeks after sowing of nursery. If deficiency symptoms of iron appear in the nursery, foliar application of 0.5% FeSO_4 should be done. For managements of weeds in the nursery, apply 1.5 kg Pretilachlore 30 EC or 3 litre Butachlore or Thiobencarb at 1-3 days after sowing of the nursery. For management of broad leaf and grassy weeds, apply 250 ml Bispyribaks sodium 10 SL at 15 days after sowing of nursery.

Transplanting: Manually transplant 2-3 seedlings per hill in lines. Maintain spacing of 15 x 15 cm and depth of should not be more than 2-3 cm. Mechanical transplanting is done by paddy transplanter in puddled and un-puddled conditions. Raising of mat type nursery is required for mechanical transplanting.

Manure and fertilizer management: Application of 15 tonnes of FYM/ha before transplanting of the crop is useful for obtaining good productivity of rice-wheat system continuously. In medium, mid-early and hybrids of rice: Apply 150 kg N, 60 kg P_2O_5 , 60 kg K_2O and 25 kg $ZnSO_4$ per hectare. In early duration varieties: Apply 120 kg, 60 kg P_2O_5 , 60 kg K_2O and 25 kg/ha $ZnSO_4$. Apply $1/3^{rd}$ N and full P, K and zinc at the time of puddling. Remaining N fertilizers should be applied in two equal splits at 3 and 6 weeks after transplanting.

Weed management: Several grassy weeds, broad leaf weeds and sedges infest paddy crop. Weeds can be managed effectively by chemical method. Apply 30 kg butachlor or 15 kg thiobencarb granuls/ ha in 4-5 cm deep water at 2-3 days after transplanting. Apply 3 litre butachlor 50 EC or thiobencarb EC or 1325 ml Anilphose 30 EC or 2 litre Pretilachlor 50 EC or 125 g oxydiazetil 80 % WP after mixing with 150 kg sand and broadcast in standing water at 2-5 days after transplanting. Apply 20 g ready mix of metsulphuron and chlorimuron or 125 g ethoxysulphuron 15 % WG or 1 kg 2,4-D ester at 20-25 DAT against broad leaf weeds. Apply 93.75 ml/ha in 500 liters of water phenoxulam 24 % SC at 8-12 DAT for control of broad leaf weeds. Apply 250 ml bispyribac sodium 10 % SL/ha in 500 litres of water at 15-25 DAT for the control of mix weeds flora in rice.

Plant protection: Coragen (Chlorantraniliprole) 18.5 SC @150 ml/ha, Fame (flubendiamide) 480 SC@ 50 ml/ha for rice stem borer and leaf folder. Ulala (flonicamid) 50 WG @ 150 g/ha and Osheen (Dinotefuran) 20 SG @ 200 g/ha for plant hoppers. Nativio (trifloxystrobin+ tebuconazole) 75 WG @200 g/ha in 500 litre of water for BLB. Awancer Glow (azoxystrobin 8.3 % + mancozeb 66.7 %) 75 WG @ 1.5 kg/ha for sheath blight Amistar Top (azoxystrobin + difenoconazole) 325 SC @ 500 ml in 500 L water per ha for blast.

Bajra

Sowing Time: The appropriate sowing time is July 1-15 but can be sown at the onset of the monsoon, even in whole month of June, if good rainfall is there.

Seed Treatment: Treat seed with biomix (*Azotobactor*+ *Azospirillum* + PSB) @ 250 g for per ha seed.

Seed Rate and sowing: Use about 4-5 kg of seed per hectare to get around 175000 plants population per ha. Sowing should be done in lines at the spacing of 45 cm, at depth not more than 2.0 cm.

Recommended varieties: HHB 50, HHB 60, HHB 67, HHB 94, HHB 117, HHB 146, HHB 197, HHB 216, HHB 223, HHB 226, HHB 234, HHB 272, HHB 299, HHB 311, HHB 67 (improved 2), HC 10, and HC 20.

Nutrient Management: For hybrid varieties in irrigated condition apply 160 kg N/ha + 62.5 kg P_2O_5 /ha. For short duration varieties apply 40 kg N/ha + 20 kg P_2O_5 /ha. Apply full phosphorus and half of the Nitrogen at the time of the sowing, remaining nitrogen may be applied in two splits, first at thinking and second at ear formation stage.

Water Management: If irrigation is available, apply water during critical growth stages: at tillering, flowering, and grain filling. This is especially important under extended dry periods to avoid moisture stress.

Plant protection: Application of Malathion 50 EC @ 0.75-1.25 ml/L of water effectively controls sucking pests like jassids, aphids, and whitefly.

Disease management: To manage bacterial blight effectively two sprays with Streptocycline @ 75g and copper oxychloride @ 500g in 500L water per ha at initiation stage of disease or at 55-60 days after sowing is beneficial.

GREEN MANURING PRACTICES

The period between harvesting of wheat crop and transplanting of rice crop (May-June) can be utilized for growing of green manure crops. Dhaincha (*Sesbania sesban*) is very suitable cover crop which can be grown for green manuring. Sowing of dhaincha @ 30 kg seed/ ha is done in first week of May. The crop is ploughed up in 3rd week of June (45-50 DAS). Transplanting of rice is done after one week of ploughing of dhaincha crop. Apply 2/3rd dose of NPK fertilizers in fields where green manuring is done.

WATER CONSERVATION PRACTICES

Transplanting of rice requires huge volume of water. It can be saved by bringing change in planting method of the crop. Direct seeding of rice crop has been found very successful technology and is helpful in conservation of water resources. Levelling of fields with laser Level Machine helps in water saving in transplanting conditions.

LIVESTOCK

Temperatures are high during this month and some areas experience severe dust storms accompanied by thunder showers. Heat related diseases in animals that can be seen to affect animals during this time are fever, dehydration, decrease in body salts, loss of appetite and decrease in productivity. Animals should be protected from the heat and strong, hot, and dry summer afternoon winds. Adequate efforts should be made for fodder collection/purchase and storage for periods of shortage. To avoid loss of essential body salts in animals ensure that a salt mixture in appropriate quantity is mixed with the feed and water and given to animals. Depending upon the season, the content of the animal feed should be changed. Currently increase the quantity of wheat chaff and Jowar in the feed. Give dairy animals a balanced feed so that their milk production capacity is enhanced. Deworming of animals should be carried out. Maize, perennial grasses and other fodder species should be harvested now. Sheep should be sheared during this month.



Use of Kera method for seed sowing in maize

Apply 10-15 t of FYM per hectare 2-3 weeks before sowing. In hybrid varieties, apply NPK @ 120:60:40 kg and 90:45:30 kg per hectare in high and low rainfall areas, respectively. In local varieties, apply NPK @ 80:40:30 and 60:30:20 kg per hectare in high and low rainfall areas respectively. Apply full dose of P & K at sowing and two split doses of N at first earthing up and one month thereafter. In case of Zn deficiency, apply 25 kg of zinc

sulphate heptahydrate (21%) or 16.25 kg zinc sulphate monohydrate (33%) per ha during sowing.

For weed control, use pre-emergence herbicides like Atrazine @ 1.5-2.0 kg a.i./ha for sole crops while alachlor @ 1.5 kg /ha for mixed cropping systems. Weeds can also be controlled post emergence by application of herbicide Tembotrione @120 g a.i./ha with surfactant 1 L/ha at 2-3 leaf stage or herbicide Topramezone 25.2 g a.i./ha + atrazine 750 g a.i./ha at 2-3 leaf stage.

In areas where cutworm and white grubs are serious pests, chlorpyrifos @ 2.0 litre per ha mixed in 60-70 kg sandy soil may be used for soil amendment before sowing or cypermethrin 10EC @ 1ml per litre may be used for foliar application. Monitor the crop regularly for fall armyworm incidence. If the leaves are damaged or insect faeces are seen, then apply flubendamide (Fame) or emamectin benzoate or chlorantraniliprol @ 0.5 ml/L and repeat application at 15 days interval if needed. Proper earthing up should be done to prevent lodging of kharif maize due to high wind, cyclone etc. If there is heavy rainfall, care must be taken for drainage of excessive water from the maize field.



Demonstration of post-emergence herbicide in maize at farmers field

Paddy

Paddy nursery sowing @ 25 kg seed/ha after seed treatment with bavistin @ 2.5 g/kg for timely sown conditions may be completed from 20th May to first week of June using high yielding varieties HPR 1068, HPR 2143, RP 2421, HPR 2720, HPR 2880 and transplanting may be completed using 25-30 days old seedlings by end of June. In case of basmati rice, high yielding and disease resistant varieties Kasturi and HPR 2612 could be used. For late sowing conditions also, high yielding short duration variety HPR 2612 can be used. Varieties Bhurghu Dhan, Varun Dhan and Naggur Dhan could be used in Kullu valley and other colder areas. Red rice varieties HPR 2795 (upland areas) and HPR 2720 for irrigated transplanted area can be sown by mid-June. University tested hybrids Super-120, Star-795, AHR-8213 can be purchased from agriculture department.



Demonstration of paddy variety HPR 2612

The transplanted rice crop should be fertilized with NPK @ 90:40:40 with 50% N and entire dose of P & K at the time of transplanting and remaining N in two equal doses at 3 and 6 weeks after transplanting. In case of zinc deficiency symptoms in previous crop, apply 60 kg of zinc sulphate heptahydrate (21%) or 40 kg zinc sulphate monohydrate (33%) per hectare at puddling. The fertilizers should be applied 10-12 cm away from

the base of the plant to avoid plant injury. Soon after application, the fertilizer on the soil surface should be covered to minimize losses.



Nursery raising of Paddy



Transplanting of Paddy

Azolla is a good biofertilizer in irrigated rice crop which not only add organic matter in the field but also supplies enough N to the rice crops & reduce 10-20 per cent of nitrogen and phosphorus fertilizer requirement. Continuous flooding of water generally provides the best growth environment for rice. After transplanting, water levels should be around 3 cm initially, and gradually increase to 5-10 cm (with increasing plant height) until the field is drained 7-10 days before harvest.

In transplanted rice, apply Butachlor 5 g @ 30 kg/ha or broadcast Butachlor 50EC @ 3L after mixing within 72 hrs of transplanting for weed management. Post emergence herbicide Bispyribac sodium @ 250 ml/ ha can also be sprayed at 20-25 days after transplanting. In case of direct seeded rice, high yielding varieties HPR 1156 (Sukara dhan), HPR 2656 and HPR 2795 could be sown by end May or first week of June @ 60 kg seed/ ha. In irrigated conditions, apply first irrigation at 25 days after sowing and schedule succeeding irrigations as per monsoon rains.

For the management of stem borer and leaf folder, apply Carbofuron granules (furadan 3g) @ 33 kg/ ha in standing water after 10 days of transplanting or spray the crop after appearance of pest incidence with Chlorpyrifos @ 3ml/ L or Emamectine Benzoate @ 0.5 g/ L water. In case of blast/ false smut incidence, spray Copper Oxychloride @ 3g/ L or Hexaconazole @ 1 ml/L water at panicle initiation stage. The harvesting of the crop is generally done when 85% of the grains turn straw coloured.

Jammu & Kashmir



JAMMU

Jammu region has three agro-climatic zones; sub-tropical (upto 800 m above msl), sub-temperate (800-1500m above msl) and temperate (above 1500m above msl). Rice is the major crop in irrigated plains, while maize is generally grown in hills under rainfed ecosystem. After Rabi harvest, deep plough fields to utilize high temperatures for killing soil-borne pathogen spores, and take preventive steps to avoid post-harvest pest infestation. The crop-wise agro-advisory for kharif season is presented as below:

CEREAL CROPS

Paddy

Soak paddy seed in carbendazim 10 g + streptocyclin 3 g/5 kg seed (in 8 L water) for 24 hrs before nursery sowing. In sub-tropical zone, use blast-resistant coarse/semi-fine varieties (SJR-5, Jaya, PR-113, PHB-71, Ratna) and superfine types (Jammu Basmati-129, Pusa Basmati 1121). For temperate/sub-temperate areas, prefer early, blast and cold-tolerant types like China-1039, Giza-14, K-84, K-343, K-39.



Blast disease



Stem Borer

Prepare nursery beds (dry or wet) with 15 kg FYM, 60 g urea, 50 g DAP per 10 m². Use 16-18 kg/acre seed for fine/coarse varieties (2+ seedlings/hill), 6 kg/acre for hybrids; treat seed with copper oxychloride @ 3 g/kg.

Control weeds manually or use butachlor 5G (1.5 kg/500 m²) or pretilachlor (20 g/500 m²). Apply 30:12:8 kg N:P:K + 8 kg ZnSO₄/acre; split N at puddling, tillering, and panicle stages. Apply full DAP, MOP, ZnSO₄, and 1/3rd N at puddling; split remaining N at tillering and panicle initiation stages, draining water before top dressing.

For chemical weed control, use butachlor 12 kg/acre after transplanting or anilophos + ethoxy sulfuron (500 + 60 ml/acre) at 10 DAT. Manage weedy rice with stale seed-bed and glyphosate (0.6 kg/acre) or paraquat (0.32 kg/acre).

For management of blast, spray the crop with tricyclazole @ 0.06% or azoxystrobin + difenoconazole @ 0.1%. For stem borer, apply Cartap hydrochloride 4G (8 kg/acre) in 5-7.5 cm standing water. Do not drain or irrigate for 72 hours. Spray Cartap hydrochloride 50 SP (240 g/acre) when there are 5% dead hearts or one egg mass/sq.m.

Maize

Grow hybrids (Ganga Safed-2, Vivek-25, HQPM-1) or composites (Mansar, C-6, C-2); use 8 kg/acre seed for line sowing, 12 kg for broadcast, and 14-16 kg in hills. Treat seed with Thiram/Captan (3 g/kg) or Carbendazim (2 g/kg) + Chlorpyrifos (5 ml/kg) in cutworm-prone areas. Apply 6 t FYM/acre 2-3 weeks before sowing; with FYM, reduce NPK by 25%; otherwise, use 24:16:8 kg NPK + 4 kg ZnSO₄/acre.



Blister Beetle



Fall army worm

Maintain spacing: 75×20 cm (hybrids), 60×20 cm (composites); use Atrazine @ 0.4 kg/acre or weed manually at 15 & 30 DAS. Keep weed-free till 40 DAS; do earthing-up at 30 DAS for stability and moisture.

For pests, apply chlorpyrifos 1.5% D (10 kg/acre) in last ploughing, use light (2/acre) & pheromone traps (6/acre), spray methyl demeton, cypermethrin or cartap hydrochloride in whorls. For Blister Beetle, Cut Worm, Stem Borer & Army Worm in maize, use acetamiprid 20SP @ 0.2 g/L or chlorpyrifos 20EC @ 480 ml/acre, apply chlorpyrifos 1.5% D @ 10 kg/acre in soil for cut worm, cartap hydrochloride 4G @ 8 kg/acre in whorls for stem borer, and adopt trap cropping (okra/arhar) and manual beetle removal for blister beetle. For smuts, treat seed with carbendazim/carboxin @ 2 g/kg; for stalk rot, drench with copper



Small ruminants

Small ruminants (Sheep and Goat)

Avoid waterlogged and parasite-infested grazing areas; rotate grazing sites to minimize overgrazing and disease risk. Provide mobile or makeshift shelters during heavy rains to prevent pneumonia and foot rot. Regularly check for ecto and endo-parasites; ensure timely vaccination against diseases like PPR (Peste des Petits Ruminants) and Enterotoxaemia, ensure deworming before/after migration, and

protect against foot rot and pneumonia during wet alpine grazing.

NUTRITION ADVISORY FOR WOMEN

Nutrition awareness is vital for women farmers during the summer season to help maintain their health, energy, and productivity. Here's a concise summary of key points:

Balanced diet

Include pulses like moong and masoor, as well as milk, curd, and paneer to support body repair and energy. For vitamins, eat green vegetables like spinach, bottle gourd, ridge gourd, and fruits such as watermelon, muskmelon, mango, and litchi to cool the body and boost immunity. Consume millets, peanuts, flaxseeds, and sesame seeds to get minerals and strengthen muscles and reduce fatigue.

Consume seasonal foods

Choose cooling foods like cucumber, bael juice, and melons. Prefer local and fresh produce for better nutrition and easy access. Also use chilled homemade buttermilk using mix curd, water, roasted cumin, black salt, and mint.

Stay hydrated

Drink 10-12 glasses of water daily. Include traditional drinks like buttermilk, coconut water, bael juice, and mango panna. Choose fresh fruit juices without added sugar for natural nutrition.

KASHMIR

CEREAL CROPS

Rice

Soil and Climate: Rice is adaptable to a wide range of soil types, from loamy to clay loams. Optimum growth and higher yield are achieved at day temperatures between 25-33°C and night temperatures between 15-20°C.

Nursery Management (Sowing/ Transplanting): Seed treatment and soaking are essential for proper sprouting. Prepare nursery beds 1 m wide and of convenient length, as per the requirement. Construct low poly tunnels using willow sticks and polyethylene sheets over the nursery beds to protect seedlings from chilling injury and to promote healthy growth. The nursery should be weed-

free, have access to adequate irrigation water, and be equipped with proper drainage facilities. Complete sowing of seeds in the nursery by the first fortnight of May. Use 50-60 kg of seed per hectare for transplanting in lower belts and 80 kg per hectare for higher belts. Sow pre-sprouted rice seeds in the nursery beds. Use pre-stored ponded water in nursery beds instead of running water to avoid chilling injury.



Nursery Preparation

Nutrient Management for Rice Nursery: Apply 275 g of urea, 450 g of DAP, and 200 g of MOP per Marla (25 m²) area of nursery bed. These fertilizers should be incorporated into the soil during bed preparation. Additionally, apply 1 kg of nitrogen (N) per 100 m² of nursery bed area 6-10 days before uprooting the seedlings. Nutrient Management for Main Field (Transplanting). Apply well-decomposed compost or farmyard manure (FYM) at the rate of 15-20 tons per hectare. Incorporate this into the soil during land preparation. Application of vermicompost @ 2.5 t/ha can replace 5 t FYM/ha and 25% of the recommended NPK dose.

Transplanting Guidelines: Transplanting should begin from the last week of May in lower belts. The field must be free from weeds, and rat burrows in bunds should be sealed to prevent water loss. FYM or compost should be incorporated into the soil during the final land preparation.

Basal Fertilizer Application (Before Transplanting): For varieties grown in lower belts the dose of urea is 80 kg/ha, DAP: 130 kg/ha, MOP: 50 kg/ha, Zinc Sulphate (if deficient): 10-15 kg/ha. For varieties grown in higher belts: Urea: 36 kg/ha, DAP: 130 kg/ha, MOP: 50 kg/ha, Zinc Sulphate (if deficient): 10-15 kg/ha. For varieties grown in waterlogged areas: Urea: 47 kg/ha, DAP: 130 kg/ha, MOP: 50 kg/ha, Zinc Sulphate (if deficient): 10-15 kg/ha.

Seedling Transplantation: Transplant 30-day-old healthy seedlings (about 20 cm tall) raised in traditional nurseries. If raised under protected nursery conditions, 25-day-old seedlings can be transplanted. Transplanting and Field Management. Transplant 2-3 seedlings per hill at a spacing of 15 × 15 cm. To encourage better tillering, shallow transplanting is recommended. Maintain standing water in the paddy field to ensure proper field levelling. A 2-3 cm water level should be maintained as needed. Avoid root damage during seedling uprooting and prevent wilting by minimizing exposure time before transplanting. Gap filling should be completed within one week of transplanting. Effective puddling and water management reduce weed pressure. Maintaining 5 cm of standing water helps suppress weed growth by creating a smothering effect. Apply Butachlor @ 1.5 kg/ha as a pre-emergence herbicide within 2-4 days after transplanting. Apply Eros @ 10 kg/ha or Erase @ 10 kg/ha within 3-5 days after transplanting to control grasses, sedges, and the first flush of Potamogeton. At the time of herbicide application, ensure a 2-3 cm water level in the field and maintain it for at least 4-5 days.

Recommended Rice Varieties: For Lower Belts of the Valley the recommended varieties are Shalimar Rice-1, Shalimar Rice-2, Shalimar Rice-3, Shalimar Rice-4 &

Shalimar Sugandh-1 while for the Higher Belts of the Valley Kohsaar, K332 & Shalimar Rice-5 are the recommended varieties.

PLANT PROTECTION

Common Diseases of Rice and Their Management

Seedling Diseases: Seedling diseases contribute to poor seedling emergence, uneven stands, and stand failure. Seedling diseases include seed and seedling decay (pre-emergence damping off) and seedling blight (post-emergence damping off).



Rice Blast: Blast is the principal disease of rice owing to its wide spread distribution and large scale destructive nature, thereby making rice cultivation impossible in many areas despite management efforts. The symptoms appear on all the aerial plant parts. Depending upon the site of symptom the rice blast is referred as 'leaf blast', 'collar blast', 'node blast' and 'neck blast'.



Leaf blast



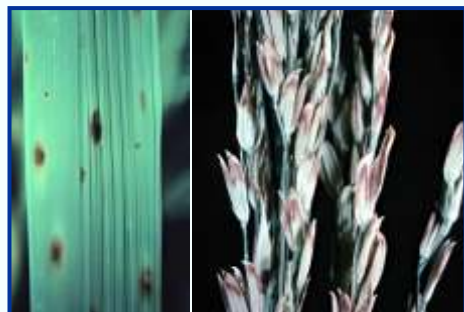
Nodal blast



Panicle blast

Brown Spot or *Helminthosporiose Leaf Spot*

The disease symptoms appear on leaves, glumes, coleoptiles, leaf sheath and panicle branches. On leaves and leaf sheaths the disease appears as typical oval or circular spots. The spots are evenly scattered all over the leaf surface. The spots when fully developed are dark brown to purplish-brown in colour with grayish white center. The disease causes black discolouration of grains so reduces seed/ grain quality as well as affects the yield.



Brown spot on leaves and Brown spot on panicles

False Smut or Green Smut: The disease appears only after flowering and the fungus infects individual grain/florets of panicle and transforms the growth into greenish spore

black in colour. The glumes show rotting symptoms. The disease intensity may range from sporadic discolouration to complete discolouration of whole glumes. Such discolorations appear externally on glumes or internally on kernels, or both. On glumes, the symptoms accordingly vary depending on the type of organisms involved and the magnitude of infection. The disease has shown increasing trend in recent years in the valley. The disease also decreases the seed quality and affects grain quality.

Integrated Disease Management: Most rice diseases can successfully be managed by use of healthy disease-free seed, proper field sanitation, application of balanced fertilizers, adherence to recommended cultural practices and, when necessary, use of biocides. Procure the seeds only from reliable registered seed agency or select only healthy and disease-free mature seeds for future use if taken from one's own crop. Before storage the seed should be cleaned, properly dried and treated with any one of the following fungicides: Tricyclazole 75 WP @ 0.6 g/ kg seed. Captan 50 WP @ 3.0 g/ kg seed. Mancozeb 75 WP @ 3.0 g/ kg seed. Mancozeb 75 WP @ 2.0 g + Carbendazim 50 WP @ 1 g/ kg seed.

Pre-sowing Seed Treatments: In order to minimize the chances of disease spread through seed, it is better to soak the seeds for overnight and at transplantation dip the seedlings roots in either of the following fungicides before sowing: Tricyclazole 75 WP @ 60 g/100 liter water, Ediphenphos 50 EC @ 100 ml/100 liter water, The sprouted seeds should not be treated with any fungicide as it may damage the plumule/ radicle.

MANAGEMENT OF DISEASES IN FIELD

Field sanitation: Stubbles and leftover straw should be collected and destroyed (either through burning or deep burial). Eradicate all the weeds from paddy field as it shall help in the removal of collateral hosts of the pathogens. The best way to control blast is to integrate the use of resistant varieties with good cultural practices (proper field selection, seeding rate, fertilization, and flooding).

Cultural Management: Strictly adhere to the recommended package of practice regarding land leveling, soil preparation, adequate puddling, plant spacing, fertilizer and manure use, crop rotation, water management and weed- and insect-pest management, etc. as these ensure better plant growth and development thereby help in minimizing the disease.

Fungicidal Management: Blast, brown spot and false smut are the most prevalent diseases of rice in Kashmir. In view of the present cropping system, it is impossible to completely eradicate the field inoculum even after taking all the precautionary measures. As a result blast incidence is inevitable. Besides pre-sowing seed treatment and seedlings dip as discussed above, the standing crop may be sprayed with fungicides no sooner the first disease symptoms appear in the field followed by three more sprays at 15-20 days interval. However, keeping in view the devastating nature of these diseases fungicidal sprays at the following four stages of crop growth are necessary to effectively control blast and other rice diseases. 1st spray at tillering stagem, 2nd spray at panicle initiation stage, 3rd spray at booting stage & 4th spray at milking or dough stage.

Integrated nutrient management : Combined application of azotobacter (seed treatment@500g/ ha) + 75% recommended N and 100% P&K (125:60:60) is an optimum nutrient combination for obtaining higher fruit yield.

Different stages of harvesting

Green stage: Fully mature green fruits are harvested to get more price of the produce at early stage. *Breaker or turning stage:* The fruits are harvested when 1/4 th blossom end of the fruit show pink colour. This is suitable to send the fruits to long distances.

Pink stage: When 3/4th of the surface towards blossom end shows pink colour. These fruits can be used to sale in the local market as well as to send distant markets.

Ripe stage: When entire fruit develop pink or red colour but fruits are still firm. This stage is suitable for table or salad purpose and to sale it at farmers field.

Over ripe stage: When fruits attain maximum colour and become soft. Such fruits are suitable for processing purposes.

Potato

Varieties recommended for Kashmir valley are Kufri Jyoti(200-250 q/ ha), Gulmarg Special(250 - 280 q/ ha), Kufri Giriraj(200-250q/ ha), Shalimar Potato- 1(350 q/ ha), Shalimar Potato-2(250-300 q/ ha),Kufri Pukhraj (30-35 t/ ha), Kufri Himalini, Kufri Ghirdhari (300-350 q/ha) and Kufri Shailja (300-350 q/ ha).potato seed tubers can be planted at spacing of 60x20cm and seed rate of 20-25q/ha. First earthing up is done when plants are 10 -15 cm long and Final earthing up is done when the crop is 6 -8 weeks old.

Methods of breaking dormancy:The cutting of tubers in pieces during the advanced stage of dormancy may help in breaking the dormancy. Cut the tubers in pieces having at least three healthy eyes and dip in 1% solution of thiourea for one hour. The planting should be done just after treatment or kept in wet gunny bags for one night.

Integrated Nutrient Management: Application of vermicompost @ 4.5t/ha along with 50% RFD (75:50:50) is recommended to obtain higher tuber yields. For supplementing Sulphur requirements Copper Sulphate@10kg/ ha is recommended. To improve shelf life in potato tubers spray Calcium Sulphate (0.2% sol) at 50,60 and 70 days after planting. Treat tubers with 1% urea and 1% sodium bicarbonate + biofertilizer (*Azotobacter + Phosphobacteria*) @ 2.5 kg/ ha and apply only 80:75:75 (NPK/ ha) as INM strategy.

Cucurbitaceous Vegetables

Raising cucurbit seedlings indoors: Seedlings of commercially important cucurbits are raised indoors in warm rooms/ locations with slightly extra care and cost, for getting seedlings early in the season. 2-3 seeds are sown in polyethylene bags of 100-gauge thickness and 12 cm x 10 cm in size filled with well crushed Dal weed or garden soil, 4 -5 weeks before the normal date of sowing. The bags before filling are provided with 4 - 5 holes on upper side for proper aeration and 2 -3 holes in the bottom to drain out excess water. When the

are available, no concentrate supplementation is required. While grazing, animals should be monitored continuously for development of bloat due to excess consumption of certain excess green fodders (e.g., clovers) and in case of any such eventuality, veterinary assistance should be sought. Sheep are seasonally migrated to subalpine and alpine pastures to take advantage of nutrient-rich forage, reduce grazing pressure on lowland areas, and support overall flock health and productivity. This traditional transhumance practice is especially important in regions like Kashmir, where high-altitude grazing improves wool quality and animal condition. Prior to migration, ensure animals are vaccinated against economically important diseases. Such as pox, PPR, Foot rot and FMD. A strategic deworming schedule should be followed to prevent development of anthelmintic resistance. This involves deworming animals based on parasite risk and fecal egg counts rather than routine blanket treatments. Rotating between classes of anthelmintics and targeting high-risk animals helps preserve the efficacy of dewormers, ensuring long-term parasite control and minimizing drug resistance in the herd. Provide salt licks and mineral blocks. This is essential to meet mineral deficiency especially at higher altitudes where soil and forage may lack essential minerals.

Backyard Poultry

Ensure proper night shelter and effective litter management, using saw dust or rice husk. Replace it every 2-3 weeks. Birds under scavenging backyard conditions or free range should be provided with supplemental feed. Additionally, vitamin C and electrolytes should be incorporated in drinking water to prevent heat stress. Deworm if it is necessary, with appropriate dewormer. Follow proper vaccination schedule to reduce early chick mortality and offset economic losses. Use pre and probiotics to improve gut health, reduce stress and improve immunity. This ensures wholesome and hygienic poultry products for human consumption.

Trout Fish

Trout fish, particularly species like rainbow trout, thrive in clean, cold water that is rich in oxygen. These fish are sensitive to water temperature and quality, with an optimal range for growth being between 50°F to 60°F (10°C to 15°C). Maintaining clean and cold water helps prevent disease and stress, ensuring healthy fish that grow at their optimal rate. Ensure a continuous water flow rate appropriate to the fish biomass. Maintain pH between 6.5 to 8.0 and dissolved oxygen above 7 mg/L. Commercial trout feed should be fed at 3-5% of body weight, divided into multiple feedings. Fish should be segregated according to size to prevent cannibalism, as larger fish may prey upon small ones. Routine inspection of the stock for signs of bacterial, fungal or parasitic infection can aid in prevention of disease. Adopt strict biosecurity measures to prevent disease outbreaks. For maximizing the returns, harvest the trout fish that have reached typical marketable size (250-300g)



CEREAL CROPS

Paddy

Farmers are advised to grow upland paddy with varieties like Vandana, Virendra, Abhishek, Swarna Shreya, Anjali, Sahbhagi Dhan, CR Dhaan-320, IR-64, etc., using the Direct Seeded Rice (DSR) method in upland areas. For direct sowing of paddy seeds in upland areas, summer ploughing should be done in May, and land preparation should be completed in the first week of June. Varieties such as CR Dhaan-320 and Swarna Shreya should be purchased in May itself. Short-duration varieties like Anjali, Sahbhagi, Vandana, and Sushak Samrat are suitable for direct sowing in the first fortnight of June.

For medium land paddy cultivation, to minimize delays due to late monsoon arrival, nurseries for paddy seedlings should be prepared at 10-day intervals in June. Improved variety seeds like Sahbhagi Dhan should be purchased at the end of May. Nursery sowing should be done in the first week of June to enable timely Rabi crop cultivation. A nursery of 100 m² is sufficient for one hectare. For this, 40 kg of OP variety seeds or 15 kg of hybrid seeds should be sown. Paddy seeds should be treated with Carbendazim @ 2 g/kg of seed. Fertilizers should be applied at a rate of 40:20:20 kg N:P:K per hectare. Of the nitrogen, 50% should be applied at sowing, 25% at 30-35 days, and the remaining 25% at 45-50 days after sowing. Entire phosphorus and potassium should be applied at sowing.

Farmers are advised to get their farm soil tested and apply fertilizers as per recommended doses. For transplanted paddy, apply fertilizers at a rate of 80:40:20 kg N:P:K per hectare. Nitrogen should be applied in split doses: 50% at transplanting, 25% at 30-35 days, and 25% at 50-60 days after transplanting. Phosphorus and potassium should be fully applied at transplanting. To prevent zinc deficiency, apply 25 kg of zinc sulphate during field preparation. For effective weed control, apply 2.5 litres of Pendimethalin 30 EC mixed in 500 litres of water



Management of paddy field

per hectare within 1-2 days of sowing. Bispyribac 10 SC should be applied at 250 ml/hectare 20-25 days after sowing. In case of chlorosis (iron deficiency), spray 1% ferrous sulphate solution (2.5 kg in 250 litres of water) two to three times at weekly intervals, and ensure adequate irrigation. For Terai paddy, seedlings should be prepared as per medium land rice guidelines. Varieties like Rajendra Mahsuri, Rajshree, and BPT-5204 should be selected and purchased in advance, preferably in May.

Maize

Farmers are advised to plough the land during May and apply 3-4 quintals of lime per hectare. Seeds should be sown in the first week of June or at the onset of the monsoon. To control termite populations and larvae, 100-150 quintals of well-decomposed farmyard manure (FYM) along with Methyl Parathion 2% dust at the rate of 25 kg per hectare should be incorporated into the soil during land preparation. Use 20 kg of seed per hectare of recommended maize varieties such as PMH-3, Orange Flint, HQPM-1, or HQPM-5. At the time of sowing, apply 86 kg of urea, 130 kg of DAP, and 68 kg of MOP per hectare. If the field remains dry, provide irrigation after 30 days. After proper weeding, apply an additional 65 kg of urea. Ensure the field has proper drainage to prevent water stagnation. Two weeks after germination, apply 10-15 granules of Fipronil 0.3G or Carbofuran 3G in each plant to manage early-stage pests. To control the incidence of Fall Armyworm in the maize crop, spray Emamectin Benzoate 5% SG at the rate of 1 gram per 3 litres of water.

PULSE CROPS

Farmers are advised to grow pulses (especially pea, chickpea, and lentil) during Kharif season. The recommended varieties for upland field include: Pigeon pea- Narendra Arhar-1, Narendra Arhar-2, Rajeev Lochan, IPA 203, Malviya Arhar-Mal-13; Black gram- WBU-109, Birsa Urd -1, PU-31; Green gram- Virat, Shikha, HUM-12, IPM 2-14, IPM 2-3, HUM-16; Horse gram- Birsa Kulthi-1, Payur-2, Madhu, VLG-19 etc. Oil seed crops like Niger- Utkal Niger-150, DNS-4, BNS-3, Pooja, JNS-29, JNC- 6, BNS-1; Sesame-RT-351, Krishna, Sekhar, RT-346, GT-4, GT-6, Punjab-1; Groundnut- K1812, TLG-45, ICGV-00350, K-6, TG-37A, TG-38 Dharni, Grinar-3, TG-51, TCGS-1043, AK12-14 etc.

Azotobacter). Ensure staking with metal wire or nylon rope and bamboo while selection of indeterminate varieties like Arka Samrat & Arka Rakshak in tomato cultivation in kharif season and planting on raised bed. Ridge and furrow method of cultivation is recommended for vegetable cultivation in kharif season along with proper drainage of water.

To control Yellow Vein Mosaic disease in okra, farmers should remove and destroy affected plants as soon as symptoms appear. After that, spray Thiamethoxam 25 WG at 100 g per hectare for effective disease management. To control red beetle in cucurbits, apply ash dust with kerosine oil on plants and near the root zone. To control red beetle infestation in cucurbitaceous crops, farmers are advised to apply Chlorpyrifos 2% dust @ 20 kg/ha in the vicinity of the root zone. Fruit fly in Cucurbits use Malathion 50% EC + Jaggery (5%) @ 2 ml/L + 50 g jaggery per spray on foliage at fruiting stage. Powdery Mildew in cucurbits use Hexaconazole 5% EC @ 1 ml/L as Foliar spray at disease onset. To control leaf spot in ridge gourd, spray 3.5 ml Indofil M-45 in 1 L water. Wilting in brinjal and tomato plants can be controlled by spraying of 1g Streptocycline + 30 g copper oxychloride in 30 L of water in the root zone of the plants.

For leaf spot in Brinjal and Tomato use Mancozeb 75 % WP @ 2.5 g/litre water. For fruit and shoot borer in brinjal & tomato use Emamectin benzoate 5% SG @ 0.4 g/litre water (Foliar spray at 10-15-day intervals after flowering) or Spinosad 45% SC @ 0.3 ml per litre water as foliar spray or coragen 20 SC @ 60 ml in 200 litre water/acre. Leaf curl virus (Vector: Whitefly) in Chilli and Capsicum and TVMV in Okra use Imidacloprid 17.8% SL @ 0.3 ml per litre. For Anthracnose / Fruit Rot use Carbendazim 50% WP @ 1gm/litre as foliar spray and repeat after 10 days. To prevent Onion tip burn, apply 400ml Hexaconazole by mixing in 200lt water and spray over 1 acre.

OTHER CROPS

Leguminous fodder crops such as Rice bean, Cowpea, and Guar bean can be grown in place of common non-leguminous fodder crops for animals during the Kharif season. For turmeric and ginger cultivation, add 25-30 tonnes of well-decomposed farmyard manure (FYM) per hectare at the time of final ploughing and complete sowing by mid-May. Lac producer farmers should spray Kusum-Jethwi on Kusum or other trees based on the infestation of enemy insects from the day of insect infestation (Jethwi). After 90-95 days of insect infestation, prepare a solution by mixing 20 ml of Spinosad (Success) 2.5% SC organic insecticide, 10 grams of Carbendazim, or 10 ml of Hexaconazole insecticide and



Management for lac cultivation

fungicide in 10 litres of water. This solution should be sprayed on the lac-coated branches using a sprayer (make sure to wear a mask while spraying). For Rangini-Baisakhi Beehan lac crops, cut the Ari Lac from the Plum trees and prune the entire Plum tree. Farmers who have not yet pruned the vacant Palas trees should complete this work as soon as possible.

LIVESTOCK

Proper vaccination schedules should be followed before the onset of the rainy season to prevent FMD and Black Quarter disease in cattle and buffaloes. Vaccination of sheep and goats against PPR and Enterotoxemia is also recommended. For dairy units, farmers should prepare feed from locally available ingredients such as maize, wheat, and pulses. It is advised to provide 20-25 grams of mineral mixture as supplementary feed to the dairy animals. Feedings should be done during the cooler parts of the day, such as early morning or late evening. Incorporating green fodder and moist feed will help with hydration. Avoid spoiled feed to prevent digestive issues. Proper sanitation and hygiene are essential. The floor of the animal shed should be cleaned regularly to prevent disease infestation, and milk utensils and equipment should be frequently cleaned. Provide adequate space to the animals to prevent overcrowding.

Piggery

For pig units, farmers are advised to grow green fodder and add 40% green matter with 60% concentrated fodder, such as rice husk, maize, wheat husk, oil cake, mineral mixture, and salt. To prepare concentrated feed, mix 60% rice husk, 20% ground corn, 20% oil cake, and 20% rice bran. Also add 10% wheat bran, 8% salt, and 1.5% vitamin and mineral mixture. Providing extra immunity boosters like zinc and selenium in the feed is recommended. Ensure pigs are vaccinated against common monsoon-related diseases like Swine Fever and Foot and Mouth Disease (FMD).

Poultry

For poultry units, cleaning the poultry shed every other day is essential. Regularly sprinkle bleaching powder or lime to disinfect the area, and all equipment and sheds should be properly disinfected. In backyard poultry farming, birds should be allowed to roam in the open yard during the day. Deworm the birds using Piperazine Hydrochloride at 5 ml per 10 birds. Vaccinate the birds against Ranikhet and fowl pox. Provide clean and fresh water three times a day. For adult birds, mix 2 grams of Electrol powder and 5 ml of Vimeral in 5 litres of water. If any symptoms such as drowsiness, cough, cramps in legs, or changes in stool color are observed, contact the doctor immediately.



Diagnostic visit to poultry farm



Conserve water in surface and sub surface horizons of soil, which in turn address drought related issues in coconut, arecanut, pepper, cocoa, nutmeg etc. Intercultural operations in uplands during the onset of south west monsoon and subsequent operations during the offset of North east monsoon will help in conserving surface moisture helping almost all crops in home steads and uplands especially in plain lands of High hills and mid land laterites. Use gypsum to manage soil acidity by blending with conventional soil ameliorants in equal proportion is beneficial in increasing the nutrient use efficiency of almost all upland crops like Coconut, Arecanut, Spices like Pepper, Cardamom, Banana and vegetables by managing the acidity in subsurface soil.

CEREAL CROPS

Rice (Wetland)

During first crop season, rice can be cultivated under two situations; wetland and upland. Under wetland land situation, two methods can be followed, direct sowing and transplanting. For direct sowing, rice is sown in April before the rains start. Wet seeding is done after the receipt of rain. Land should be prepared well by thorough ploughing. Use 32-40 kg seeds per acre. Seed treatment with *pseudomonas* @ 10 g / kg seed should be adopted before sowing. Basal dose of lime @ 140 kg / acre should be applied at the time of first ploughing. Organic manure @ 2 t / acre should be applied while ploughing. Weeds are a major problem in direct sown rice. Hence herbicides like Butachlor 50 EC @ 1 l /acre or Pretilachlor 540 EC @ 600 ml / acre or Pretilachlor + Bensulfuron methyl @ 4 kg / acre may be applied 6 - 9 days after sowing in case of wet seeded rice and 0-6 days in case of dry seeded rice.

For transplanted paddy use 24 - 34 kg seed per acre. Seeds should be treated with *Pseudomonas fluorescens* @ 10 g/ kg seed. For transplanted rice, nursery should be prepared. Wet method can be adopted in areas here water is available. Otherwise, dry nursery can be raised. It is better to adopt dry method in first crop season, if the time of planting is uncertain and depends on the receipt of rains. Nursery beds of 5 -

10 cm height and 1.5m width are recommended. The height of the beds can be increased to 15 cm in dry nursery. One tenth of the main field area should be taken for nursery. Organic manure should be applied at 1 kg / m². If nitrogen deficiency symptoms are noticed, urea can be applied @ 1 kg / 100m².

Short duration varieties should be transplanted at an age of 18 days, medium duration at 20 -25 days and long duration at 30 days. In first crop season, transplanting can be delayed up to 35 days in medium duration varieties and up to 25 days in short duration varieties, if growth is slower. Paired row planting with a spacing of 35 cm between the pairs and 15 cm within the pair will improve the growth of the plants, make intercultural operations easier, reduce pest and disease incidence and increase yield. Date of planting may be adjusted so as to avoid synchronisation of the critical stages of maximum tillering or heading with the usual flood period in the tract.

If water shortage is experienced in the season due to unexpected reduction in monsoon showers, irrigation to maintain 5 cm water level two days after the disappearance of ponded water is sufficient. Apply fertiliser dose of 70: 35: 35 kg NPK / ha for high yielding short duration varieties, 90: 45: 45 kg NPK / ha high yielding medium duration varieties and 40: 20: 20 Kg NPK / ha for local varieties. Phosphorus should be applied fully as basal dose at the time of transplanting. Nitrogen should be applied as 2-3 split doses as basal dose, at tillering and 5 - 7 days before panicle initiation. Potassium fertilisers should be applied as two equal splits as basal and 5- 7 days before panicle initiation. In case of direct seeded rice, nitrogen should be given as three equal splits. Second dose of lime @ 100 kg / acre may be applied one week before first top dressing.

A broad-spectrum weedicide, Pretilachlor + Bensulfuron Methyl 0.66 WP @ 4 kg/ acre may be applied at 0-6 days after transplanting. To manage pest and diseases, release egg parasitoid, *Trichogramma japonicum* and *Trichogramma chilonis* @ 2 cc/ acre for six weeks starting from one week after transplanting and three weeks after transplanting respectively for the management of rice stem borer and leaf folder. In areas endemic to Brown plant hopper, avoid closer planting, provide alley ways after every 3 m and apply nitrogen in split doses. Chances of incidence of bacterial leaf blight is higher during first crop season.

Biocontrol agent *Pseudomonas fluorescens* should be applied as seed treatment @10g/ kg seed, seedling root dip @ 1 kg/ acre during transplanting and foliar spray @ 20g / litre one month after transplanting. When incidence is noticed, apply bleaching powder @ 2 kg/ acre in small pouches in standing water to avoid spread of the disease. Foliar spray with cow dung slurry @ 20g/ l will manage the disease. In severe cases, apply streptocyclin @ 40 g/ acre. False smut is likely to occur if humidity is high during inflorescence emergence. A prophylactic spray of propiconazole 25 EC is recommended @ booting stage to prevent false smut.

Rice (Upland)

Rice is grown in upland situation in certain areas during rainy season. Local varieties such as modern varieties and high yielding short duration varieties such as Harsha, Annapoorna, Aiswarya, Swarnaprabha, Mattathriveni and Vaishak are suitable for

mortality. During breeding season, one male against every 50 females should be used for breeding to avoid inbreeding. Ensure deworming of sheep and goats every 6 months using a broad spectrum anthelmintic. Vaccinate your animal against FMD, PPR, Goat Pox and Sheep Pox. Give enterotoxaemia vaccination @ 2 ml s/c to pregnant Ewe/Doe before one month of expected date of lambing/kidding. Provide 25-50 g molasses per day to each pregnant Ewe/ Doe to prevent pregnancy toxemia. Follow rotational grazing, with a grazing period of 8 hrs during winter and 10 hours in summer. Give amprolium as a drench in kids to prevent coccidiosis in young ones. Herders are advised to take care of nutrition of pregnant and lactating ewes and does especially during winter and supplementary feeding with local barley or concentrate feed should be given @ 200-300 gms/day for adult and 150-200gms/day for young ones.

Treatment with Ivermectin for ticks and mites in sheep and goats should be done during winters and Dipping with Butox or N-cidol during summer months soon after combing/shearing. Stock enough dry fodder/ concentrate feed/ barley for the herds to feed then during heavy snow days. Practice Drum silage making using locally available ingredients (developed by KVK- Leh) to meet the nutrient requirement and to meet scarcity of fodder

Back Yard Poultry

The poultry farmers are recommended, disinfect the premises of poultry houses with 1% sodium hypochlorite and inhibit the entry of outsiders to the poultry houses and premises. The litter materials in poultry shed need to be changed to prevent coccidiosis infestation. If red diarrhoea symptoms are noticed in the birds, immediately administer anti- coccidial drugs (in the drinking water for 3 -5 days.)

Poultry birds may be fed with green leaves along with grinded maize/rice. The saw dust should be dried completely in sun before using as bedding material. Apply hydrated lime powder @ 1-2 kg per 10 sq.ft area in the shed before birds are placed in the shed. Add 8% sugar solution containing 1% sodium chloride during first 15 hours after placement. Maintain proper shed temperature (35 C) during the first week of brooding. Provide good ventilation in the shed during the first three weeks of brooding. Provide minimum 23 hours light in the brooder continuously. The adult birds should be allowed to move out in the open yards during daytime. Vaccinate the adult birds against Ranikhet disease @ 0.5ml intramuscular after 2 months of age. Provide Layer mesh @ 20gms/bird in addition to free range feeding to the layers for good egg production. Deworm the birds using piperazine hydrochloride @ 5ml per 10 birds. Provide a minimum light of 16 hours for adult birds. Use anti-stress vitamins such as vimeral @ 10ml per 10 birds for 3 days in a week.



CEREAL CROPS

Rice

Land preparation: During the summer months, broadcast the seeds of green manuring crops such as dhaincha, sunhemp, black gram, or green gram, and incorporate them into the field just before flowering stage to maintain soil health. Land preparation for dry nursery may be done with the onset of pre-monsoon showers. Perform deep ploughing to destroy pest larvae, and level the land to ensure uniform water distribution. Considering the current weather conditions, farmers are advised to prepare rice nurseries. For transplanting in one hectare, approximately 800-1000 m² of nursery area is sufficient. The nursery soil should be well pulverized and divided into beds 1.5 meters wide and of convenient length. During nursery preparation, an adequate quantity of well-decomposed FYM should be incorporated into the soil.

Varieties and seed treatment: Use high-yielding varieties such as Pusa Basmati 1718, PB 1985, PB-1847, PB-1885, PB-1886, Danteswari, JR 206, Pusa Basmati 1121, and Pusa Sugandha 5, Mahamaya, MTU-1318, Swarna Sub-1 (flood tolerant) for better production. Before sowing, seeds should be treated with 1 g of *Streptocycline* per 10 liters of water for every 5 kilograms of seed and Vitavax @ 2.5 g/Kg seed or Carbendazim 12% + Mancozeb 63% @ 2.5 g/kg seed. The ideal time for transplanting is from June 15th to July 15th, with a row-to-plant spacing of 20*10-15 cm.

Weed management: Paddy should be sown in line and pre-emergence herbicide like Pretilachlor 50% EC @ 1500 ml/ha or Pyrazosulfuron 10% WP @ 200 ml/ha should be applied within 3 days after sowing. Followed by post emergence application of Bispyribac sodium (10%) @ 250 ml/ha.

Pest and disease management: For management of stem borer and leaf folder, release *Trichogramma* cards @ 50,000 to 75,000 per hectare at 21, 30, and 45 days after transplanting. To effectively control stem borers, apply Profenofos 40% + Cypermethrin 4% @ 1000 ml/ha or Chlorantraniliprole 18.5 SC @ 200 ml/ha.

Additionally, Install pheromone traps @ 5 traps/ha to monitor and manage stem borer populations in the rice field. Regularly monitor rice fields for brown plant hopper infestations. If the pest population exceeds the economic threshold level, spray Imidacloprid 17.8% SC at 250 ml/ha. In case of severe infestations, spray Dinotefuran 10% + Pymetrozine 40% WG @ 250 g/ha. To control blast disease in paddy, treat the seeds with Tricyclazole @ 2 g/kg and spray Propiconazole 25 EC @ 500 ml/ha. If large, muddy-colored spots with purple margins appear on rice sheaths above the water surface, indicating sheath blight, spray Trifloxystrobin 25% + Tebuconazole 50% @ 0.5 gm/L or Carbendazim 50% WP @ 2 gm/L on the affected areas.

Repeat the spray after 12-15 days, if necessary. To conserve rainwater in fields, construct bunds that are high and wide to maximize water retention.

Maize

Use of high-yielding varieties such as Vivek Hy-17, Vivek Hy-43, Pusa Vivek QPM-9, VLQPM-45, Hybrid Maize-1, HQPM-1, Vivek Makka Hy-51, Pusa HM-4, DMRH-1308, IMH 230, and IMH 231. A germination test is essential to ensure an optimum number of healthy plants in the field. This should be followed by bio-priming or seed treatment using Carbendazim 50% WP @ 2 g/kg seed, Imidacloprid 600 FS @ 2 g/kg seed, *Trichoderma* @ 10 g/kg seed, and *Azotobacter* @ 10 g/kg seed to enhance productivity. The optimum sowing time is from June 15th to June 30th, with a spacing of 60 x 25 cm (row x plant). Apply balanced fertilizers, including zinc-based fertilizers, for better crop growth.

For weed control, apply Tropicamezone 33.6% SC @ 30 ml per acre as a post-emergence spray at 20-25 days after sowing. The pupae of Fall Armyworm and stem borer, as well as the spores of *Rhizoctonia solani* (the causal agent of sheath blight), remain in the soil. Therefore, deep summer ploughing is crucial to reduce the inoculum potential. The infestation of Fall Armyworm can be significantly minimized by dry sowing or



Integrated Weed Management

sowing at the onset of the monsoon, practicing crop rotation, and alternating maize with other crops that are not affected by Fall Armyworm. Additionally, apply a basal dose of nutrients and use the ridge and furrow method of bed preparation. These practices should be completed by the last week of May or the first week of June. Monitor regularly to prevent Fall Armyworm infestations. If necessary, spray Spinetoram 11.7 SC @ 350 ml/ha or Emamectin Benzoate 5 SG @ 200 g/ha. To control leaf blight disease, take up spray of Mancozeb 75 WP @ 1000 g/ha.

PULSE CROPS

Green gram

Use of high-yielding varieties such as IPM 410-3 (Shikha), IPM 205-7 (Virat). Before sowing, mix 2.5 kg *Trichoderma* with 1 quintal of FYM and incorporate it into the soil, to control soil borne diseases. Use of 20-25 kg/ha of improved green gram seed for sowing. To control yellow mosaic virus, treat the seeds with Imidacloprid 48% FS at 2 ml/kg of seed.



The Kharif season, stretching from June to October, is a crucial agricultural period for the state of Maharashtra, significantly influencing the livelihoods of millions of farmers. Characterized by the onset of the southwest monsoon, this season supports the cultivation of key crops such as rice, cotton, soybeans, pulses, and millets. However, climate variability, erratic rainfall, and changing weather patterns pose continuous challenges to sustainable agricultural production. In the light of these challenges, the Kharif 2025 Agro-Advisory aims to provide timely, region-specific, and science-based recommendations to empower farmers with the knowledge needed to make informed decisions.

REGIONAL ADVISORY

Maharashtra is divided into several agroclimatic zones, each with unique characteristics influencing crop selection, sowing windows, and input management. The following region-wise advisories aim to support farmers with tailored recommendations for improved productivity and resilience.

VIDARBHA REGION

The Vidarbha region of Maharashtra is composed of Bhandara, Chandrapur, Gondia, Gadchiroli, Nagpur, Amravati, Akola, Yavatmal, Buldhana, Wardha, Washim and possesses state agricultural university viz., Dr. Panjabrao Deshmukh Krishi Vidyapeeth, Akola.

Soil Testing & Preparation (All Crops)

Soil testing is crucial for effective fertilizer management, optimizing input use, and maintaining soil fertility. For proper soil sampling, the recommended depths are 15-20 cm for crops like sorghum, rice, groundnut, and wheat; 30 cm for cotton, sugarcane, and banana; and 30 cm for fruit trees, sampled from a distance of 1-1.5 feet from the trunk. Additionally, applying decomposers to crop residues like straw and leaves enhances organic matter content, improves soil structure, and boosts water retention, contributing to sustainable agricultural practices.

FODDER MANAGEMENT

Use Maize African Tall and new varieties of hybrid napier like Phule Gunwant, DHN-6, BNH-10 and Super Napier for round the year availability of green fodder. Excess green fodder should be converted into silage.

MILCH ANIMALS

Provide clean, hygienic and plenty amount of drinking water to milch animals. To reduce the stress of heat in farm animals, provide roughages by mixing with solution of 1% jaggery and 0.5% salt separately. Milch animals should be offered 50 gm of mineral mixture daily in two divided doses to maintain milk yield and to overcome the problem of anoestrous cycle. Cow teats should be dipped in 2 % solution of Potassium Permanganate after every milking to avoid Mastitis in milch animals. Regular deworming should be done after every two months for one year for good health and weight gain of calves.

Buffalo

Allow buffalo for wallowing in water as they have fewer sweat glands compared to cattle to maintain their body's thermo-equilibrium. Water-soaked gunny bag or other cloth should kept on a body to avoid occurrence of heat stroke.

Goat

Animals should be vaccinated before monsoon season for Enterotoxaemia and Peste des petes (PPR) ruminants' diseases to avoid great losses. Improved breeds like Osmanabadi should be used for higher income. Regular deworming should be done after every month to goat kids for six months for good health and weight gain.

Poultry

In the summer the poultry shed should be kept cool. Plants more trees around the shed. The roof of the shed should be covered with grass, so that the temperature remains under control. Exhaust fan should be kept in the shed, so that the hot air inside shed is thrown out. Use foggers, so the shade will cool down. Improved backyard poultry breeds like Grampriya, Shrinidhi, Kaveri, Giriraja and Vanaraja should be used to boost up the income of small and marginal farmers.

WESTERN MAHARASHTRA

Western Maharashtra is comprised of Ahmednagar, Pune, Satara, Sangli, Kolhapur, Solapur. The region has moderate rainfall and Black to medium black, well-drained soils. This region has state agricultural university viz., Mahatma Phule Krishi Vidyapeeth, Rahuri, Ahmednagar.

CEREAL CROPS

Rice

In Western Maharashtra, nursery sowing for Kharif rice generally begins in late May to mid-June, aligning with the onset of monsoon. Recommended varieties include Phule Radha, Indrayani, Sahyadri, and Kolhapur-1, chosen for local adaptability and consumer preference. About 40-50 kg of seed per hectare of main field is required for a nursery area of 800-1000 m². Seeds are soaked in water for 24 hours, then incubated for another 24-36 hours to promote uniform germination. Treat seeds

with Carbendazim or *Trichoderma* to prevent fungal diseases. Use raised beds (1-1.5 m wide, 15-20 cm high) with good drainage to prevent waterlogging, common in clayey soils during monsoon. The soil should be well-pulverized and enriched with FYM (Farm Yard Manure) at 5-10 t/ha. Rice nurseries are typically raised in medium to heavy black soils with good water retention. Adequate but controlled irrigation is necessary to avoid overwatering.

Fertilizer Application: Apply urea, SSP, and MOP as a basal dose (e.g., 1 kg urea, 0.5 kg SSP per 100 m²). Top-dress with urea 10-12 days after sowing.

Weed Management: Hand weeding after 10-15 days or pre-emergence herbicides (like Butachlor) can be applied carefully depending on weed load.

Pest and Disease Control: Monitor for early signs of blast, sheath blight, and stem borers. Use recommended insecticides/fungicides only when needed to avoid resistance. Use of Mat-Type Nursery (for Machine Transplanting) is recommended. In areas using mechanical transplanters, mat-type nurseries are promoted using plastic trays or polythene sheets.

Nursery production: The seedlings are usually ready for transplanting in 20-25 days when they reach 3-4 leaf stage and 15-20 cm height. Drain water 2-3 days before uprooting to harden seedlings and reduce transplanting shock. Use of organic amendments, biofertilizers like *Azospirillum* and phosphate solubilizing bacteria is gaining ground among progressive farmers. For transplanted rice, 30-35 kg seed per hectare is used, and for direct seeding, 75-100 kg per hectare is recommended. For dibbling method use the seed rate @50 to 60 kg/ha. Seeds should be treated with fungicide (like Thiram @ 2.5 g/kg seed) and biofertilizers such as *Azotobactor* and PSB @ 25 gm/kg of seed for better seed germination. The recommended dose is 120-150 kg Nitrogen, 60-75 kg Phosphorus (P₂O₅), and 50-60 kg Potassium (K₂O) per hectare. Zinc sulphate (25 kg/ha) and sulphur (15 kg/ha) may also be applied. Use pre-emergence herbicide Pendimethalin (1.0 kg a.i./ha) and post-emergence herbicides like Bispyribac-sodium. Manual weeding can be done if necessary.

Rice field should have a depth of 5-7 cm water during the growing period, especially during tillering, flowering, and grain filling stages. to prevent grain shattering or bird damage. For the management of stem borer, apply Carbofuron granules (Furadan 3G) @ 33 kg/ ha in standing water after 10 days of transplanting. Soaking of 1 ha seed (20-30 kg seed) in 25 litre water solution with 25 g bavistin and 2.5 g Streptocycline is done for 24 hrs. Treated seed should be covered with wet gunny bags for 24-36 hrs for getting germination of seeds. For organic control insect pest, apply need based biopesticides, *Trichogramma*, yellow sticky traps etc. Biocontrol agents like coccinellids, spiders, damsel flies should be conserved. Management of Stem borer is observed @ 4-5% in paddy field, spray with Quinalphos @ 32 ml in 10 litres of water.



Grape

After pruning, grapevines are treated with Hydrogen Cyanamide (30-40 ml/L) on the top 2-3 buds to encourage growth. GA sprays of 10 ppm are applied 17-18 days after pruning, followed by 15 ppm after 4-5 days. Mealybug control in September involves removing bark and spraying Methomyl 40 SP at 1 g/L

Banana

Banana is planted on raised beds with a paired row system, ideally after 2-3 monsoon showers. Varieties like Grand 9 and Basarai are recommended. Green manuring and proper drainage are crucial during heavy rains. Preventive soil applications of *Pseudomonas* and *Trichoderma* help manage diseases. Practices like de-navelling during fruiting, irrigation at fruit development, and spraying Propiconazole for leaf spot management are recommended for healthy crop growth. During flowering/fruiting stage, carry out de-navelling (removal of male inflorescences for nutrient diversion) practices for better growth of banana fruits. Apply irrigation at fruit development stage. At the time of planting treat the sucker with Mancozeb (3g/l) and Monocrotophos @ 2ml/l, apply *Trichoderma viride* @ 5 kg) + FYM @ 100 kg for management of soil borne diseases. For management of Sigatoka leaf spot disease in banana orchard, spraying of 0.5 ml Propiconazole 25 EC + Mineral oil 1% per litre of water.

Onion (Kharif)

For effective nursery preparation, raise beds enriched with FYM, copper oxychloride, and a balanced mixed fertilizer. Prior to sowing, treat seeds with Thiram or Carbendazim at 2-3g/kg to prevent fungal infections. Apply Basalin at 25ml per 10 L of water as a pre-emergence herbicide for effective weed control. At planting, use a fertilizer dose of 100:50:50 NPK, with 50:50:50 applied at sowing and the remaining 50?N at 45 days after transplanting (DAT). To manage fungal diseases, integrate *Trichoderma* along with regular fungicide sprays every 10-15 days. For thrips control, install sticky traps and use Profenofos, Fipronil, or Spinosad as required based on pest observation.

KONKAN REGION

Konkan Region of Maharashtra is comprised of Mumbai, Mumbai Suburban, Thane, Palghar, Raigad, Ratnagiri, Sindhudurg). The region has high rainfall (3000-4000mm) and has Lateritic, acidic, high organic matter. This region possesses a state agricultural university viz., Dr. Balasaheb Sawant Konkan Krishi Vidyapeeth, Dapoli.

CEREAL CROPS

Rice

Rice nursery sowing should begin with the onset of monsoon showers, typically in the first to the second week of June. It is crucial to select suitable varieties like Ratnagiri-1, Karjat-3, Karjat-4, and Sahyadri based on local climatic conditions and market demand. For one hectare of the main field, approximately 60-75 kg of seed is required, and a nursery area of around 1000 m² should be prepared. Soak rice seeds in water for 24 hours, followed by incubation for 24-36 hours. Treat the seeds with fungicides like Carbendazim (2g/kg seed) or biocontrol agents such as *Trichoderma*

to prevent seed-borne diseases. In regions with heavy rainfall, wet bed nurseries are recommended, especially on low-lying lateritic or clay loam soils. Where drainage is poor, raised beds should be prepared. Select well-drained and fertile fields near water sources, avoiding waterlogged or saline areas. Apply 2 kg of urea and 1 kg of SSP per 100 m² of nursery, and incorporate well-decomposed FYM or compost (5-10 t/ha) during bed preparation.

Pre-germinated seeds should be uniformly broadcasted on nursery beds with light pressing to ensure proper soil contact. Maintain a shallow water level of 2-3 cm during the early growth stage, avoiding stagnation to prevent diseases. Weed pressure is generally low, but light raking or manual weeding should be done 10-12 days after sowing if required. To manage weeds effectively, spray Butachlor 50 EC @30 ml/litre water 2-3 days after sowing. Transplant 21-day-old seedlings with a spacing of 20×15 cm for normal varieties and 15×15 cm for short-duration varieties. At transplanting, apply 10 t/ha of Gliricidia leaves or 8 t + 50 kg of nitrogen for better puddling and root establishment. Give the second nitrogen dose (1 kg urea/guntha) 15 days after transplanting to boost growth. Replant damaged or gap areas promptly with healthy seedlings to maintain uniformity

Fertilization and Pest/Disease Control: Apply 3 kg of compost per m² of nursery bed before sowing. For mechanized farming, mat nurseries on plastic sheets or trays are recommended, with fertilizers applied at 2.2 kg urea, 3 kg SSP, and 0.8 kg MOP per guntha. For effective pest control, monitor the nursery for stem borers and use Quinolphos (150 ml/guntha) or Carbofuran (165 g/guntha) as needed. Crab control can be managed using toxic bait prepared with Acephate or Carbaryl mixed with cooked rice. Monitor for diseases like blast, brown spot, and root rot, and apply fungicides and insecticides as per local recommendations.

MILLET CROPS

Finger Millet (Nagli) & Proso Millet (Vari)

For effective nursery preparation, sow the rice seeds in rows on 10 guntha nursery beds using a seed rate of 6 kg per hectare. Fifteen days after sowing, apply Quinolphos at 15 kg/ha or Carbofuran at 16.5 kg/ha to manage early pest infestations. To prevent blast disease, spray Zineb at 2 kg or Copper Oxychloride at 1.25 kg mixed in 500 liters of water per hectare. Transplant healthy 4-5-week-old seedlings at a spacing of 15×20 cm, and ensure proper nutrient application with a recommended dose of 40:40 N:P₂O₅ kg/ha to promote vigorous growth and better yield.

OILSEED CROPS

Groundnut

Incorporate 5-7.5 t/ha of well-decomposed FYM during the final tillage to enhance soil fertility. At the time of sowing, apply 25 kg of Nitrogen and 50 kg of Phosphorus per ha using boronated SSP for better root development. Use 100-110 kg of seed per hectare from recommended varieties like Konkan Bhuratna and TAG-24. For effective disease control, treat seeds with Thiram at 5 g/kg along with Rhizobium culture at 25 g/kg before sowing. Follow a sowing pattern of 30 cm paired rows with



CEREAL CROPS

Paddy

Select high-yielding, disease- and insect-resistant rice varieties like RC Maniphou-13, RC Maniphou-14, RC Maniphou-15, and RC Maniphou-16. Treat seeds with *Trichoderma* or *Pseudomonas* (5-10 g/ha) to control seed or soil-borne diseases. Soak seeds in a recommended fungicide solution to prevent seed-borne diseases. The optimum age of seedlings for transplanting is 18-22 days for short-duration varieties, 25-30 days for medium, and 35-40 days for long-duration varieties. Transplanting should be done when seedlings are free from weed seedlings. Puddle fields with 5-10 cm of standing water after ploughing to break up clods and churn the



IPM in rice for management of leaf folder pest

60 kg N, 30 kg P_2O_5 , and 30 kg K_2O per hectare for optimal crop nutrition. Millets can be intercropped with soybean. Conduct three hoeings with a hand hoe during early growth stages for effective weed suppression. Spray Mancozeb (0.2%) at early crop stages if Blast, Brown Spot, or Rust symptoms appear. Rogue out diseased plants to manage spread and for armyworms, cutworms, and leaf-scraping beetles, apply Thiamethoxam @ 5 g/15 litres of water for effective protection.

PULSE CROPS

Blackgram

Conduct deep summer ploughing to expose and destroy the eggs, larvae, and pupae of major pests like pod borer, caterpillars, and weevils. Mix FYM/compost into the soil 10-15 days before sowing. Ensure the field is free from weeds for optimal crop growth. Use high-yielding varieties like PU 31 and LBG-20 (Teja). PU 31 is tolerant to Yellow Mosaic Virus (YMV) and well-suited for Kharif cultivation, while LBG-



Varietal demonstration in blackgram

20 performs well in all seasons. Use a seed rate of 10-15 kg/ha (or 2.5-3.75 kg/sangam). Sowing can be done by broadcasting or dibbling at a spacing of 15 cm × 30 cm. If required, apply lime in the furrows at sowing, preferably every alternate year, based on soil test recommendations to manage soil acidity. Treat seeds with Carbendazim @ 2g/kg of seed before sowing to protect against seed-borne diseases and promote healthy germination. Apply 50 kg potash/ha to the crop raised with closer spacing (20X10cm) to reduce stem fly and pod borer incidence. Treat the seeds with Carbendazim 50WP or Thiram @ 2g/kg of seeds or with Trichoderma viride @ 4g/kg of seeds before sowing. Install yellow sticky traps @ 10 traps/ha in the field to manage aphids. Spray Dimethoate 30EC @ 1ml/litre of water or Imidacloprid 17.8SL @ 0.5ml/litre of water for managing sucking pests.

OILSEED CROPS

Groundnut

Perform one ploughing with a soil-turning plough followed by two harrowings to achieve fine tilth up to 12-18 cm. Sow from 1st week of May to 1st week of June for optimal growth. Use ICGS 76 (semi-spreading, matures in 120-125 days) and JL 27 varieties. Treat groundnut seeds with Dithane M-45 or Thiram @ 3 g/kg to protect from seed and soil-borne diseases. For both soybean and groundnut, seed treatment is crucial under dry conditions. Keep the groundnut field weed-free for up to 50 days after sowing, as weed pressure is high due to monsoon rains. Timely inter-cultivation and weeding are essential for higher yields. Perform earthing up at 30-35 days after sowing in groundnut to promote pegging and pod development. Promote groundnut and soybean in foothill areas with limited irrigation, where rice cannot be grown, as climate-resilient and economically viable alternatives.



Stocking fingerling



Breeding



Water Quality Management

to disinfect the pond and stabilize the pH, with half the dose applied initially and the remainder in monthly installments. Fill the pond with freshwater to a depth of 1-1.5 m and fertilize using a combination of organic manure (e.g., Raw Cow Dung and Poultry Manure) and inorganic fertilizers (urea and Single Super Phosphate). For breeding, start with common carp in February until April, and Indian Major Carp (IMC) from April to June. In nursery management, eradicate weeds and pests, check water quality, and ensure adequate natural food by applying lime and fertilizers. Stock 30-50 lac fry per hectare and follow a structured feeding schedule to maximize survival and growth, gradually increasing feed intake from 400% to 800% of initial body weight over the first 15 days, with harvesting on day 17. When stocking fingerlings, ensure the proper species ratio in composite fish culture, with carp fingerlings at a density of 8,000-10,000 per hectare. Regular feeding should be based on total fish biomass, with a daily feed mixture of rice bran and oil cake or sinking/floating pellet feed. Monitor water quality, maintain ideal pond conditions with frequent manuring, and always observe for signs of disease, keeping a detailed pond diary for all activities and inputs. Staying vigilant for abnormalities in fish behavior and taking prompt action is essential for maintaining healthy fish stocks.

quintals per hectare and maintain a spacing of 30 cm × 30 cm. Before sowing, treat rhizome seeds by dipping them in a solution of *Trichoderma viride* @ 4 g/litre of water for 5 minutes or *Trichoderma harzianum* @ 10 ml/litre to prevent rhizome rot. For already sown crops, carry out proper weeding and earthing up to promote healthy growth. Ensure planting in ridges and furrows with good drainage. Farmers may also opt for intercropping with crops like soybean, rice bean, French bean, finger millet, or maize for better resource use and income diversification. To conserve soil moisture and control weeds, mulch the beds using dry paddy straw or green leaves. For effective disease management, especially soft rot, spray *Trichoderma viride* or *Trichoderma harzianum* @ 10 ml/litre or neem oil (10,000 ppm) @ 2 ml/litre as a preventive measure.

LIVESTOCK

Piggery

Farmers are recommended to select pig breeds like Lumsniang or Hampshire cross; supplement mineral mixture (50-60g/day) and calcium for lactating/pregnant sows; use local feed alternatives like banana pseudo stem, colocasia, and sweet potatoes. Ensure adequate clean drinking water and proper feed. Deworm pigs with Albendazole/Febendazole (7.5mg/kg body weight every 3 months); vaccinate against Swine Fever; regularly provide 20g of mineral mixture per pig with feed; and isolate sick pigs in a quarantine shed until fully recovered. Maintain dry, clean pig sheds with proper roofing and walls; avoid waterlogging; disinfect farm entrances and exits with 1% formaldehyde or 2% NaOH; and ensure footwear and equipment are disinfected. Quarantine newly purchased pigs for 4 weeks before introduction. Implement strict measures for African Swine Fever (ASF) by quarantining infected farms, prohibiting the movement of animals and personnel between farms, and ensuring the proper disposal of carcasses (deep burial with lime/bleaching powder). Avoid swill feeding unless properly boiled; maintain proper records of animal health and visitors; and prohibit animal movement between farms to reduce disease risk. Report suspected ASF cases to veterinary authorities immediately.

Poultry

Farmers should maintain clean, ventilated poultry sheds with proper space (1 ½ sqft/bird); avoid overcrowding and heat stress in birds. Regularly change litter materials to prevent coccidiosis and maintain hygiene. Provide clean drinking water 3-4 times a day, occasionally adding electrolytes. Ensure the water is free from contamination and provide proper feed to maintain bird health. Backyard poultry



Vaccination in poultry

breeds like Vanaraja, Shrinidhi, and Kuroiler can be reared in low-cost housing systems. Ensure the houses are clean, dry, and properly ventilated to prevent infections and diseases. Administer anticoccidial drugs in drinking water for 3-5 days

if red diarrhoea symptoms are observed. Clean the poultry house regularly with phenol (1 ml/l of water) to prevent infections. Vaccinate birds against diseases like Ranikhet, Marek's, Gumboro, and Fowl pox as per the vaccination schedule. Report any unusual deaths or illness to the nearest veterinary center immediately. Avoid waterlogging near poultry houses to prevent flies and mosquitoes, ensuring a clean and dry environment. Turn litter materials weekly to maintain hygiene and reduce disease risks.

Fishery

Farmers are advised to carry out de-silting of ponds to facilitate further fish farming, followed by the application of lime and repairs to ponds, dykes, and inlet-outlet systems. Pre-stocking management for composite fish culture includes clearing aquatic weeds and regularly netting out predatory and weed fish to maintain a healthy ecosystem. Any surplus common carp brood fish should be disposed of accordingly. It is also the breeding season for exotic carps and Indian major carps, so breeders should begin production of fish fingerlings. Farmers are advised to avoid exceeding a stocking density of 10,000 fingerlings per hectare. Fish stocking should follow a ratio of 3:4:3 (surface feeder: marginal feeder: bottom feeder) for optimal growth. Additionally, fish growers can prepare fish feed at home by mixing mustard cake, rice polish, and cooked rice. For paddy-cum-fish farming, land preparation should begin to ensure proper conditions for both crops.

MUSHROOM

Growing oyster mushrooms (*Pleurotus spp.*), especially *Pleurotus djamor* (pink oyster mushroom), indoors using paddy straw can provide additional income for farmers and improve the nutritional status of their diet.

SELF-HELP GROUP/FPOs

Women members in SHGs should establish Nutrition Gardens in each household to provide fresh vegetables rich in vitamins and minerals, enhancing immunity. Vegetables such as brinjal, capsicum, chili, and cucurbitaceous vegetables can be grown in the kitchen garden. Additionally, SHG members can make masks, hand gloves, and cotton clothes for their family members and the neighbouring community, ensuring safety while generating additional income. FPOs should begin their business by producing high-quality seeds and planting materials through organic cultivation of ginger, turmeric, and pineapple, along with value addition.

Harvest physiologically mature cobs (intended for seed) as early as possible to avoid damage from predicted moderate to heavy rainfall.

Upland paddy

For rainfed upland paddy cultivation, farmers should begin by treating seeds with Carbenazim (2-3g/kg seed) to prevent fungal diseases and ensure healthy germination. The dibbling method is strongly recommended over broadcasting to optimize seed usage and improve germination rates. Farmers should select from improved drought-tolerant varieties such as Fazu-MZ UPR 4, Maibasa-MZ UPR 25, Bhadepui-MZ UPR



Upland rice cultivation

23, Biroyin-MZ UPR 24, Sazukthau-MZ UPR 6, Lalron-MZ 28, Idaw-MZ UPR 11, Buhban-MZ UPR 2, and MZ UPR 33, which are specially developed for Mizoram's upland conditions. Implement critical weed management with first weeding by mid-May followed by a second weeding between the first week of June to second week of July. For soil conservation in jhum fields with slopes exceeding 30%, construct log wood or bamboo bunds across slopes and dig recharge pits (1×1×1 ft) to prevent severe soil erosion and retain topsoil.

Lowland Paddy

Nursery Preparation(2nd Week of June): Prepare raised nursery beds with finely tilled soil and apply 2 t/ha of Farm Yard Manure (FYM) along with 2 kg of *Azospirillum* per 1000 square meters. Two to three days before sowing, incorporate 10 quintals of well-decomposed FYM per 1000 square meters into the nursery beds. Use a seed rate of 3.5-4 kg per 100 square meters of nursery area to raise sufficient seedlings for transplanting in one hectare. For improved germination, treat the seeds with Zinc sulphate priming before sowing. After sowing, cover the seedbed lightly with a thin layer of soil or compost to ensure proper seed establishment.

Land Preparation (From 1st week of June): Begin puddling with the onset of monsoon and ensure proper levelling for uniform water distribution and to prevent water stagnation.

Transplanting (1st-3rd Week of July, Based on Variety): Uproot 25-30-day-old healthy seedlings for transplanting to ensure vigorous growth. For optimal plant density, follow these spacing recommendations: under the normal method, maintain 20 cm × 15 cm spacing with 2-3 seedlings per hill; for Integrated Crop Management (ICM), use 20 cm × 20 cm spacing; and for the System of Rice Intensification (SRI), adopt a wider 25 cm × 25 cm spacing to promote higher yields in smaller areas. Adjust seed rates accordingly-35-40 kg/ha for ICM and just 3-5 kg/ha for SRI-to maximize efficiency and productivity while conserving seeds. Root dip treatment with SSP/ Rock phosphate + Microbial consortium should be done overnight with uprooted rice seedlings prior to transplanting of rice to address phosphorus deficiency in acidic soils. Recommended high yielding varieties for irrigated lowland such as CAU R-1, CAU R-3,

crops. Prepare the field by thoroughly tilling the soil and incorporating 10-15 t/ha of farmyard manure. Apply balanced fertilizers at 60:60:90 kg NPK/ha, using urea (80kg), DAP (130kg), and MOP (150kg), with urea split into 2-3 applications at two-week intervals. Implement pest control by spraying Diamethoate (0.05%) for leaf hoppers and drenching soil with Chlorpyrifos (0.1%) for cutworms. Expect yields of 15-30 t/ha, though fresh carrots maintain quality for only 3-4 days under normal storage conditions.

LIVESTOCK

Poultry

Farmers are encouraged to begin rearing improved dual-purpose chicken breeds starting from the second week of May, which can contribute to both egg and meat production. For effective disease prevention, administer Ascal feed supplement (20-60 ml per 100 birds) for 10 days. Follow a strict vaccination schedule, starting with the Marek's and Infectious Bursal Disease (IBD) vaccines from May 1st, and the Ranikhet (Lasota strain) vaccine at 4-8 weeks of age (1,000 doses in 20 liters of drinking water). Additionally, maintain biosecurity by regularly disinfecting equipment and the poultry environment using Kohrsolin or Germex spray. For proper nutrition management, provide broilers with starter feed containing 22-23% protein for the first six weeks, followed by finisher feed with 18-20% protein until they reach market age. To enhance feed efficiency and reduce costs, incorporate 30% *Azolla* with 70% concentrate feed. It is also essential to store feed properly to prevent spoilage and fungal contamination.



Improved Housing Structure

Piggery

To maintain healthy and productive pig herds, farmers should implement strict biosecurity measures by regularly cleaning pens every 2-3 days using 1% sodium/calcium hydroxide or 2% caustic soda solution, while disinfecting all equipment and vehicles with 0.03% sodium hypochlorite, 1% formaldehyde, or Virkon S/B-904 (1:100 dilution). Farm workers must disinfect their shoes, clothes, and tools before contact with pigs, and animals should be provided with cooling water baths during peak heat along with clean drinking water refreshed at least thrice daily. Critical health interventions include vaccinating against Swine Fever by the 3rd week of May, administering Ivermectin (200 mcg/kg BW) and Albendazole (5-10 mg/kg BW) for deworming, giving piglets 1ml iron injections on days 4 and 14, and supplementing feed with Ascal (10ml twice daily for 10 days). Nutritional management requires age-appropriate rations: starter (20-22% protein), grower (17-18%), and finisher (14-16%) feeds, ensuring freshness by avoiding mold and not storing beyond 3 months. Additional measures include weekly cypermethrin/deltamethrin sprays (1-2 ml/L) for ectoparasite control, maintaining

proper drainage, and providing immune boosters with cool baths to mitigate heat stress. Regular monitoring and these integrated practices will enhance productivity while preventing disease outbreaks in your piggery operations.

Fisheries

Farmers are advised to prepare their ponds by maintaining a maximum depth of 1.75 meters, which is ideal for growing fry into healthy fingerlings (10-15 cm). Begin by applying an initial dose of cow dung manure at 10,000 kg/ha, followed by regular applications of urea and superphosphate (40-80 kg/ha every 15 days) to promote plankton growth, while maintaining optimal water quality through lime application at 250 kg/ha. For stocking, introduce fry at a density of 1 lakh/ha using recommended polyculture ratios of either 1:2:2 or 3:4:3 for Catla, Rohu, and Mrigal species, progressing to fingerling stocking at 5,000-8,000/ha in a 30:30:40 ratio as the fish mature. Farmers are advised to practice Happa breeding for common carp between June 1st and July 31st, and begin stocking of locally produced fingerlings from the first week of July to ensure better survival rates and disease-free stock. For feeding, provide commercial pellets at 2% of body weight, or alternatively, use a nutritionally balanced 1:1 mixture of rice bran with either mustard oil cake or palm oil cake when pellets are unavailable.

CALENDAR OF OPERATION FOR REJUVENATION OF KHASI MANDARIN

Trunk borer control - carefully clean infested holes with an iron wire before applying treatment either by inserting a cotton swab soaked in petrol or Dichlorvos, or by injecting 5 mL of 0.2% Dichlorvos solution (prepared at 2ml/L of water) and sealing the hole with mud to effectively eliminate grubs. As spring arrives, continue proper nutrient management by applying the annual dose of 30 kg FYM per tree along with balanced chemical fertilizers (650 g urea, 1220 g SSP, and 450 g MOP per tree). Implement comprehensive plant protection measures including: application of Bordeaux paste on tree trunks up to 60cm height; foliar spraying of Multiplex (2.5g/L) on new leaves; a protective spray of Bavistin (1g/L) combined with Monocrotophos (1mL/L) on new flushes; Micronutrient supplementation through foliar application of Zinc Sulphate (0.5%), Magnesium Sulphate (0.2%), Copper Sulphate (0.4%), and Manganese Sulphate (0.4%). For management of bark-eating caterpillars, inject 0.1% Dichlorvos directly into their tunnels. Complete these spring operations by mulching the basin area to conserve soil moisture and regulate soil temperature.

CEREAL CROPS

Paddy

Lowland Paddy: Prepare a nursery for kharif paddy on an area of 500-1000 sq. meters to raise seedlings for one hectare with a seed rate of 40 kg/ha. Use Farm Yard Manure (FYM) and Vermicompost in nursery beds for healthy seedling growth. Choose hybrids or high-yielding varieties that are tolerant or resistant to insect pests and diseases, such as Sahsharang, RC Maniphou-14, RC Maniphou-15, and RC Maniphou-16. Prepare the paddy field with 2-3 ploughings using a power tiller, incorporating 5t/ha of FYM. Transplant 20-25-day-old lowland rice seedlings by mid-June at a spacing of 20 x 15 cm. Ensure timely nursery sowing in May and transplanting in June for better grain quality, water efficiency, and reduced stem borer buildup. For effective weed management, conduct regular manual weeding operations as needed. Rogue out off-type plants during flowering and at maturity, and remove plants affected by disease. At least two manual weedings are required. Apply *Azolla* to enhance organic matter and supply nitrogen. To control zinc deficiency, apply 60 kg of zinc sulphate heptahydrate (21%) or 40 kg of zinc sulphate monohydrate (33%) per hectare during puddling, especially if the previous crop in the field showed zinc deficiency symptoms. If deficiency is noticed in the growing crop, apply zinc sulphate as soon as possible.

Upland Paddy: Treat seeds of direct-seeded upland rice with Carbendazim @ 2-3 g/kg of seed to protect against seed-borne diseases. Adopt the dibbling method of sowing instead of broadcasting to ensure assured germination and a uniform crop stand. Carry out at least two intercultural operations or manual weedings-first at 25 days after sowing (DAS), ideally from mid-May, and second between the first week of June and the second week of July. Construct logwood or bamboo bunds across slopes and dig recharge pits (1x1x1 ft) in jhum fields with slopes over 30% to reduce soil erosion and enhance water retention. Rogue out off-type and diseased plants during

for optimum crop performance. Harvest mature cobs promptly to prevent spoilage. For enhanced yield and soil health, adopt intercropping with soybean, kholar, or other legumes.

Management of Fall Armyworm: Sowing maize after the onset of rains reduces insect infestation. Regularly hand-pick and destroy egg masses and larvae by crushing or immersing in kerosene water. Apply sand or ash into the whorls of affected plants. Use pheromone traps @ 15/acre to trap male moths, and spray neem-based formulations (NSKE 5% or neem @ 4 ml/L) at early infestation stages. Apply a whorl spray of *Metarhizium anisopliae* @ 5 g/L at 15-25 DAS, followed by 1-2 sprays at 10-day intervals. For effective control of late instar larvae, use *Bacillus thuringiensis* var. *kurstaki* @ 2 g/L, Spinosad @ 0.3 ml/L, or prepare poison bait using rice bran, jaggery, and Thiodicarb.

MILLET

Prepare the field with 2-3 harrowings followed by leveling. Sow the crop in March to utilize early soil moisture or in August to benefit from residual moisture after the kharif season or late monsoon rains. Use a seed rate of 8-10 kg/ha, sown at a depth not exceeding 3 cm, and maintain a line spacing of 25 × 8 cm for optimal plant population and intercultural operations. Farmers are suggested to select local varieties such as Shongtak, Langjai, Buyan, Tansho, Senyak, Anong, Yempang, Shonye, and Phuknyak, or recommended improved varieties like SiA 3085 and SiA 3156 for higher productivity and adaptability. Apply 10-15 tonnes of well-decomposed FYM per hectare. Keep the crop free from weeds through one or two manual weedings during early growth stages. Treat seeds with Bavistin or Captan @ 2-3 g/kg to control seed-borne diseases. Rogue out off-type and diseased plants at flowering and maturity stages to maintain genetic purity and reduce disease spread. For neck and finger blast, spray *Pseudomonas fluorescens* twice-once at flowering and again 10 days later. Foxtail millet generally does not face major insect issues; however, field sanitation and timely removal of diseased plants will help maintain crop health.

OILSEED CROPS

Soybean

Select high-yielding soybean varieties such as JS-95-60, JS-97-52, JS-2093, RVSM-1135, and MACS-1460 for better yield potential. Sowing should be completed between mid-June and the first fortnight of July in well-drained loamy or sandy loam soils, free from water stagnation. Treat seeds with *Rhizobium* and *Phosphotika* @ 20 g each per kg of seed to improve nodulation and nutrient uptake. Use line sowing with a spacing of 45 cm between rows and 5 cm between plants for optimal plant population and crop management. Maintain a weed-free crop up to 60 days after sowing (DAS). Conduct at least two intercultural operations-first at 25 DAS and the second at 45 DAS-to support healthy crop growth and suppress weed competition.



Varietal demonstration in soyabean

Large Cardamom

Regular weeding should be carried out in June and again in August to effectively manage large cardamom fields. Plant healthy disease-free large cardamom suckers and replace the old diseased plants. A mature tiller with 2-3 vegetative buds are advised to plant during May-June. For insect pest management, spray neem oil (1500 ppm) at a rate of 3 ml/L every 20 days, with a minimum of four applications. Suckers are treated with *Pseudomonas fluorescens* at a concentration of 0.5% to prevent blight diseases

VEGETABLE CROPS

Chilli/Naga King Chilli

Transplant seedlings at the 4-5 leaf stage ($\frac{1}{2}$ ft tall) during May-June when soil moisture is adequate, irrigate immediately after transplanting, carry out regular weeding, and monitor fields for timely rouging and removal of diseased plants. Gap filling is advised.

Tomato

Grow seedlings on raised nursery beds or pro-trays to prevent damping off, use disease-resistant, high-yielding varieties like Arka Rakshak and Arka Samrat. Protect nursery beds with polythene covers and prefer poly-house cultivation of tomatoes during peak monsoon to prevent rain damage. Intercultural operations such as manual weeding, earthing up and particularly staking at 25-30 days after transplanting (DAT) are very important. Subsequent weeding and earthing up as and when required.

Cucurbits

Construction of chungs/ pandals (bamboo made structure) for the cucurbit seedlings. Optimum moisture level should be maintained in the field by providing proper irrigation as dry conditions may lead to poor pollination and thus drop in yield of the crop.

Control of Red Pumpkin Beetle: Collection and destruction of the adults and larvae during early stage of the crop helps in reducing the population of the insects. Incorporation of neem cake during land preparation is effective in controlling the larvae. Treatment with entomopathogenic fungi *Beauveria bassiana* @ 2g/litre of water helps in the reduction of the beetle infestation.

Control of Fruit Fly: Installation of sex attractant cue lure for monitoring and mass trapping for fruit fly @ 25 traps /ha. Field sanitation and removal of damaged fruits daily to minimise pest infestation and burying them deep into the soil to break the reproduction cycle and population increase.

Control of Downy Mildew: Avoid overhead irrigation and spraying in the evening hours as it creates leaf wetness favourable for pathogen development. Promote good air circulation by adopting wider spacing. Spray of Bordeaux mixture 1% or Metalaxyl or Mancozeb 0.4% and reapplied at 5-7 days interval.

Citrus Plants (ORANGE/ACID LIME/LEMON)

Legumes (Pea, Cowpea, Green gram, Rice bean etc) and vegetable crops can be raised as an intercrop during pre-bearing stage to get additional income. Monitor for Trunk borer infestation during this month. Advise to inject petrol/diesel into the tree trunk holes and plug it with mud balls.

Use suitable litter material such as sawdust or paddy husk, spread to a 5 cm depth. Stir the litter regularly to prevent caking. Due to high humidity and frequent rainfall, change the litter frequently to prevent Coccidiosis infections. Store poultry feeds properly to avoid fungal growth, especially in hot and humid weather conditions that promote mold. Deworm poultry with Piperazine (100 mg/kg body weight) for roundworms, Albendazole (10 mg/kg body weight) for other worms, and Praziquantel (5 mg/kg body weight) based on the bird's age. Clean the poultry house regularly with a phenol solution (20 ml/l of water) to prevent infections. Backyard poultry breeds such as Vanaraja, Srinidhi, Rainbow Rooster, and Kuroiler can be reared in low-cost housing. Ensure that the shed is designed for easy management and proper ventilation. Follow a proper vaccination schedule and implement biosecurity measures to prevent diseases. Vaccinate poultry against Ranikhet disease, Marek's disease, Gumboro, and Fowl pox as per the recommended schedule.

Rabbits

Farmers are advised to select breeds like New Zealand, Grey Giant and Soviet Chinchilla suitable for NEH Region. Construct rabbit houses in elevated areas to ensure proper drainage. The area should be shaded to reduce heat stress and provide protection from wind, promoting a comfortable living environment for the rabbits. Mate rabbits in the morning or evening for optimal breeding success. It is recommended to mate the female rabbit twice for higher chances of conception. Provide extra bedding materials like gunny bags and paddy straw inside the rabbit house before kindling to ensure a comfortable and safe environment for the doe and her kits. Supplement pelleted feeds with a mineral mixture, vitamins, and salt. Additionally, provide fresh green fodder to support optimal weight gain. Avoid storing feed in damp conditions, and discard any moulded feed to prevent aflatoxicosis. Offer fresh green plants like beans, alfalfa, berseem, grass, and root vegetables such as cassava, yams, carrots, beets, and turnips, as well as inedible portions of cabbage and cauliflower, to ensure a balanced and nutritious diet for the rabbits.

Mithun

Mithun farmers should begin feeding concentrate mixtures at the rate of 2 kg per mithun per day to enhance body weight gain. The concentrate feed should be prepared using locally available ingredients. To prevent deficiency diseases, farmers should provide 40-50 grams of mineral supplements per day and ensure adequate water availability, especially during the lean season. Paddy by-products such as rice bran and straw can be used as alternative feed resources. Establishing an agroforestry system with fodder trees is recommended to avoid fodder scarcity. Regular deworming is essential to improve productivity. Administer Ivermectin injections to control ectoparasite infestations effectively. Follow a proper vaccination schedule to prevent disease outbreaks. Mithuns should be vaccinated every six months against Foot and Mouth Disease (FMD) to avoid significant economic losses. Implement rotational grazing in forest areas to optimize forage use and ensure sustainable land and fodder management. Farmers are also encouraged to adopt Mithun-based Integrated Farming Systems (IFS) to maximize resource utilization and improve profitability. Additionally, milking mithuns for their highly nutritious milk and processing it into products like paneer, curd, and lassi can enhance income and promote value addition.

Agro-advisory services are very crucial for Odisha state as the state is heavily reliant on agriculture and allied sectors. Moreover, due to its geographical location, Odisha is vulnerable to climate change like droughts, floods, cyclones, hail storms, uncertain rainfalls etc. Agro-advisories can help timely management of different agri-operations and thus, can protect agri-produces ensuring food security and farmers' livelihoods.

Be updated with the regular weather advisory on thunderstorm associated with high rainfall and accordingly, take appropriate crop management practices for standing field and horticultural crops during pre-monsoon season. Take up summer ploughing with the receipt of summer rain. For transplanted kharif rice, dhaincha green manure seeds may be broadcast about 45-50 days before transplanting. Transplanting of vegetables should be done in the afternoon hour to avoid high temperature. Dry out the fish ponds and plough down the pond bottom, apply lime and sow dhaincha seeds for green manuring. It is the time for planting of root crops, tuber crops and rhizomes before onset of monsoon. Be aware of weather advisory and accordingly, avoid spraying plant protection chemicals during rainy days. Use protective covers while undertaking spraying of pesticides as safety measures. Adopt soil test-based fertilizer scheduling for improving nutrient use efficiency and reducing cost therein.

CEREAL CROPS

Rice

Drill seeded direct seeded rice (DSR) method may be adopted for which seeds may be sown during last week of May till onset of monsoon. For transplanted rice, timely sowing will result higher productivity. Rice nursery may be prepared with onset of monsoon using certified seeds. For medium to shallow lowland transplanted rice, varieties like Hasanta, CR 1009 sub-1, MTU 1075, CR Dhan 310, DRR 44, CR Dhan 801, CR Dhan 404, Upahar, Swarna, Pooja, Swarna sub-1 and BPT

5204 are suitable. For coastal saline soils, farmers are advised to grow salt tolerant varieties such as CR Dhan 405, CR Dhan 403, CR Dhan 412 and Lunishree. Suitable varieties for upland situation are-Khandagiri, Vandana, Nabanna, Kolab, CR Dhan 202 and Sahbhagi dhan. Use herbicides for control of weeds for which pre-emergence herbicides such as Pretilachlor with safener@1500 ml/ha, pyrazosulfuron@200 g/ha, Pretilachlor 6% + Bensulfuron methyl 0.6% GR @ 10 kg/ha at 0-3 days of sowing or transplanting and post-emergence application of Fenoxaprop-ethyl @ 1300ml/ha + Ethoxysulfuron @ 100 ml/ha, Bispyribac sodium @250 g/ha, Triafamone + Ethoxysulfuron@90 g/ha, cyhalofop butyl+penoxsulam@2250 ml/ha at 15-20 days after transplanting/sowing may be used. Algal bloom is a major problem in transplanted rice. To control it, spray Copper oxychloride@ 3 g/litre of water or apply 10 kg CuSO₄/ha in soil. Install pheromone traps with scirpolure (5 no./ha) to monitor incidence of yellow stemborer. Tricho cards (50000 eggs/ha releas for 4 times) are recommended to manage the pests like yellow stemborer and leaf folder.

PULSE CROPS

Use suitable varieties of green gram- Virat, Sikha, IPM 2-14; black gram- PU 9, Sashi and IPU 2-43; and arhar- LRG 52 and BRG 2. For management of yellow mosaic vein virus in green gram/black gram, seed treatment with Thiamethoxam 30 FS or Imidachloprid 600 FS @5 ml/kg seed, installation of yellow sticky traps 50 no./ha 30 DAS along with need-based spray of neem oil 1500 ppm @1500 ml/ha alternately with Flonicamide@120 g/ha may be undertaken. Treat the seeds with *rhizobium* and *PSB* biofertilizers @25g each/kg seeds along with ammonium molybdate@ 3 g/10 kg seed mixing with jaggery and water before planting. To manage weeds Pendimethalin 30% as pre-emergence herbicide @2500 ml/ha or Quizalofop ethyl 5% @1000 ml/ ha as post-emergence herbicide may be used.

OILSEED CROPS

Groundnut

Cultivate suitable groundnut varieties like Kadiri Lepakshi, Dharani and Smruti. Soil application of 20 kg N, 40 kg P₂O₅, 40 kg K₂O/ ha at sowing and Gypsum @ 250 kg/ha at the time of hoeing should be applied to improve productivity. Tobacco caterpillars can be controlled with the spray Indoxacarb + Nuvaluron @ 625 per ha or Chlorpyrifos 50% + Cypermethrin 5% EC @ 1000 ml/ha or Fipronil 5SC @ 1000 ml/ha. Harvest the crop when plants turn yellow and leaves start drying. If no rain is received, apply light irrigation 2-3 days before harvesting. Dry the pods properly after harvest so as to reduce the moisture content to 8-9 % before storing.

COMMERCIAL CROPS

Sugarcane

Cultivate high yielding sugarcane varieties like Sabita, Nilamadhava and Charchika. Carry out wrapping and propping as the crop attains 4-5 months age in medium and late duration varieties. Spray Lamba cyhalothrin @2 ml/L of water to minimize infestation of top shoot borer. Avoid ratooning in red rot affected areas and sett treatment to be done with Azoxystrobin 18.2% + Difconazole 11.4% @ 0.1% per kg sett.

supplement azolla @ 1-1.5 kg in daily ration of cattle to reduce the feed cost by 20%. Provide supplementary feeding such as broken rice, paddy, cooked rice, azolla etc. to the backyard poultry birds during continuous rain. It is preferable to expose the feed ingredient sacks/bags under sun every week to prevent the feeds getting more moisture. Wash the cattle shed with phenyl (dilution 1:20 ratio with water) and sprinkle bleaching powder around the shed at weekly intervals. Mastitis is a common problem in high yielding cows in rainy season. Keep the cattle shed clean and dry. Clean the udder in soap water daily before milking. Apply Mastilep® ointment over the udder after each milking.

Fishery

Raise the height of the pond dykes, repair the damaged dykes, if necessary and keep proper outlet to drain out excess water. Put net in the lower dyke or outlet to avoid escape of fish from the pond. Test the water quality parameters mainly pH and alkalinity, and apply lime, fertilizers and manures consulting fishery officers, scientists and extension functionaries. In weed infested ponds, stock some grass carp yearlings (500 no/ha) along with Catla, Rohu and Mrigal to control weeds. Stock yearlings of Indian Major Carps @ 5000 per ha at a ratio of 1500 Catla, 2000 Rohu /Jayanti rohu and 1500 Mrigal at a ratio of 3:4:3 having 2 metres water depth. Adopt sample netting using cast net at every 2-months intervals to check the growth and health of fish. Utilize the seasonal /shallow ponds in villages for fish fingerling production. Rise in water temperature expedites decomposition of organic matter which accumulates at the pond bottom and generates gases. These gases are toxic to the fishes. To overcome the problem, zeolite @ 20 kg per acre should be applied after mixing it with one bag (30 kg) of dry river sand.



Punjab's agriculture is characterized by the cultivation of wheat and rice, which together dominate over 80% of the state's total cropped area. Other significant crops include sugarcane, maize, oilseeds, cotton, and various fruits and vegetables. The state has a high crop intensity, with many fields being sown multiple times annually, contributing to its status as a major food producer in India. Punjab's agriculture faces several challenges, including monocropping, depletion of water resources, excessive use of pesticides and fertilizers, etc. These issues contribute to environmental degradation, economic stress for farmers, and ultimately impact crop production. There is utmost need to educate the farmers with agro-advisories regarding advanced techniques in cultivation of field crops, vegetable & fruit crops, agroforestry and livestock management in order to promote the scientific cultivation along with conservation of natural resources. The following agro-advisory highlights the important points to be followed by the farmers during kharif season to sustain the productivity of field crops, horticulture crops, agroforestry and livestock.

SOIL TESTING

Soil testing is the best tool to ensure optimum and balanced use of nutrients. Improper use of fertilizers leads not only to imbalanced nutrition but also deterioration of the environment. For collection of soil sample scrap away surface litter and make a V-shaped cut with a spade or a khurpa to a depth of 6 inches. Remove about 1" thick uniform slice of soil from one side of the cut. Similarly, collect samples from 7 to 8 places in the field of uniform texture and uniform fertility. Put the samples in a clean bucket, tray or cloth and mix it thoroughly. Take approximately half kg of soil in a cloth bag and label it with information such as field number, name of the farmer, address, date of sampling etc. If the samples are wet, dry them in shade before putting into the cloth bag. Soil and water samples are tested by Soil Testing Laboratory, Deptt. of Soil Science, Punjab Agricultural University, Ludhiana and Soil Testing Laboratory of all KVK's.

Use of farmyard manure (FYM) in kharif crops

Farmyard manure (FYM) is a valuable resource for improving soil health and enhancing crop production, especially for kharif crops. FYM helps in improving soil structure and biomass, and it increases soil organic carbon, nitrogen, phosphorus, and potassium, crucial nutrients for plant growth. It also enhances water-holding capacity and microbial activity, contributing to a healthier soil ecosystem. Apply farmyard manure (FYM) in adequate quantities to the soil ensures good tilth and improves soil water-holding capacity.

CEREAL CROPS

Rice

Selection of varieties: To save water and for ease in management of paddy straw, grow PAU recommended short duration varieties. i.e. PR 132, PR 131, PR 130, PR 129, PR 128, HKR 47, PR 127, PR 126, PR 122, PR 121, PR 114, PR 113.

Time of nursery sowing: Restrict to timely sowing of nursery (20 May-20 June) and timely transplanting (20 June-10 July) schedule for better grain quality, water saving and low build-up of stem borers.

Transplanting: Use 30-35 days old nursery for medium duration varieties and 25-30 days nursery for short duration variety (PR 126). Transplant seedlings in lines at 20 x 15 cm (33 hills/sq m) and put 2 seedlings per hill.

Water management: Keep water standing in paddy field only for 2 weeks after transplanting and thereafter apply irrigation 2 days after the ponded water has infiltrated into soil.

Fertilizer management: Apply fertilizers as per soil test. Omit phosphorus application in rice following wheat that received recommended dose of phosphatic fertilizer. Use nitrogen judiciously based on Leaf Colour Chart (LCC). To obtain higher yield of PR 132, apply 25% less nitrogenous fertilizers than other varieties

Zinc deficiency: Due to zinc deficiency the lower leaves of crop become rusty brown near the base and ultimately dry up. To control this malady, apply 25 kg of zinc sulphate heptahydrate (21%) or 16 kg zinc sulphate monohydrate (33%) per acre at puddling in case previous crop in this field had shown the symptoms of zinc deficiency.

Plant protection: To control false smut disease, initiate the spray of recommended fungicides at boot stage of the crop. Plant hoppers feed at the base of rice plants and are often overlooked. Their damage is noticed only when the crop is hopper burnt. Hence, regular monitoring of the insect population is necessary. Use of synthetic pyrethroids leads to increase in the population of rice planthoppers. Hence, these insecticides should not be used for the control of rice insect-pests.

Basmati Rice

Selection of varieties: Pusa Basmati 1121, Pusa Basmati 1509, Punjab Basmati 5, Punjab Basmati 7, Pusa Basmati 1718, Pusa Basmati 1847 and CSR 30 are the most suitable varieties.

Time of nursery sowing and transplanting: For Basmati varieties Punjab Basmati 7, Punjab Basmati 5, Pusa Basmati 1847, Pusa Basmati 1121 and Pusa Basmati 1718,

the optimum time of nursery sowing is 1st fortnight of June and transplanting time is 1st fortnight of July, whereas, for CSR 30 and Pusa Basmati 1509, the optimum time of nursery sowing is 2nd fortnight of June and transplanting time is 2nd fortnight of July.

Seedling Treatment: To control Foot rot, smear the seeds with talc formulation of *Trichoderma harzianum* @ 15 g per kg of seed immediately before sowing and seedling root dip for 6 hrs with *T. harzianum* @ 15 g per litre of water before transplanting. Alternatively, treat the seed with Sprint 75 WS (carbendazim + mancozeb) by making slurry of 3 g fungicide formulation in 10 ml water for one kg seed before sowing to manage other seed borne diseases of rice.

Method of Transplanting: Transplant two seedlings per hill in lines at 20 x 15 cm (33 hills/sq. metre) during the optimum period in a well puddled field.

Nutrition management: Skip phosphorus application if the recommended dose of phosphorus has been applied to the preceding wheat crop. However, in phosphorus deficient soils, apply 75 kg of superphosphate per acre before last puddling. Apply 45 kg/ha urea for CSR 30, 90 kg/ha urea for varieties Punjab Basmati 5, Punjab Basmati 7, Pusa Basmati 1718 and Pusa Basmati 1121, and 135 kg/ha urea for Pusa Basmati 1509. Apply urea in two equal splits at 3 weeks and 6 weeks after transplanting.

Direct Seeded Rice

Suitable soils: Sow DSR only in medium to heavy textured soils. It cultivation is not successful in light textured soils due to severe iron deficiency and lower crop yields

Suitable varieties: Short and medium duration varieties of rice and Punjab Basmati 7, Pusa Basmati 1718, Pusa Basmati 1121 and Pusa Basmati 1509 for basmati

Sowing time: Sowing should be done in the first fortnight of June for rice and second fortnight of June for basmati rice

Seed rate and seed treatment: Use 8-10 kg seed per acre. Imbibe the seed by dipping in 2% potassium nitrate solution (prepare solution by dissolving 200 g KNO_3 in 10 litre water for 8 kg seed) for 12 hours. Dry the seed in shade and treat with 3 g Sprint 75 WS (mencozeb + carbendazim) per kg seed.

Direct seeding in tar-wattar fields: Divide the field into kiyaras of desirable size after laser levelling and apply Rauni irrigation. When the field comes to tar-wattar (sufficiently high but workable soil moisture) conditions, prepare field kiyara-wise with shallow cultivation followed by 2-3 plankings with load of 3 sandbags on planker and sow immediately. Seed should be placed at 3-4 cm depth in 20 cm spaced rows. Avoid field preparation and sowing during noon hours for better results.

Direct seeding in dry fields: Treated seed should be placed 2-3 cm deep in rows spaced at 20 cm in a dry field and irrigation is applied immediately after sowing.

Weed Control

Pre-emergence: Spray 1.0 litre per acre Stomp/Bunker 30 EC (pendimethalin) or PEPE 25 SE (penoxsulam 1% pendimethlin 24%) in 200 litres of water for control of annual grass weeds and some broadleaf weeds.

Post-emergence: If fields are infested with Swank, paddy moths, Nominee Gold 10 SC (bispyribac sodium) @ 100 ml per acre should be sprayed at 15-25 days after

sowing (2-4 weed leaf stage). For swank, chinigha, broadleaf weeds, paddy mothas, gandiwalamotha, Novlect 12 EC (fenoxaprop-p-ethyl) @ 500 ml per acre should be sprayed at 20-25 days after sowing (2-4 weed leaf stage). For madhana, chinigha, chirigha, takrigha, Ricestar 6.7 EC (fenoxaprop-p-ethyl) @ 400 ml per acre should be sprayed at 20-25 days after sowing (2-4 weed leaf stage). For broadleaf weeds, paddy mothas, gandiwalamotha, Almix 20 WP (chlorimuron ethyl 10% + metsulfuron methyl 10%) @ 8 g per acre should be sprayed at 20-25 days after sowing.

Fertilizers management: Apply 130 kg urea per acre in three equal splits at 4, 6 and 9 weeks of sowing and for basmati rice, apply 54 kg urea per acre in three equal splits at 3, 6 and 9 weeks after sowing.

Irrigation

Direct seeding (flat/raised beds) in tar-wattar fields: Apply first irrigation at around 21 days after sowing. After that, apply irrigations at 5-7 days interval depending on soil type.

Direct seeding in dry fields: Apply first irrigation immediately after sowing and second irrigation at 4-5 days after sowing. Subsequent irrigations should be applied at 5-7 days interval depending on soil type.

Maize

Improved varieties: PMH 17, DKC 9144, Bioseed 9788, PMH 14, PMH 13, ADV 9293, JC 12, PMH 11, PMH 1, JC 4, PMH 2, JC 12 & JC 4 variety is recommended for kandi areas of the state.

Time of Sowing: Last week of May to end of June. In fields, which are prone to damage through water stagnation, sow the crop in end of May to early June, so that the crop gets firmly established before the rains.

Seed rate and Method of Sowing: Use 10 kg seed per acre for other varieties. Sow the seed 3-5 cm deep in lines with a maize planter or seed-cum-fertilizer drill, provided with a planting attachment. The row to row spacing should be kept at 60 cm whereas plant to plant spacing to be maintained at 20 cm. A four-row pneumatic planter with provision of bed making can be used for maize sowing.

Weed management: Give two hoeings about 15 to 30 days after sowing with khurpa/ kasaula/ wheel-hoe/ triphali/ tractor-drawn cultivator. Spray 800 g per acre Atrataf/ Atragold/ Masstaf/ Atari/Traxx 50 WP (atrazine) on medium to heavy textured soils and 500 g per acre in light soils within ten days of sowing, using 200 litres of water. Alternatively, spray 105 ml per acre Laudis 420 SC (tembotrione) in 150 litres of water at 20 days after sowing provides effective control of mixed weed flora. For the control of dila/ motha, apply 400 mL per acre 2,4-D amine salt 58 SL as post emergence 20-25 days after sowing in 150 litres of water.

Fertilizer management: Apply 110 kg urea/acre, 55 kg DAP/150 kg SSP per acre and 20 kg MOP/acre. If maize follows wheat, which had received the recommended dose of phosphorus, omit its application to maize. When 27 kg of DAP is used, reduce the urea dose by 10 kg and when 55 kg of DAP is used, reduce the dose of urea by 20 kg. To all recommended varieties, drill one-third of nitrogen and the entire quantity of phosphorus and potassium at the time of sowing. Top dress one-

Control of ectoparasites: Topical acaricide (Butox, Taktik etc.) used for spray on animal body should be sprayed in all cracks/orifices in shed too (@2ml/ltr water). Take care that feed the animal well before application of acaricide and apply muzzle to avoid licking of the drug. Topical acaricide application on animal and in shed should be repeated thrice at a gap of one week.

Screening: Screening for diseases like mastitis (through California Mastitis testing), brucellosis, TB should be carried out regularly, this increases the success rate of prevention and treatment.

Heat Stress Monitoring: Be vigilant for indicators of heat stress, including excessive panting, drooling, or decreased appetite. Administer prompt care if any symptoms manifest. Avoid bathing of animals in common village ponds as this may lead to spread of infections.

Wound care: Use ointments like Himax/Lorexene/Ectosep for infestation of flies on wounds and facilitate healing.

Hygiene and Sanitation

Regular Cleaning: Daily maintenance of shelters, feeding zones, and water troughs is essential to mitigate the risk of aflatoxin contamination.

Manure Management: Properly dispose of manure to minimize the risk of disease, greenhouse gas emissions, and odour issues. Alternatives such as vermicomposting and biogas plants are also effective methods for waste management.

Fly Control: Utilize fly repellents or traps to reduce the annoyance and health hazards associated with flies.

Breeding Management

Avoid Heat Stress: Plan breeding activities during the cooler times of the day or season to enhance conception rates. Ensure a gap of 12 hrs between signs of heat and insemination.

Pregnancy Care: Ensure that pregnant animals receive exceptional care, which includes a diet rich in nutrients and an environment free from stress.

Disaster Preparedness

Flood Management: In case of heavy rainfall, it is essential to provide livestock with access to higher ground to prevent waterlogging.

Emergency Feed Stock: Store dry fodder and concentrates as a contingency for unexpected weather disruptions.

Rajasthan is predominantly a kharif cropping state due to vast area under rainfed cropping and crops. The diverse crops are grown during Kharif season. During May and June preparations for sowing of different crops like maize, bajra, green gram, cowpea, guar, moth bean are completed by the farmers. The sowing and nursery raising of cotton, groundnut and rice are done under varying situations and climatic conditions. To harness the potential of the crops, an appropriate technology along with management practices are inevitable. Therefore, a proper agro-advisory is need of the day to be issued for the use of the farmers for different agricultural operations. It will help in increasing the productivity and profit from agriculture and empower the farmers while helping in sustainable management of natural resources and applied inputs. The agro-advisories related to different activities and operations in agriculture conducted by the farmers during May-June are discussed in following para:

Soil Analysis

After the harvesting of rabi season crops in March and April and before sowing of kharif crops is the best time for taking soil samples from the farmers' fields for their analysis for physical, chemical and biological properties. The farmers are advised to take soil samples from 5 places of their fields, mix it properly and take them to nearby KVK or district soil testing labs for analysis. The sampling should be done from 0-30 cm soil depth. Attempt should be made get the soil sample analysed before sowing of kharif crops. The nutrient status as evidenced from soil sample analysis will help to decide right and balanced application of fertilizers and avoid extra application. It will also help in enhancing the nutrient use efficiency and saving of cost of cultivation while maximising the profit to farmers.

Summer Deep Ploughing and Application of Gypsum

May and June are the best time for summer deep ploughing and application of Gypsum in the field. The summer deep ploughing is process of solarization of the soil

Water Management: Adequate water-supply is essential throughout the crop season particularly during the pre-tasselling, silking and grain filling stages, if no rain during this stage.

Weed management: Spray 1-1.5 kg a.i. atrazine 50 WP (atrazine) per ha on medium to heavy textured soils immediately after sowing using 500 litres of water followed by Tembotrione 42 SC 120 g a.i. per ha in 500 litre of water after 25-30 days after sowing hoeing/interculture at 15 to 30 days after sowing. For the control of broad leaf weed, apply 900 g per ha 2,4-D ester 38 EC in 450 litres water at 30-35 days after sowing.

Plant Protection: Fall armyworm: Spray the crop with chlorantraniliprole 18.5 SC @ 200 ml or spinetoram 11.7 SC 250 ml or emamectin benzoate 5 SG 200 g in 500 litres of water per ha, for crop up to 20 days old. Maize stem borer: Apply Phorate 10 G or carbofuran 3% granules @ 7.5 kg per ha in the shoot of the plant after 16 to 30 days after sowing. Alternatively, spray 10% solution of neem leaves/kernels after 10-20 days of germination. Armyworm and grasshoppers: Apply Methyl Parathion 2% power @ 25 kg per ha, summer ploughing in fallow land is advised. Aphids: Apply methyl demeton 25 EC @ 1 liter per ha in 500 litre of water at the time tasseling. Maydis leaf blight: Spray mancozeb or propinab or carbendazim 0.2 percent solution, repeat the spray in 10-15 days.

Rice

Time of Nursery Sowing: The ideal window for paddy nursery sowing in Rajasthan is 20 May to 20 June.

Seed Rate and Seed Treatment: 20-25 kg/ha seed is sufficient for transplanting. Treat the seed with Sprint 75 WS (carbendazim + mancozeb) by making slurry of 3 g fungicide formulation in 10-12 mL water for one kg seed (24 g fungicide in 80-100 mL water for 8 kg seed) before sowing.

Nursery Raising: Prepare 1.0-1.5 m wide nursery beds. Generally, 400-500 m² nursery area is sufficient for transplanting in one ha area. Paddy nursery can be raised in second fortnight of May and 24-25 kg seed is sufficient for transplanting in one ha area. In nursery paddy seed should be covered properly with sand or with mixture of sand and manure, nursery bed should be always moist without standing water.

Suitable Varieties: PR-126, PR-106, PR-1121, BK-190, PB-1718, PB-1509, PB-1885. These varieties offer a mix of high yield, stress tolerance, and market preference.

Transplanting: Manual transplanting: Transplant the seedlings in lines, 2-3 seedlings per hill at a spacing of 15 x 15 cm, Depth of transplanting should not be more than 2-3 cm. Mechanical transplanting: Done by paddy transplanter in puddled and un-puddled conditions. Raising of mat type nursery is required

Fertilizer Management: Medium duration varieties & Hybrids (e.g., PB-1718, PB-1509): 150 kg N, 60 kg P₂O₅, 60 kg K₂O, 25 kg Zinc sulphate per hectare. Early Duration Varieties (e.g., PR-126): 120 kg N, 60 kg P₂O₅, 60 kg K₂O, 25 kg Zinc sulphate per hectare.

Application Schedule: Apply the full dose of P, K, Zn, and one-third of N at the time of puddling. The remaining two-thirds of N should be top-dressed in two equal splits at 3 and 6 weeks after transplanting. Incorporate 15 tons of well-decomposed FYM/ha before transplanting.

Weed Management: Apply 30 kg butachlor or 15 kg thiobencarb granules/ ha in 4-5 cm deep water at 2-3 days after transplanting. Apply 3.0-liter butachlor 50 EC or thiobencarb EC or 1325 ml Anilophose 30 EC or 2.0-liter Pretilachlor 50 EC or 125 g oxydiazon 80 % WP after mixing with 150 kg sand and broadcast in standing water at 2-5 days after transplanting. Apply 20 g ready mix of metsulphuron and chlorimuron or 125 g ethoxysulphuron 15 % WG or 1 kg 2,4-D ester at 20-25 days after transplanting for the control of broad leave weeds. Broad leaved weeds can also be managed by application of 93.75 ml/ha in 500 liters of water phenoxulam 24 % SC at 8-12 days after transplanting. Apply 250 ml bispyribac sodium 10 % SL/ha in 500 litres of water at 15-25 days after transplanting to manage mix weeds flora in rice.

Plant Protection: Coragen (Chlorantraniliprole) 18.5 SC @150 ml/ha, Fame (flubendiamide) 480 SC@ 50 ml/ha for rice stem borer and leaf folder. Ulala (flonicamid) 50 WG @ 150 g/ha and Osheen (Dinotefuran) 20 SG @ 200 g/ha for plant hoppers. Natio (trifloxystrobin+tebuconazole) 75 WG @200 g/ha in 500 L of water for BLB. Awancer Glow (azoxystrobin 8.3 % + mancozeb 66.7 %) 75 WG @ 1.5 kg/ha for sheath blight. Amistar Top (azoxystrobin + difenoconazole) 325 SC @ 500 ml in 500 L water per ha for blast.

Jowar

Time of Sowing: Optimum period of sowing is mid-June to mid-July; Early green fodder- 2nd fortnight of March- 1st fortnight of April.

Improved Varieties: Hybrids: CSH 1,5,6,9,10,11, Varieties: SPV 96, 245, 462, 475, CSV 15, 17, 23, Pratap Jowar 1430, Fodder purpose: (i) Multi-cut: MP chari, SSG 59-3, (ii) Single cut: Raj Chari 1, 2, and Pratap Chari 1080

Seed Rate & Spacing: Grain purpose: 12-15 kg/ha at 45 cm line spacing. Fodder Purpose: 35-40 kg per ha at 25-30 cm line spacing.

Fertilizer Application: Apply 50 kg of N and 20 kg P₂O₅ per hectare in rows at the time of sowing under rainfed and low rainfall conditions. Apply 50 kg of N and 20 kg P₂O₅ per hectare at the time of sowing and another 50 kg N per hectare about one month later under irrigated conditions.

Plant Protection: Seed treatment with Slayer (thiamethoxam) 30 FS @10 ml/kg seed for the control of shoot fly. Seed treatment with Sulphur dust @ 4 g/kg seed for grain smut.

PULSE CROPS

Mungbean

Sowing Time: Last week of June to first week of July depending upon the rainfall and moisture in the soil profile.

Seed Rate & Spacing: Seed rate of 15-20 kg/ha is recommended. Row to row spacing should be maintained at 25-30 cm and plant to plant at 10 cm.

Seed Treatment: Prior to sowing treat seed with Thiram @3g/kg, Imidacloprid @5ml/kg. Also treat with *Trichoderma* or *Pseudomonas*@5-10g/kg seed, then inoculate with *Bradyrhizobium*@25 g/kg seed.

of Chloropyriphos 20 EC (1 ml/L) help to manage it, destroy alternate host plants growing nearby. Shoot Borer: Prune and burn affected branches. Spray branches and trunks with a solution of Chloropyriphos of 2 ml/ L of water. Clean the cavities with a pesticide-coated stick or cotton, then seal with mud and cow dung paste. Root knot nematode: To manage practice soil solarization, use healthy planting material, apply organic manures, inter crop marigold plants, use resistant varieties and apply 5-10 kg neem cake per plant to suppress nematodes. Wilt Disease: control is difficult but drenching the soil around the root zone with 20-30 liters water per plant of carbendazim 50 wp 2 gram/L fungicide solution can help. Mix 1 kg of *Trichoderma viride* per 10 q of compost or farmyard manure and apply around the root zone, after keeping it 15-20 days under shade. Use 1-2 kg *Trichoderma* mixed compost per plant. *Anthracnose*: To manage this remove and destroy the infected twigs and plant parts. Spray Mancozeb 2 gm per liter of water every 10-15 days until fruiting begins. Fruit Rot: Spray Mancozeb or Zineb 2 gm per liter of water every 15 days at least twice. For already infected plants, spray preventively before rainfall.

Harvesting and Yield: For local markets, harvest when fruits are fully ripe. For long-distance market fruit should be harvested when mature but still firm, when fruits change from green to reddish in 24 hours or more, harvest them promptly. A single plant can produce around 40-50 kg of fruits, and mature trees (8-10 years) can yield up to 100 kg per tree. Good practice of grading and sorting, packing, transport and marketing helps to fetch better prices.

VEGETABLE CROPS

Brinjal

Improved Varieties: Pusa Purple Long, Punjab Sadabahar, ARU-2, H-4, H-7, PB-91-2, K-202-9, DBR-8, AB-1, Hybrid varieties- Arka Navneet, Pusa Hybrid-6, Pusa Hybrid-2

Nursery: 400-500 grams of seeds are required for one hectare. For one hectare of seedlings, 15-20 beds of 1 meter width and 3 meters length and 10-15 cm height are required. Seeds should be treated with thiram or captan 2 g/kg. In the nursery, seeds should be sown in rows at an interval of 5 cm.

Field Preparation: The field should be ploughed 3-4 times deep. First ploughing should be done by mixing 120-150 quintals of decomposed cow dung manure/hectare in the soil. During the last ploughing, 40 kg nitrogen, 80 kg phosphorus and 60 kg potash/hectare should be mixed in the soil.

Sowing Time: Nursery Preparation: June-July; Transplanting in the main field: July-August

Planting: Saplings should be transplanted in the evening when they are 10-15 cm and 30-40 days old. Distance between rows should be 60-70 cm and distance between plants should be 60 cm.

Manure and Fertilizer: 20 kg/hectare nitrogen should be sprayed when flowers appear.

Irrigation: Irrigation should be done every 4-5 days or as per requirement in rainy season.

fruits should be removed. Capsicum is a very sensitive to root rot, LCV and anthracnose, etc. They should be properly inspected and treated from time to time. Due to continuous cropping in poly house, the deficiency of micro nutrients develops in the soil, especially of calcium. This causes black brown spots on the fruits which start rotting resembling with a fungal disease. In such a situation, gypsum should be applied @ 10-15 kg per 1000 m² area. In case of coloured capsicum, only fully developed and matured fruits should be harvested with the help of a knife or secateurs. Sometimes, price of coloured capsicum in the market rules below than that of green capsicum. In such a case the colored capsicum fruits should be plucked at the initial stage. The branches of capsicum plants are generally weak. They break due to the weight of the fruits. They should be tied with the help of a string from time to time. The shoots coming out of the stem of the plants should be removed from time to time. The old leaves should also be removed.

Tomato

Varieties: Badshah, Dev, Abhinav, Himshikhar, Subhrano, Naveen, DT-1, DT-7, ARTH-4, Nun-7711 and 646. Cherry tomato varieties: BR-124 and HA-818

Planting Time and Spacing: Tomato can be planted in greenhouse from July to September. The crop lasts up to April to June or July next year. The period of growing depends on the size and type of greenhouse and the climate. The row to row spacing is kept at 60-70 cm and plant to plant at 50-60 cm.

Pruning and Training: In polyhouse, usually 2 to 3 main branches of the plants are allowed to grow and the rest are removed continuously. This process should be repeated at every 10-15 days. Tomato plants are trained upwards supported by plastic ropes which are tied to main iron wire at a height of 9 to 10 feet.

Irrigation and Fertilizer: Regular irrigation is recommended along with fertigation with fertilizer solution which is a mixture of nitrogen, phosphorus and potash in the ratio of 5:3:6. From transplantation to flowering, 2000 to 2500 litre of water is given/1000 m² of area. Along with this, nitrogen is given @ 1 g/liter, phosphorus @ 0.5 g/liter and potash @1 g/liter of water. From flowering to fruit setting, 3000 to 4000 litre water is applied per 1000 m² area along with N, P, and K @ 2 g, 1 g and 2 g/liter of water through drip system. The fertigation should be done at 3-4 days interval in summer and 6-8 days interval in winter.

Plant protection: The affected plants by viral disease TMV should be uprooted and destroyed immediately.

Harvesting of fruits: Harvesting should be done with scissors or sharp knives so that tomato plants and other fruits are not damaged. Fruits are harvested only when they are fully ripe (red in colour). They should be graded according to colour, size and weight for getting better price.

Storage: If the fruits have to be sold after a day or two, they should be kept at a temperature of 8-10^o C in summer. In winter, they can also be kept at normal room temperature.

Yield: Generally, 200 to 220 t of tomato yield is obtained per hectare from a good greenhouse. About 100 to 120 t of yield can be obtained from cherry tomatoes.



CEREAL CROPS

Rice

Farmers can select the medium duration varieties like Sikkim Dhan 1, CAU-R1, Abhishek, Attey etc. Farmers are suggested to start main field preparation for paddy transplantation through repairing old bunds by maintaining 20-30 cm height in low land situation. Well decomposed FYM should be applied 2-3 days before sowing @10 q/1000 m². or vermicompost @ 3-6 t/ha either alone or in combination 15 days before



Cultivation of Rice Variety Sikkim Dhan 1

transplanting. Sow the seeds @35-40 kg/ha for transplanting, 60-100 kg/ha for direct seeding, 5 kg/ha for SRI method. Seed should be treated with Azospirillum @ 20 g/kg seed. Farmers of mid and high altitude are suggested to complete the transplanting by month end of May and in low altitude it should be completed by last week of June. An area of 1000 m² nursery area is sufficient for planting in one hectare of land. Uprooting of 25-30 days old rice seedlings with 20 cm x 15 cm spacing and 2-3 seedlings/hill is considered best for timely transplanted crop. Neem based formulation (15ppm) @ 0.5% or 5 ml/l of water two to three times can be used for pest management.

Maize

Promising varieties are Pahlenlo makkai (local variety) and CP-333 (hybrid variety). Seeds should be inoculated with Azospirillum, Azotobacter etc., and phosphorus solubilizing bacteria (PSB) @ 20 g/kg seed before sowing. A seed rate of 15-20 kg/ha is optimum for maize. Sowing should be done in furrows at a depth of 2-3 cm and

Ginger/Turmeric

Local cultivar Bhaise, Gorbuthane, Sano Aduwa. The best time to sow ginger is in the month of April to May when there is sufficient moisture in the soil. A healthy rhizome free from diseases of about 30-45 g size is to be made and sown in furrows in a raised bed. Spacing of 30cm x 25cm is considered ideal for ginger sowing. The rhizomes should be placed at a depth of 4-5 cm in furrows and covered with soil. Nutrients should be properly managed through organic sources with the use of different nutrient rich manures, FYM @ 20 t/ha, Neem Cake @ 2 t/ha, Vermicompost @ 2.5t/ha. Weeding needs to be carried out twice or thrice for checking the weed growth. Mulching with locally available mulch materials like dried leaves and straw may be used for covering the beds. Rhizome rot/ Soft rot is the major disease and can be controlled by spraying with 1% Bordeaux mixture. Drenching of affected and surrounding beds with 0.3% Copper Oxychloride is recommended. Shoot borer can be managed by spraying Neem oil @ 3ml/l or Beauveria bassiana@ 5ml/l or spinosad @ 0.2 ml/l.

VEGETABLES CROP

In high altitude, for raising nursery it is advised to use polythene covers/shade above the nursery to protect seedling from untimely extreme weather condition. Provide support of bamboo structure for climbing upward the plants like bottle gourd, bitter gourd, cucumber and bean. Apply bio mulch @ 5 t/ha to cover the ridges and furrows of the field. Start spraying copper oxychloride @ 0.25 % at weekly interval to control fungal and bacterial diseases. Apply plastic mulch/agro jute textiles as mulching for soil & water conservation and weed control. Regular monitoring for increase in pest and disease of large cardamom should be done. Shade management should be done with the proper lopping of trees. Farmers are advised to spray neem-based formulation (1500 ppm) @ 3-5 ml/litre of water or *Bacillus thuringiensis* @ 2g/l or *Metarhiziumanisopliae* @ 5ml/l and Copper oxychloride @ 0.25% to avoid any of pest and diseases.

LIVESTOCK

For poultry, variety like broilers & Vanaraja is recommended. Due to prevailing weather conditions of high humidity and frequent rainfall it is utmost necessary to change the litter materials of poultry shed frequently to prevent Coccidiosis infection. Feed materials of poultry should be stored properly to avoid possible mould growth (Aflatoxins) due to prevailing hot and humid weather condition.



Use of Clean & fresh litter in poultry rearing



Deworming in Pig

Tamil Nadu & Puducherry



GENERAL ADVISORY

Zinc Solubilizing Bacterial Biofertilizer: Carrier or liquid bioformulation that improves Zn availability and increases yield by 17-20%. Soil Respiration Indicator (SRI) Gel Probe: Visual colorimetric tool for rapid microbial activity testing in soil (results in 6-8 hours). Vetiver Grass for Tannery-Polluted Soils: Used with organic composts to reduce chromium (21.5%) and enhance carbon (25%) with by-product value. Sesuvium portulacastrum: Salt-tolerant species for reclaiming saline soils and improving physical and chemical soil properties. TNAU Panchagavya: 3% foliar spray of organic growth promoter containing nutrients and hormones; boosts yield by 25-30%, improves quality, and acts as pest repellent.

For Horticultural Crops:

Fruit Fly Management: IIHR's Arka Dorsolure-F and kairomone traps targeting *Bactrocera dorsalis* (gravid females).

Seed Treatment: Use TNAU seed coating formulation, dry, and sow accordingly.

Drought Mitigation: Apply Nano Revive @ 10 ml/L at drought onset.

CEREAL CROPS

Paddy

Foliar application of Rice Reap (5 kg/ha) at booting and 10 days later, and Rice Bloom (5 kg/ha) at heading and grain filling stages with wetting agent for increasing spikelet fertility, grain filling and yield up to 15%. *Zinc Solubilizing Bacteria* (2 L/ha) + Zinc Sulphate (25 kg/ha) + RDF; *Bacillus altitudinis* FD48 improves drought resilience. Water management: Alternate Wetting and Drying from 21 DAS to 10 days before harvest + 125% RDF. Stubble management: Incorporate with TNAU Biomineralizer (2 kg/t) + urea. Disease control: Azoxystrobin + Propiconazole @ 500 ml/ha at tillering and booting stages. FD48 applications: Seed treatment (125 ml/ha) and foliar spray (500 ml/ha at 30 & 50 DAS) enhance yield under stress.

Stage-Specific Inoculants: Biofertilizer pack with *Azospirillum*, *Azotobacter*, *Phosphobacteria*, *Potash bacteria*, *Pseudomonas*, and PPFM for different crop stages; enhances yield by 10-15% and reduces chemical fertilizer use by 25%. **Zinc Solubilizing Bacterial Biofertilizer:** Carrier-based or liquid form; enhances Zn nutrition and yields by 17-20%, adding 250-400 kg/ha. **TNAU Micronutrient Mixtures:** Enriched FYM-based micronutrient mixtures for wetland and rainfed rice; alleviates deficiencies, improves quality, and enhances yield by 11-20%. **Nanosci Lure for YSB:** Nano-pheromone lure for eco-friendly pest monitoring and mass trapping of yellow stem borer. **Stage-Specific Inoculants:** Biofertilizer pack tailored to growth stages; increases yield (10-15%) and reduces chemical fertilizers by 25%. *Bacillus altitudinis* FD 48: Dual-use bioagent (seed+foliar) enhancing root traits and yields under both irrigated and stress conditions. **TNAU Biomineralizer for Stubble:** Microbial product (2 kg/tonne) for in-situ rice straw decomposition to reduce burning and enrich soil.

Maize

Nutrient & Soil Management: Neem cake @ 250 kg/ha; Zn citrate (0.5%) + Fe citrate (1%) foliar sprays at 30, 40, and 50 DAS in deficient soils. **FAW Management (Technology Capsule):** Seed treatment: Thiamethoxam 30 FS @ 10 ml/kg; maintain spacing (60 × 25 cm) + rogue rows every 10 rows. **Border cropping:** Cowpea, gingelly; **Intercrop** with blackgram; install pheromone traps (50/ha) and solar traps (1/ha). **Insecticides by stage:** Early whorl (15-20 DAS): Emamectin benzoate 5 SG, Chlorantraniliprole 18.5 SC, Flubendiamide 480 SC, or Azadirachtin. Late whorl (35-40 DAS): Metarhizium anisopliae, Emamectin benzoate, Novaluron, or Spinetoram. **Tasseling/cob formation** (60-65 DAS): Use any insecticide not used earlier. **Biologicals:** Release *Telenomus remus* @ 1.25 lakh/ha.

COMMERCIAL CROPS

Cotton

High Density Planting: Closer spacing (90 × 15 cm) for 29,629 plants/acre using suitable varieties like Suraksha, CO-17, and VPT-2.

Fertilizer: Apply 100:50:50 kg NPK/ha; weed control using Pendimethalin 38.7% CS @ 700 ml/acre within 24-48 hours of sowing.

Growth Management: Spray NAA @ 40 ppm or Cotton Plus @ 2.5 kg/acre; apply Mepiquat Chloride (0.015%) at square/flowering stage; use sodium chlorate (0.9%) or Drop Ultra as defoliant at 60% boll burst.

Intercropping: Paired row planting (45/90 × 30 cm) of Bt Cotton with two rows of cluster bean; application of Mepiquat Chloride and Sodium Chlorate enhances suitability of CO 17 for mechanical harvesting.

Coconut

Pest Traps: Install yellow sticky traps (20/ha or 8/ac), trunk banding, and spray jet water on fronds. **Biocontrol:** Release *Apertochrysa astur* (1000/ha), *Encarsia spp.* (250/ha or 100/ac). **IDM Package:** FYM (50 kg) + neem cake (5 kg) + fertilizers; CuSO_4 (200 g) + MgSO_4 (1 kg) alternately; bio-agents (*Trichoderma*, *Bacillus*, *Phosphobacteria*, *Azospirillum*, VAM); TNAU Coconut Tonic (40 ml twice/year);

COCOCON drench @ 2 L/tree every 2 months; crown spray of hexaconazole if needed. *TNAU Coconut Tonic*: Nutrient-rich tonic applied via root feeding (200ml/tree every 6 months); increases photosynthetic efficiency, nut yield, and palm vigour. *IIHR Arka Decomposer*: Microbial mix that rapidly composts coconut residues, enhancing organic carbon and soil health. *Copra Guard*: Polymeric nano-formulation (10 ml/litre) replacing sulphur fumigation to control fungal growth on copra.

FRUIT CROPS

Banana

Insect Management: Apply emamectin benzoate 5 SG @ 0.4 g/L at 5th, 6th & 7th months via pseudo stem injector. Seedling Bio-hardening: *Bacillus velezensis* + *Brachybacterium paraconglomeratum* (1% consortia) twice fortnightly; soil drenching @ 0.5 L/plant at 2, 4, 6 & 8 months.

Papaya

Vector & Disease Management: 3-row maize border; insect-proof nursery; neem oil spray (3%); humic acid drenching @ 2 ml/L every 2 months; foliar spray of ZnSO₄ (5 g), boron (1 g), and urea (10 g)/L at 4th and 7th month. Jeevamruth: Drenching (200 L + oilcake extracts) @ 1 L/plant on 3rd, 5th & 8th months; Flonicamid 50 WG @ 75 g a.i./ha on need. Hi-Tech Farming Practices: Raised bed planting with drip+fertigation and foliar micronutrients (Zn, B) improving yield and fruit quality by 25-30%.

VEGETABLE CROPS

Chilli

Seed treatment: *Bacillus subtilis* @ 10g/kg; use three-row maize barrier; install yellow sticky traps @ 12/ha. Nutrition: Apply and spray a micronutrient mix (Fe, Zn, Cu, Mn, B). Virus & pest management: Rogue infected plants till 45 DAT; apply Imidacloprid 17.8% SL @ 3ml/10L and Pyriproxyfen 10% EC @ 1ml/L alternately. Fungal disease management: Spray Azoxystrobin 18.2% + Difenoconazole 11.4% SC @ 0.1% thrice at 15-day intervals. IPM module for chilli: Includes border cropping with cowpea/lab-lab/coriander, bird perches (20/ha), sticky traps (20/ha), pheromone traps (12/ha), need-based foliar sprays with Azadirachtin 1%, NSKE 5%, 3G extract 5%, five-leaf extract 10%, and *Bacillus subtilis* 0.5%-effective in reducing thrips population and increasing yield and B:C ratio.

Cassava

IPM module for mealybug: Application of Azadirachtin 0.15% (1.5 L/acre) on 60 DAP followed by release of *Anagyrus lopezi* and *Apertochrysa astur*; reduces mealybug population and increases tuber yield. *TNAU Cassava Booster*: A mixture involving cow dung slurry enriched with neem cake, bioagents, and nutrients; enhances starch content, reduces CMD, and increases yield by 20-25%.

Nutrients for Vegetables: Bio NPK Formulation: Multi-functional biofertilizer (NPK bacteria) improving nutrient uptake and reducing chemical fertilizer dependency. BIOGROW Liquid (Tomato): Microbial liquid with *Bacillus* and *Pseudomonas* boosting yield (25-30%), lycopene, -carotene, and plant health.



Farmers are advised to follow the sowing rule to take-up sowing of rainfed crops like Cotton, Soybean, Maize, Jowar, Redgram, Greengram etc., only after receipt of cumulative rainfall of 50-60 mm in light soils and 60-75 mm in heavy soils or the soil should be wet up to top 15-20 cm depth after enter of South-West monsoon rains into the state.

CEREAL CROPS

Paddy

Under assured irrigation farmers are advised to raise rice nurseries of long duration (140-145 days) varieties from May 20th to June 20th. Under assured irrigation farmers are advised to raise rice nurseries of medium duration (130-135 days) or short duration (120-125 days) varieties from June 20th onwards. Month of June is not optimum for raising paddy nurseries using Telangana Sona (RNR15048) variety. Take the Greengram as catch crop to preceding rice in areas where the release of water into the canals and tanks is delayed. Green manuring with Sesbania/ Crotalaria / Pillipesara / grain legumes, residues of blackgram / greengram and in situ incorporation before flowering. Varieties: cultivation of long duration (BPT-5204, WGL-44, RNR 11718 etc.), medium duration (WGL-14, JGL-11470, JGL-384, RNR-28361, JGL-33124 etc), short duration (RNR-15048, KNM-1638, KPS-6251, WGL-362, KNM-118, JGL-24423, RNR-29325, JGL-28639 etc.) varieties.

Direct seeding of rice with Seed-cum ferti drill wherever suitable. In Telangana state, dry converted wet and wet direct seeded rice are popular, and the sowing can be started from June ending. Treat seeds with carbendazim @ 3 g/ kg (dry seed treatment) or 1 g/L (wet seed treatment). Farmers are advised to apply the Carbofuran 3CG granules @ 800 g per acre of nursery one week before pulling of the nursery. Root dipping: prepare the slurry with 5 packets (1000 g/ha) of Azospirillum and 5 packets (1000 g/ha) of phosphobacteria or 10 packets of (2000 g/ha) of Azophos, inoculant in 40 lit. of water and dip. the root portion of the seedlings for 15-

Greengram and Blackgram

Sowing time is generally from June 15th fortnight- July 15th during Kharif, Sept 15th to 30th October during Rabi, January last week to March 15th during Summer. Greengram varieties MGG-295, MGG-385, MGG-351, TM-96-2, WGG-37 and WGG-42, Virat, IPM-2-14, Sikha. Blackgram YMV-resistant varieties MBG-1070, PU-31, GBG-1 and TBG-104 are the preferred varieties. Treat seeds with imidacloprid @ 5g or thiamethoxam @ 5g/kg. Intercropping can be done with cotton (2:1 or 1:1 ratio) or redgram (6:1)/(8:1). Greengram can be taken as a preceding crop to Rice and Chilli as green manure for incorporation in the soil.

OILSEED CROPS

Groundnut

Seed rate recommended is 237 kg/ha and practice seed treatment with imidacloprid 600 FS @ 2 ml/kg followed by tebuconazole 2 DS @ 1g or Trichoderma viride @ 10g/kg to control early and late leaf spot diseases, sucking pests, collar rot, PSND in the early stages. Application of Gypsum @ 500 kg/ha at pegging stage near podding zone for pod development. Weed management with Diclosulam 84% WDG @31 gm/ha as pre-emergence herbicide followed by post-emergence application of Imazethapyr @ 750 ml/ha at 20 to 25 days after sowing.

Soybean

Sow varieties ASB-22, JS-335, AISb-1, JS-93-05 during 15th June 1st week of July using seed drill/ gorru/ desi plough/ by tractor drawn seed cum fertilizer drill / broad bed furrow planter. Sowing of Soybean by Broad bed Furrow (BBF) planter or Raised bed method facilitates the crop to withstand heavy rains and moisture stress conditions and to obtain higher yields. Treat the seeds with carbendazim @ 1g or thiram or captan @ 3 g/kg followed by 1.5 ml of imidacloprid (48 %) and Rhizobium culture @ 200 g/10 kg seed. Apply 50 kg DAP + 26 kg MOP at the time of sowing and top dressing of urea @ 25 -30 kg/acre at 30 DAS. Pre-emergence herbicides like pendimethalin 30 % @ 1 - 1.5 L/ acre should be applied within 24-48 hours after sowing with the help of hand sprayer for better control of weeds. Post emergence application of quizalofop ethyl @ 400 ml/ acre for control of grasses and imazethapyr + imazomax 40 g /acre or imazethapyr 10 % @ 250 ml/acre to control the grasses and broad-leaved weeds. To control stem fly and stem gridler as a prophylactic application of carbofuran 3G granules @ 8 kg/acre is recommended at 10 DAS. To control of pod blight spraying of tebuconazole + sulphur @ 2.5 g/L is recommended. Dry the seed and maintain the below 12 % moisture during seed storage to avoid incidence of pest and disease.

Castor

Sow high yielding castor hybrids viz., ICH-5, ICH-66, PCH-111, DCH-519 & ICH-6, PCS-262 from 15th June to 15th July in rainy season under rainfed conditions. Intercrop with redgram in 1:1 or greengram in 1:3 ratio, open furrows between rows to conserve moisture in light soils.

LIVESTOCK

Animal Husbandry & Fisheries

Agriculture and allied sectors play a vital role in the lives and livelihoods of the majority of Telangana's population. The state is rich in livestock resources and

livestock production, according to national Livestock census 2019, the total livestock population in the state is 3.26 crore. The livestock sector is a major driver of primary sector growth. The sectors contribution to Telangana's Gross State Value Added (GSVA) at current prices increased from Rs. 96,908 crores in 2023-24 to Rs. 1,02,835 in 2024-25. Across India, inland fisheries where farmers build ponds on their landholdings to breed fish, is emerging as an important source of additional income to the farmers (Socio Economic Outlook, 2025). Scarcity of feeds and fodder resources is a big problem limiting animal production. Among all, flood and drought are the major devastating natural calamity leading to a massive loss of vegetation.

General Guidelines for Livestock Management: To protect animals from the weather fluctuations keep them in well-ventilated shelters and under the shade of trees. Restrict the movement of visitors to the livestock farms and practice biosecurity measures in case of disease out breaks. Keep sufficient feed, fodder and medicine in hand. Vaccinate animals against Foot and Mouth Disease and Hemorrhagic septicemia (Cows and buffaloes), PPR (goats). Except pregnant animals, deworm all animals for external and internal parasites. Keep clean drinking water available at all times. Monitor water intake by animals to protect from heat stress. Supplement herbal immune boosting products in ration to boost immunity. Observe and detect oestrus and inseminate on time. Splash/bath buffaloes especially pregnant animals to prevent adverse effect of summer on oestrus and conception and prevent early embryonic mortality/ abortion due to heat waves.

Offer ample milk or electrolyte water to new born calves to prevent dehydration, heat stroke and death of neonatal stocks. Maintain clean, well-ventilated shelters with proper drainage to prevent diseases during floods. Ensure adequate space: Overcrowding causes stress and disease outbreaks. Local authorities may be informed about the outbreak of the diseases. All animals should be untied/evacuated immediately in case of floods to avoid mortality. All dead animals should be disposed of hygienically and should not be thrown in nearby drainages, water bodies etc. Burial Pit should be of 8X9 feet. Sprinkling of disinfectants like bleaching powder, quick lime etc. in the affected areas helps in reducing the spread of diseases. Remove all the flammable items from the area in and around the animal shed. Always maintain a sufficient stock of feed, fodder, water and medicines and store them in a safe place. Proper Care should be taken after reallocation of animals after natural disasters. During Drought bajra fodder varieties and in water logging areas Para grass may be cultivated. Clean, safe, hygienic water should be provided to the animals. Restrict the transportation of animals to consumer's doorstep for milking.

Dairy

Disinfect sheds regularly with Phenol 1-2% solution and Washing Soda (4% solution). Deworm all adult stock with broad spectrum antihelmintic, Albendazole (Dose: 10 mg/ kg body weight) during last week of May. Vaccinate animals with Haemorrhagic septicaemia (HS) and black quarter in June @ 5ml per animal subcutaneously. Segregate animals in milk production, dry, pregnant, non- pregnant and heifers. Allow animals to graze early in morning or later in evening. Follow body weight-based feeding regimen. Provide 1 kg concentrate for every 2.5 kg milk for

liquids & probiotics in drinking water during summer. Utilize broken rice or bajra (up to 30-40%), cotton seed meal (10-12%) and rapeseed meal (5 -8%) in poultry when there is shortage of maize during natural calamities. Keeping in view of disease outbreak follow biosecurity measures and mass vaccinate backyard poultry birds against Newcastle and viral diseases to prevent outbreak during stress periods. For hatching purpose of fertile eggs keep it in fridge at temperature of 4°C. To improve the growth rate 1 per cent protein level (soybean meal) may be added in the feed. Suitable coccidiostat should be added in the feed continuously by consulting the local veterinarian to prevent coccidiosis. 3-5 g of oyster shell/limestone/grit per day/bird shall be given to laying hens to avoid leathery eggs.

Value Addition: Convert milk into ghee, paneer, curd, khoa for better returns. Explore meat processing and packaging for hygienic sale. Utilize manure for biogas and compost.

Fisheries

Failed Monsoon: Use bore well or alternative water sources to maintain minimum pond water levels. Reduce stocking density to lower water demand. Adopt partial harvest strategies to minimize losses. Use water-saving practices like reduced water exchange. Avoid new stocking if water availability is uncertain.

Flood: Strengthen pond bunds and spillways to prevent overflow and escape of fish. Install protective netting around pond perimeters. Avoid feeding during peak flood to prevent water quality issues. Harvest fish early if flood risk is imminent. Apply disinfectants like Benzalkonium Chloride 80% (BKC) @ 2.5 L/ha and Rock Salt @ 75-125 kg/ha/5ft depth/ha after floodwater recedes to control disease outbreaks.

Drought: Reduce biomass load in ponds by early or partial harvesting. Avoid use of organic manure if water level is too low to prevent oxygen depletion. Apply aeration measures (e.g., paddle wheel aerators) to maintain DO levels. Maintain good water quality by regular monitoring and minimum exchange.

Delayed Monsoon: Postpone seed stocking until adequate water level is available. Increase application of manures/fertilizers once rains begin to promote plankton growth. Avoid breeding activities during uncertainty to protect brood stock. Maintain brood stock in protected tanks.

Advanced Monsoon: Prepare ponds early for stocking, complete liming 200 kg/ha and fertilization in advance. Ensure brood stock is not disturbed during monsoon (15th June to 31st August). Avoid overfeeding as heavy rains can cause feed waste and water quality degradation. Watch for sudden temperature or pH fluctuations due to early rains.

High Temperature: Maintain higher water levels (5-6 feet) to buffer temperature fluctuations. Avoid feeding during peak heat hours, feed early morning and late evening. Use shade nets or plant trees around ponds to reduce direct sunlight. Monitor dissolved oxygen (DO) and apply aeration during hot afternoons.

Low Temperature: Reduce feeding rate and switch to highly digestible feed. Avoid handling fish during early mornings and nights. Provide windbreaks and increase pond depth to retain heat. Avoid use of antibiotics and strong chemicals during cold periods unless essential.



CEREAL CROPS

Paddy

Farmers should begin nursery preparation by late May using recommended varieties like CR Dhan 310, CR Dhan 311, Gomati, or Swarna Masuri (135-150 day duration). Treat seeds 24 hours before sowing with Carbendazim/Captan (2g/kg) or Trichoderma viride (5 g/kg) to prevent soil-borne diseases. Transplant seedlings based on variety duration: 20-25 days for short/medium (?120 days), 25-30 days for medium-long (120-130 days), and 30-40 days for long-duration (130-160 days) varieties. To enhance soil fertility, incorporate green manure crops (dhaincha, cowpea, moong) or Gliricidia leaves during field preparation, along with Azotobacter application (12kg/ha soil or 1.5L/ha seed treatment). For weed control, use post-emergence herbicide Metsulfuron Methyl + Chlorimuron Ethyl (50gm/ha). Farmers currently harvesting boro rice should watch for gundhi bugs (control with Ethofenprox 10EC/Imidacloprid sprays) and blast disease (apply Hexaconazole at flowering), harvesting when 85% grains turn straw-colored.

Maize

Farmers are advised to cultivate high yielding varieties include RCM 1-1, RCM 1-2, RCM 1-3, Vijay Composite, DA 61-A, HQPM-1, and HQPM-2. Perform three deep ploughings to ensure proper soil aeration and seedbed preparation. Apply atrazine herbicide 2-3 days after sowing to effectively manage weeds for up to 30 days. Proper earthing up is essential to prevent lodging due to high winds or cyclonic conditions, ensuring stable maize growth. Ensure a well-maintained drainage system during the kharif season to avoid waterlogging, as excessive moisture can negatively impact maize yield. To control maize stem borer, apply Dimethoate 30 EC (660 ml/ha) or Carbaryl 50 WP (1.0 kg/ha) for effective management. For Fall Armyworm (FAW) control in kharif maize, use Emamectin Benzoate 5% SG (600-700 g/ha) as a systemic insecticide to protect the crop.



CEREAL CROPS

Direct Seeded Rice

DSR is suitable for sandy loam to heavy clay soils, but should be avoided on light-textured soils like loamy sands and sands. Choose a well-levelled field (which is best achieved using laser-assisted land levelling) with good drainage and reliable irrigation, as DSR needs precise water control during early growth stages. Select high-yielding, early to medium duration varieties that are resistant to lodging and local pests/diseases. Good choices include MTU-7029, Pusa Sugandha 4, Pusa Rice Hybrid 10, NDR- 9930111, Sambha Sub-1 and NDR-9436. DSR can be sown in dry soil followed by light irrigation for germination, or in moist soil after pre-sowing irrigation or rain. Planking after sowing in moist soil conserves moisture and improves establishment. While both methods suit conventional tillage, moist soil sowing is preferred for ZT-DSR due to softer soil.

Plan to sow seeds between mid-May and mid-June, before the monsoon sets in, to take advantage of early rains and reduce weed problems. Sowing with primed seeds in moist soil promotes rapid crop establishment. To prime, soak seeds in water for 10-12 hours in a gunny bag, then air-dry for 1-2 hours to ensure smooth flow through the seed drill. For pre-germination (moist soil sowing), incubate the soaked seeds for an additional 8-12 hours. Seeds should be sown soon after priming or incubation. Avoid sowing pre-germinated seeds in dry soil, as this can reduce emergence.

Treat seeds with fungicides (e.g., Carbendazim 2g/kg seed) to prevent seed-borne diseases. Use a seed rate of 20-30 kg per hectare, depending on the sowing method (drum seeder or broadcasting) and type of grain (fine or coarse grain) at a depth of 2-3 cm in rows spaced 20-25 cm apart to ensure uniform germination and allow easy weed control later.

Apply fertilizers based on soil test results. If a test is not available, use a general dose of 80-120:60:60 kg/ha of N:P:K along with 25 kg/ha of zinc sulphate (33%). Apply all

fertilizers except urea at the time of sowing, placing compound fertilizers like DAP or NPK as basal. Apply the remaining nitrogen as urea in 3-4 equal splits, starting 2-3 weeks after sowing and finishing at panicle initiation. Ensure nitrogen is applied just before irrigation.

Immediately after sowing, apply a pre-emergence herbicide such as Pendimethalin (1.0 kg a.i./ ha) within 1-2 days to control early germinating weeds. Monitor weed growth, and if needed, spray a post-emergence herbicide like Bispyribac-sodium (25 g a.i./ha) around 20-25 days after sowing, depending on weed type and density.

Keep the field moist, especially during critical stages like germination, tillering, flowering, and grain filling. Avoid standing water in early growth stages, but ensure timely irrigation. Adopt alternate wetting and drying method to save water. Allow the soil to dry for 2-3 days before irrigating again.

Regularly inspect crops for pest attack, especially stem borer, leaf folder, and brown plant hopper. Use insecticides like Chlorantraniliprole, Flubendiamide, or Imidacloprid as per recommendations when needed. For diseases such as rice blast, sheath blight, and bacterial leaf blight, grow resistant varieties, improve drainage, and spray appropriate fungicides (e.g., Tricyclazole, Propiconazole) in early stages if symptoms appear.

Harvest when about 80-85% of the grains on each panicle turn golden yellow. Ensure the grain moisture content is around 20% at the time of harvest to avoid damage during threshing. Dry the harvested paddy to 13-14% moisture to prevent fungal growth in storage.

Puddled transplanted rice

Select suitable rice varieties: Early maturing (110-115 days): Malviya Dhan 2, Pusa 2-21, Pusa 33, Pusa 834, Govind, Saket 4, Narendra 97. Mid-late maturing (120-125 days): NDR 2065, VL Dhan 61, Pusa 169, Pusa 205, Pusa 44, Sarju 52, Pand Dhan 10, Pant Dhan 12. Long duration (more than 130 days): Malviya 36, Narendra 359. Use certified, disease-free seed; seed rate: 30-40 kg/ha; seed treatment with Carbendazim or Trichoderma @ 2 g/kg seed to prevent fungal diseases; for insect protection, treat seed with Imidacloprid @ 5 ml/kg seed.

Nursery area: about 800-1000 m² per hectare of main field, nursery sowing time: mid-May to mid-June. Apply FYM or compost to nursery @ 5-10 t/ha and apply 1 kg urea, 2.5 kg SSP and 0.5 kg MOP as basal for 100 m² nursery. Maintain proper moisture; avoid water stagnation in early stages; use light irrigation and drain excess water during heavy rainfall in nursery.

Transplant 20-25 day-old seedlings during mid-June to early July, preferably in the morning or evening hours, avoiding continuous rain, using 2-3 seedlings per hill with a spacing of 20×15 cm for HYVs and 20×20 cm for hybrids. Maintain 2-3 cm standing water for the first 3-5 days after transplanting, followed by 2-5 cm during tillering and 5 cm during panicle initiation and flowering. Apply the recommended dose of fertilizers, and in case of zinc deficiency, spray 0.5% zinc sulphate mixed with 2.5% urea solution at 15-day intervals, repeating 2-3 times as needed.

Perform weeding manually or mechanically at 20 and 40-45 days after transplanting; for chemical weed control, apply Bispyribac-sodium @ 25 g a.i./ ha at 15-20 days after transplanting. Monitor for pests such as stem borer, leaf folder, gall midge, and grain bugs; apply Cartap hydrochloride 4 G @ 10 kg/ha for stem borer; spray Chlorantraniliprole @ 150 ml/ha for leaf folder and for grain-sucking bugs, apply Acephate 75 SP @ 1 g/liter. Monitor for diseases such as blast, sheath blight, brown spot and false smut, and manage them with appropriate fungicides- spray Tricyclazole @ 0.6 g/liter for blast, Mancozeb @ 2.5 g/liter for brown spot, and Tilt (Propiconazole) @ 1 ml/liter for false smut.

Avoid nitrogen topdressing at flowering/grain-filling stage to reduce disease risk. Harvest when 80-85% grains are golden yellow and hard. Stop irrigation 10-12 days before harvest. Delay in harvesting increases shattering and lodging risk. Thresh soon after harvest and dry grains to 13-14% moisture

Basmati Rice

Cultivation practices for Basmati scented rice differ slightly from general rice, primarily in aspects like nitrogen application, lodging and specific pest/disease management. Basmati, particularly tall and semi-tall varieties, require less nitrogen and prone to lodging. For pest and disease management, basmati rice requires specific strategies for diseases like neck blast and sheath blight. Variety of scented basmati rice: Azad Basmati, Pusa



Paddy blast disease

Basmati 1, Pusa Basmati 6, Pusa Basmati 4, Pusa Sugandh-2, Pusa Sugandh-3, Pusa Sugandh-4, Pusa Sugandh-5, Ballabh Basmati 22, Malviya Sugandh 105, Malviya Sugandh 4, Malviya Basmati 1, Pusa Basmati 1509, PRH 10.

Optimum sowing time for DSR is the first fortnight of June. For nursery sowing and transplanting: Pusa Sugandh-4, Pusa Sugandh-5, Ballabh Basmati 22, Malviya Sugandh 105, Malviya Sugandh 4, Malviya Basmati 10, Pusa Basmati 1509: Nursery sowing in 1st fortnight of June; transplanting in 1st fortnight of July. CSR 30 (for pH >8.5 soils), Basmati 370, Basmati 386: Nursery sowing in 2nd fortnight of June; transplanting in 2nd fortnight of July.

Treat seeds with *Trichoderma harzianum* @ 15 g/kg seed before sowing to prevent foot rot. Dip seedling roots for 6 hours in *T. harzianum* solution (15 g/litre) before transplanting. Transplant 25-30 day-old seedlings (5-6 leaf stage), using 2-3 seedlings per hill at 20×15 cm spacing (33 hills/m²); reduce to 15×15 cm (44 hills/m²) for late planting to avoid yield loss. Prefer green manuring with 45-55 days-old dhaincha/sunn hemp or incorporation of summer moong straw. Skip urea application if green manure is used.

Apply fertilizers based on soil test and crop rotation. Avoid phosphorus if already applied in the preceding crop. For Pusa Sugandh-4, Sugandh-5, Ballabh Basmati 22, and Malviya varieties, apply 90 kg urea/ha in two equal splits at 3rd and 6th weeks

CEREAL CROPS

Rice

Important rice varieties for Tarai, Bhabhar & Plain areas are Pant Dhan 10,12, 18,19, 22, 24, 26 & 28, Narendra 359, PR 113 & 114 in bold rice, scented rice-Pant Sugandha Dhan-15,17, 25 & 27, Pusa Sugandha-4 and 5, basmati rice-Pant Basmati 1 and 2, Pusa Basmati 1121, Pusa Basmati 1509, Pusa Basmati 1692, Pusa Basmati 1718, Pusa Basmati 1728, Pusa Basmati 1847, Pusa Basmati 1885, Pusa Basmati 1886 and traditional basmati rice are Type 3, Basmati-370 and Taraori recommended for cultivation. Time of nursery sowing is 2nd fortnight of May to 1st week of June and transplanting time is June end to 1st week of July. Rice varieties for irrigated valleys & lower hill (<900 msl)- Govind, Pant Dhan 6, 10, 11& 12, VL Dhan 65, 81, 85 & Vivek Dhan 82, for mid hills (900-1500msl)- Pant Dhan 6,10 & 12, VL Dhan 61, 65,85, Vivek Dhan 62 & 82. Nursery sowing time is May 2nd fortnight to June 1st week in valley & lower hill and May 1st fortnight for mid hills, transplanting time is June last week for valley and lower hill and 2nd fortnight of June for mid hills. Varieties of Jethi rice (rainfed, direct seeds) are Vivek Dhan 154 & VL Dhan 221 for sowing during 4th week May - 1st week of June, Chetaki rice (rainfed) varieties are VL Dhan 207, 208 & 209 are suitable for sowing in mid-March to 1st fortnight of April. Seed rate in transplanting rice are 40 kg (bold grains) and 25-30 kg/ha for fine grains is sufficient. Raising the healthy nursery in basmati rice 1.0kg of seed should be sown in 25 m² area. Planting geometry should be maintained as 20*10-20cm in transplanting to obtaining higher productivity. To reduce/avoid the crop weed competition, transplant at least two healthy seedlings per hill early in the season to optimize plant population.

Fertilizer Application: The transplanted rice crop in Tarai, Bhabhar & Plain area should be fertilized with NPK @ 150:60:40 Kg/ha, hybrid rice 150:60:60, scented and basmati rice 100:60:40, old basmati rice 60:30:20 with 50 % N and entire dose of



HYV of Rice Pant Dhan-28



Basmati rice cultivation in Valley of Dehardun

P & K at the time of transplanting as basal and remaining N top dressing in two equal splits at 3rd and 6th weeks after transplanting. In hilly area dwarf rice should be fertilized with 100-120:60:40 and desi rice 60:30:30 kg/ha of NPK. In case of zinc deficiency symptoms in previous crop, apply zinc sulphate (21%) 25 kg/ha at puddling.

Weed Management: Pendimethalin 1.0 kg /ha at 0-3 DAS fb metsulfuron methyl 4.0 g/ha at 25-30 DAS and one hand weeding (40-45 DAS) is recommended for DSR. Post emergence application of bispyribac sodium 10 SC 25g/ha or penoxulam 24 SC 22.5g/ha or ready mixture of penoxulam 1.02 % + cyhalofop-butyl 5.1% (120- 135 g/ha) or cyhalofop-butyl 10%EC 80 g/ha used for effective control of weeds in direct seeded rice. In transplanting rice, maintaining at least 3-4 cm standing water in main field during early stages is recommended to check weed emergence. Application of butachlor 1.5 kg/ha, or pretilachlor 1.0 kg/ha as pre-emergence (0-3 DAP) and if required 2,4-D 500 g/ha or metsulfuron methyl 10% + chlorimuron ethyl 10% WP 20g/ha as post-emergence to control sedges and broad leaf weeds or one manual weeding is effective for weed control in transplanted rice.

Organic Pest Management: Application of Trichoderma enriched FYM compost @100kg/ha (Trichoderma 2kg+100kg FYM composting) at field preparation. Seed treatment with *Pseudomonas fluorescens* 1.5% LF @ 4.5 ml/kg seed for leaf and neck blast management. Seedling treatment with *Pseudomonas fluorescens* 1.5% LF @ 10 ml/liter of water dip seedlings for 30 minutes for management of BLB. Two spray of *Pseudomonas fluorescens* 1.5% LF @ 6 L/ha, 1st at vegetative growth and 2nd at 25% flowering initiation can be done for blast control. Spray of Azadirachtin 0.03% EC 5.0L dissolved in 1000 liter of water/ha for leaf roller, stem borer and brown plant hopper .Application of *Bacillus thuringiensis* var rustaki 2.5% @1.5 kg in 500-750 liter of water for stem borer and leaf folder management in rice.

Chemical Pest management: Spray of fipronil 5% SC @1000-1500ml/ha in 500 liter water for management of stem borer, BPH, green leaf hopper, rice gall midge and white backed plant hopper. Spray of thiamethoame 25% WG @100 g/ha in 500-750 liter water for stem borer, gall midge, leaf folder, white backed plant hopper, brown plant hopper, green leaf hopper and thrips control. Spray of imidacloprid 6% + lambda-cyhalothrin 64% SL @ 300 ml/ha in 500 liter of water for stem borer, hispa,

plant hopper and gandhi bug management. Spray of hexaconazole 5%EC 1000ml or azoxystrobin 18.2%+ difenconazole 11.4% SC 500ml or mancozeb 75WP 1.5-2.0 kg or isoprothiolane 40% EC 750ml/ha for of blast and sheath blight disease management in rice.

Maize

Recommended sankul varieties for Tarai, Bhabhar & Plain area are Pant Sankul Makka-3, Sweta, Bajora Makka-1 & Vivek Sankul-11. Hybrids maize are H M 4, 10, 11, 13, HQPM 1, 4, 5, Pant Sankar Makka-1, 4, 5, 6, Sartaj, P-3522 and Vivek Makka-51. Best sowing time is mid-June in Uttarakhand plains. Popcorn varieties: Pant Popcorn-1 and VL Amber Popcorn, fodder maize variety is African Tall. Sowing time is mid-June to mid-July. Sankul varieties suitable for hills: Vivek Sankul Makka-11, 37, 31, 35 and Pant Sankul Makka-4, hybrids maize are Vivek Sanker-9, 39, 45, 53, 55, 57 and DMH-109. Time of sowing in lower hill (1st week June to mid-June), mid hill (May end to mid-June) and higher hill during April end to mid-May is suggested. Optimum seed rate of 18-20 Kg for sankul, 20-22 Kg for hybrids, 12-14 Kg for popcorn, 40-45 for baby corn and 8 kg/ha for sweet corn is recommended. Spacing for maize may be maintained at 60*20-25 cm in line sowing at a depth of 5 cm. Inter-cropping of cowpea/soybean/ginger in maize would be effective to achieve the higher productivity, profitability and also control the soil erosion in hills.

Fertilizer Application: The maize crop in Tarai, Bhabhar & Plain area should be fertilized with NPK @ 120:60:40 kg/ha. Full P and K, $\frac{1}{4}$ N apply at sowing and rest of N applied in 3 equal splits at 3-4 leaf stage, knee stage and flowering time. In case of zinc deficiency symptoms in previous crop, apply zinc sulphate (21%) 25 kg/ha as basal application.

Weed Management: For proper weed control minimum two hand weeding at 20-25 and 40-45 days after sowing is recommended. Application of atrazine 2.5 kg/ha as pre-emergence for broad leaf weeds fb tembotrione 42% SC 120 g/ha post-emergence at 3-4 leaf stage (15-20 DAS) for grassy and broad leaf weeds or alachlor 2.0 kg/ ha as pre-emergence for both grassy and BLW is recommended. To control the sedges, application of halosulfuron methyl 70WG 67.5g a.i./ha as post-emergence at 3-4 leaf stage (15-20 DAS) is recommended. For control of broad leaf weeds in maize crop application of 2, 4 -D dimethyl amine salt 58% SL 0.5 kg/ha or 2, 4-D ethyl ester 38% EC 0.9 kg/ha as post-emergence at 15-20 DAS.

Plant Protection: Banded Leaf and Sheath Blight disease in maize can be effectively managed through an integrated approach. Soil application of Trichoderma (Pant Bioagent-1) at the rate of 1 kg multiplied in 100 kg of vermicompost (VC) or farmyard manure (FYM), applied 10-15 days prior to sowing, has shown good results. Additionally, seed treatment with Trichoderma (Pant Bioagent-1) at 20 g



FAW infestation in Maize

setts in furrows at planting for checking the infestation of shoot borer. Collection and destruction of infested shoots at periodic intervals from March to May on campaign basis is also recommended. Whitefly can be managed effectively by removal of lower leaves followed by foliar application of Imidacloprid 0.005% + 2% urea is recommended. Urea solution should be prepared first and then Imidacloprid is to be added to avoid coagulation of the mixture. Spraying of 0.2% carbendazim or 0.2% Blitox-50 or 0.2% copper oxychloride are the most effective fungicides for pokkah boeng disease management. Propineb @ 0.25% and mancozeb @ 0.20% are found to be effective against sugarcane rust and it should be sprayed on the foliage just after the appearance of rust pustules, thrice at 15 days interval.

PULSE CROPS

Urdbean varieties are Pant Urd 19, 31, 35, 40, Pant Urd 10, Pant Urd 7, Pant Urd 8 & Pant Urd 9 for Tarai, Babhar and lower hills and sowing time is 3rd week of July to 1st week of August in plain areas and 2nd fortnight of June in valleys. Mungbean varieties for Tarai, Babhar and hills are Pant Mung 2, 5, 8 & 9 Pusa Vishal and Virat sowing time is last week of July - 2nd week of August in plain areas and 2nd fortnight of June in valleys. Pigeopea varieties for hill and plains area are Pant Arhar-3 & 291, Pant Arhar-6, VL Arhar 1 and UPAS-120, time of sowing in hills mid-April to mid-May and in plains till mid-June. Horsegram varieties are VLG-15 and 19 for hilly area, sowing time 1st fortnight June. The seed rate for urd and mung 12-15kg/ha is sufficient; line sowing maintains the 30-45 cm R to R space at 3-4 cm depth. Arhar crop prefer to sow in lines using seed rate of 15-20 kg/ha, spacing of 60-75*20 cm row to row and plant to plants obtaining desired plant population. Prior to sowing seed treatment with carbendazim 2.0 g/kg of seed, then seed bio fortification with Rhizobium and PSB and Trichoderma culture should be done.



Pulses cultivation in Uttarakhand

Fertilizer Application: Kharif pulses should be fertilized with NPK @ 20:40:30 kg/ha as basal. It is better to supply the P with sulphur (SSP) containing fertilizer. Foliar spray of 2% urea in pulses at flowering stage will be beneficial.

Weed Management: For proper weed control, 2 hand weeding at 15 and 30 days after sowing is recommended. For effective weed control in pulses viz., urd, mung, cowpea and pigeonpea pre-emergence application of pendimethalin 30%EC 1.0 kg/ha at 0-3DAS or pendimethalin 30%+ imezathapyr 2% 750+50 g/ha at 3-4 leaf stage is recommended. Application of imazethapyr 35%+imazamox 35% or quizalofop 5% EC 37.5-50 g/ha or propaquizafop 10% EC 75g/ha as POE at 3-4 leaf stage is recommended in such crops.

Plant Protection: Application of flubendiamide 20% WG @ 300 g per ha in 500 L water to control the Spodoptera litura and Maruca spp. Application of chlorantraniliprole 18.5% SC @ 100 ml/ha in 500 L water for pod borer control. Application of indoxacarb



Performance of kharif crop is highly affected by uncertainty of weather like depleting rainfall pattern, late onset of monsoon, uneven and erratic rainfall, intensive rainfall in very short span causing flood like situation or short/ long dry spell. So, farmers are advised to be remained updated about weather forecast being publicized through different electronic media. Crop wise advisories for farmers of the state of West Bengal are mentioned below in summarized form.

CEREAL CROPS

Rice

Select varieties depending upon land situation, cropping pattern, soil character and climatic vulnerabilities usually occurs in the state. In medium to upland condition medium to short duration high yielding (<125 days) like Gontra Bidhan-1, Gontra Bidhan-3, MTU-1153, MTU-1010, IET-4786, IET-4094, CR Dhan-802, Uttar Sona etc. or short duration varieties like Annada, Anjali etc. are usually recommended depending upon the next crop preference. Long duration (>135 days) paddy varieties like MTU-7029, CR Dhan-800, Dhiren and Swarna Sub-1 are preferred in medium-to-medium low land where as in low land having temporary water stagnation problem submergence tolerance rice varieties like Swarna Sub-1, CR Dhan 801 and Ranjit Sub- 1 are recommended. In low land having permanent water stagnation problem deep water rice varieties like Sabita is recommended. In order to cope up with drought like situation drought tolerant varieties like CR Dhan 801, CR Dhan-802, Sahabhazi etc. are to be cultivated. For saline lands, select the improved salt tolerant rice varieties Chinsura Nona 2 (Gosaba 6), Nona Sampad, Nona Suvarna or local variety Dudheswar.

Seed bed preparation (between May 20 to June, 5) and transplanting (June 10 to June 25) should be accomplished in time. Ideally, 25-35 days age old seedlings should be transplanted depending upon duration of the variety used. Seeds should be treated

structure. Control weed applying pendimethalin @ 2 ml/L water as pre-emergent herbicide or oxyfluorfen @ 1 ml/L at 10-15 DAT and/or oxygold @ 2 ml/L water, if weed infestation is heavy and yellowing of tip occurs. Select upland or medium up-land with good drainage facility. Saplings or vine cutting should be planted in broad raised bed to avoid or minimize damage of planted seedling during heavy rain. Use farm yard manure as much as possible. Application of ZnSO_4 @ 6 kg/ac as basal and spraying of chelated zinc @ 0.5 g/L of water and boron @ 1-1.5 g/L of water is recommended for better crop performance. In order to control thrips & aphids use Dimethoate @ 2 ml/L or Imidachloprid @ 1.5 ml/L of water. Tip burn is very common in kharif onion and to control it spray hexaconazole @ 2 ml/L water/acre.

Early Cauliflower and Cabbage

Farmers may sow seeds @ 250-300 gram/ac in the month of July and August under the low-cost shade house or net house for best result to get early seedling. Raised sapling may be planted in broad raised bed in open. However, for better result, one can cultivate it under poly-tunnel also.

Elephant Foot Yam

Keep rhizosphere clean and ensure good drainage facility to reduce chance of collar rot disease. If symptoms appear then application of FYM and neem cake mixture (10:1) inoculated with *Trichoderma harzianum* @ 2.5 kg/t of FYM-neem cake mixture (KAU POP 2024) or spraying of Carbendazim + Mancozeb @ 2 g/L water is recommended.

SPICE CROPS

Betel Vine

Treat soil with formalin at 1:50 ratio at least 30 days prior to planting. Treat vine cuttings by soaking in Streptocyclin solution (1 g/4 L of water) for 25 min followed treatment with 0.5% Bordeaux mixture. Select upland or medium up-land with good drainage facility. Saplings or vine cuttings should be planted in broad raised bed to avoid or minimize damage of planted seedling during heavy rain. Use 50% agro-shade net to maintain favourable micro-climate. Prune vines at 2 m height. Maintain boroj height of 2.5 m and apply soil rings at base after 2 m growth.

Maintain the boroj organically, as much as possible, with only organic manure, bio-pesticides and life saving irrigation. Avoiding inorganic fertilizers will increase plant defence against biotic and abiotic stresses. Spray micronutrient mixture having Ca, Zn, Mn, B and Mo @ 2 g/L of water at 2 months interval. Tricontanol (0.05%) should be sprayed thrice monthly during post-monsoon. To control whitefly in the main field installation of 60 yellow sticky traps/ha & spraying pyrifluquinazon of @ 1 g/L or Dimpropyridaz @ 0.5 ml/L at 15-day intervals is advocated or alternatively moringa root extract (10%) + Afidopyropen / Flupyradifurone in 10:1 ratio can be used. Pre- and post-monsoon soil drench with 1% Bordeaux mixture followed by soil amendment with *T. harzianum* and *P. fluorescens* mixed compost/mustard oil cake during the month of May can prevent collar rot and during June to October can control *Phytophthora* foot rot and leaf rot.

Ginger

Weeding and earthing up should be done within 45 to 60 days after planting. Crop may be affected by rhizome rot, blight, anthracnose disease which can be controlled by soil application of Trichoderma and Pseudomonas @ 7.5 kg/ha or spraying of Carbendazim + Mancozeb @ 2 g/L of water. Pests like rhizome borer, leaf roller, aphid, scale insect may damage the crop and these can be controlled by apply Neem oil 10000 PPM @ 2 ml/L of water.

Turmeric

Weeding and earthing up should be done within 45 to 60 days after planting. Crop may be affected by rhizome rot, blight, anthracnose disease which can be controlled by soil application of Trichoderma and Pseudomonas @ 7.5 kg/ha or spraying of Carbendazim + Mancozeb @ 2 g/L water. Pests like rhizome borer, leaf roller, aphid, scale insect may damage the crop and these can be controlled by apply Neem oil 10000 PPM @ 2 ml/L of water.

FRUIT CROPS

Mango

To control fruit drop spray NAA 4.5% W/W (Planofix) @ 0.2 ml/L at 10-12 days interval. For better fruit development, spray n-triacontanol (Miraculan) @ 0.5 ml/L at 15 days interval. Keep sufficient moisture in the orchard basin. Mulching with straw or grass may be practiced for better soil moisture conservation. Apply Emamectin benzoate @ 0.5 g/L or Spinetoram @ 0.6 ml/L to control fruit and shoot borer. Install methyl eugenol pheromone traps @ 30/ha and change lure every 15-20 days interval for controlling fruit fly of mango. In order to control anthracnose disease in mango spray Trifloxystrobin + Tebuconazole @ 0.5 g/L water.

Guava

Farmer are advised not to take full crop but to take only 30-40 % of present fruit stand and rest 60-70% should be sacrificed to obtain a good harvest in next winter (when the market price of the produce usually remains high). To remove/shed major portion of flower or small fruit, spray 15% urea solution (150 g/L of water) or 50 ppm NAA (Planofix @ 1.1 ml/L) and stop irrigating the crop. To control further fruit drop, spray NAA 4.5% W/W (Planofix) @ 0.2 ml/L at 10-12 days interval.

Wrapping of fruits should be done with transparent polythene packets with a paper piece inside it to protect the fruit from direct sunlight. For better fruit colour and quality, apply ammonium sulphate as source of N-fertilizer. For better fruit growth, soluble fertilizers like 19:19:19 @ 3 to 5 g/lit and any mixed micronutrient @ 1.5 to 2 g/L may be applied at 10-12 days interval. Spray N-triacontanol (Miraculan) @ 0.5 ml/L at 15 days interval. Install methyl eugenol pheromone traps @ 30 nos./ha and change lure every 15-20 days interval or spray poison bait (10 g jaggery + 2 ml dichlorvos in 1 L of water) for controlling fruit fly of mango. Apply *T. harzianum* to the root zone, mixed with compost, to prevent Fusarium wilt.

Comfortable and ventilated housing should be there and practice regular deworming after every quarter. Provide daily molasses-based anti-heat drink (molasses, wheat bran, salt, water) or jaggery-gram flour beverage to reduce heat stress. Check feed for fungal contamination. Start cultivating J-1006 maize post-flood for fodder. Use enriched straw (with molasses, urea, minerals), tree leaves, aquatic weeds, and silage for nutritional improvement. Frequent cleaning and sanitation of milk utensils and equipment's should be practised with suitable detergent. Maintain hygiene during feeding and milking; wash hands, teats, and feed utensils. Conduct timely vaccinations against FMD, HS and BQ, and Deworm animals with proper anthelmintics. Contact Veterinarian posted at the nearby Block under State Govt.

Sheep, Goat and Pig

The floor of the animal shed should be clean and dry to prevent foot rot and other diseases. In case of foot rot of goat and sheep, wash affected feet (foot bath) with 10% ZnSO₄ or CuSO₄ daily until lesions subside. In severe case, provide antibiotic Enrofloxacin @ 7 mg/kg b wt. Construct elevated low-cost shelters for goats and pigs in flood-prone areas with proper ventilation. Provide plenty of clean water along with supplementation mineral mixture @ 20-30 g/day in extreme hot season to combat with the heat stress. Provide daily molasses-based drink (molasses, wheat bran, salt, water) or jaggery-gram flour beverage to reduce heat stress. Practice regular deworming once in pre-monsoon and post-monsoon each. Give enterotoxaemia vaccination @ 2 ml s/c to pregnant Ewe/Doe before one month of expected date of lambing/kidding. 25-50 g molasses per day may be provided to each pregnant Ewe/ Doe to prevent pregnancy Toxaemia.

Poultry

Adopt hardy breeds like Vanaraja or Hit CARI for better heat tolerance. Provide electrolyte-enriched water frequently and maintain a balanced diet. Follow vaccination schedules to prevent disease outbreaks. Poultry birds may be fed with green leaves along with grinded maize/rice. Poultry sheds should be properly cleaned and disinfected with 60-70 g of KMnO₄ mixed with 120-150 ml formaline solution per 10 cubic feet space. Place the birds in the sheds after 24 hours of fumigation. Disinfect the premises of poultry houses with 1% hypochloride + bleaching powder (7 g/L of water). The saw dust should be dried completely before using as bedding material. Apply hydrated lime powder @ 1-2 kg per 10 sq. ft area in the shed before birds are placed in the shed. All the equipments in the shed should be disinfected using hot water.

Fisheries

Strengthen/repair pond dykes and avoid deciduous plants like bamboo. Prefer planting banana, coconut, arecanut, cucurbits or solanaceous crops. Remove pond bottom mud to prevent gas buildup and ensure proper de-weeding while retaining some Eichhornia for shade and evaporation control.

Ensure 1-1.5 m water depth and use polythene lining or dyke plantation to reduce seepage. Lime dry pond before stocking (500-1000 kg/ha) and monitor water pH (7.5-8.5), DO (>5 mg/L), and salinity as per species requirement. Ponds which

Field Crops

India has one of the largest agricultural sectors worldwide, which is the primary source of livelihood for ~55% of India's population. India has the world's largest cultivated land under wheat, rice, and cotton. India is the largest producer of milk, pulses, and spices in the world. It is the second-largest producer of fruits, vegetables, tea, farmed fish, cotton, sugarcane, wheat, rice, cotton, and sugar. The agriculture sector in India holds the record for second-largest agricultural land in the world, generating employment for about half of the country's population. Thus, the farming community comprise an integral part of this sector to provide food security with a means of sustenance.

During the last 9 years (2014-2023), a total of 2279 varieties of field crops have been released in India under the aegis of ICAR, which include 1092 of cereals, 343 of oilseeds, 343 of pulses, 137 of forages crops, 279 of fibrecrops, 68 of sugarcane and 17 of potential (underutilized) crops. These varieties include 1888 climate resilient varieties (83%) tolerant to one or more biotic and/or abiotic stresses. Out of these 1888 varieties tolerant to one or more biotic and/or abiotic stresses, 407 varieties have been developed specially for extreme climate including flood/ water submergence/ water logging tolerance (73), drought/moisture stress/ water stress tolerance (220), salinity/ alkalinity/ sodic soils tolerance (52), heat stress/ high temperature tolerance (49), cold/ frost/ winter chilling tolerance (13) using the precision phenotyping tools. In horticultural crops also, 22 climate resilient varieties have been developed. This includes varieties of potato (5), tomato (2), spinach (1), radish (1), cassava (4), sweet potato (2), greater yam (1), white yam (1), coconut (3) and taro (2).

Further, to tackle the widespread problem of malnutrition in globally, concerted efforts have been made to develop varieties rich in various essentially required micro-nutrients (Protein, Fe, Zn, Vit. A, Vit. C, oleic acid, anthocynin, lysine, tryptophan, removal of anti-nutritional factors like erucic acid, glucosinolates, KTI) for proper growth and development of the body and as a result 124 field crops and horticultural crops varieties have been released and notified during past nine years [Rice (10), Wheat (41), Maize (20), Pearl millet (9), finger millet (4), little millet (1), Lentil (2), Mustard (8), Linseed (1), Soybean (7), groundnut (3) and 18 Horticultural crops]. Concerted efforts for seed production have been made to take the biofortified varieties in general cultivation, which have resulted in coverage of more than 10 million hectares during 2022-23. Biofortified varieties have more than 50% share in total wheat breeder seed indents.

1. Implementation of GAP

The ministry of Agriculture and Farmer's Welfare and Processed Food Products Export Development Authority (APEDA), Govt. of India implements the GAP. APEDA adopts the approach of developing product specific clusters in different agro- climatic zones of the country to help in dealing with various supply side issues through Good Agriculture Practices addressing issues such as, soil nutrients management, higher productivity, adoption of market-oriented varieties etc.

institutes (Crop or commodity based) and 71 agricultural universities spread across the varying agro climatic regions of the country, NAREES (National Agriculture Research Education and Extension) is one of the largest national agricultural systems in the world. Many INM technologies including micro irrigation/fertigation methods and bio-control strategies to reduce use of chemicals in crop production were developed and popularized. ICAR has developed the production protocols of GAPs for several crops, vegetables, fruits and spices and medicinal crops which are ready to be adopted.

The Agricultural & Processed Food Export Development Authority (APEDA) has focused on collaborative approach for capacity building of various stakeholders involved in the development of export enhancement of fruits by creating Agri export clusters for different commodities in India. Thirteen Clusters are identified for district(s) like potato in Punjab and Uttar Pradesh; isabgol in Rajasthan; orange, pomegranate and grapes in Maharashtra; banana in Tamil Nadu; mango in Uttar Pradesh; rose and onion in Karnataka and fresh vegetables in Uttar Pradesh. The Quality Council of India (QCI) in partnership with the India's leading industry bodies, - the Confederation of Indian Industries (CII), the Federation of Indian Chambers of Commerce and Industry (FICCI) etc has launched accreditation and certification schemes for the identified sectors such as Food Safety Management Systems, Product Certification, Inspection, INDGAP certification, India-GHP Certification, India-HACCP Certification etc.

6. Good Agricultural Practices for Rice production

A. Water

- Water for growing rice shall be from source that environment is safe from hazardous substance contamination.
- The construction of separate channels to move water to and from each field greatly improves the control of water by individual farmers.
- Field channels allow water to be delivered to the individual seed beds separately and the main field does not need to be irrigated until it's time to plant in the main field.
- In addition, the ability to control water to your field is important when you need to retain water (especially after applying fertilizer so nutrients are not lost) or when you need to drain the field for harvest.
- Construction of individual field channels is the recommended practice in any type of irrigation system.
- Continuous flooding of water generally provides the best growth environment for rice. After transplanting, water levels should be around 3 cm initially, and gradually increase to 5–10 cm (with increasing plant height) and remain there until the field is drained 7–10 days before harvest.
- For direct wet seeded rice, keep the soil moist but not saturated from

sowing till emergence, to avoid seeds from rotting in the soil.

- field should be flooded only once the plants are large enough to withstand shallow flooding (3-4 leaf stage).

B. Planting Area

- Timely planting into a well-prepared seedbed will help produce a fast growing, uniform crop that will have higher yields and better competition against weeds and other pests. The best time to plant depends on locality, variety, weather, water availability, and the best harvest time. Planting at the same time (or within a two- week window) as the neighbouring fields can help to minimize insect, disease, bird, and rat pressure on individual fields.
- A well-prepared land controls weeds, recycles plant nutrients, and provides a soft soil mass for transplanting and a suitable soil surface for direct seeding.

C. Seeds and Sowings

- Use healthy seeds of a locally adapted variety to get a good crop that has high yield potential and a good market price.
- Seed must be grown, harvested, and processed correctly for best yield and quality results. The number of plants established and its seedling vigor will affect the competitiveness of the crop against weeds, and determine the final yield potential.
- Direct seeding requires 60–80 kg of seeds per ha, while transplanting only requires 40 kg per ha, at 2 plants per hill.

D. Planting

- Rice crops can be either direct seeded or transplanted. It can be manual with labour or with transplanters.
- The seedlings should be planted at an ideal depth of 4-5 cm at spacing of 15cm x 20cm or at random at 15 cm spacings.
- It is ideal to leave alleyways every 2 m for interculture operations, movement of labour, controls pests and the better aeration results in more productive tillers.

E. Intercultural Operations

- Ploughing and harrowing in fallow should be undertaken at least 10–14 days apart or after rain.
- First weeding begins within 2–3 weeks after establishment and the second in another 2–3 weeks. Weeding before fertilizer application is ideal.
- Identify the weed correctly and use the appropriate herbicide as recommended on the label and Spray when the weeds are small.

F. Fertilizer and Pesticides

- Use organic fertilizer (manure, compost, straw, husk, plant leaves) whenever possible, especially in nurseries.
- Apply fertilizer according to soil type and expected yield. As a guide, a 2 t/ha yield on clay loam soil will require 20 kg N and 5 kg P. Sandy soils may require another 10–15 kg K. Double these recommendations for a 3 t/ha expected yield.
- Apply all P, K, and 10% N evenly and incorporate just before seeding or transplanting. For direct seeded broadcast crops, it is okay to apply 10–14 days after establishment when there is water in the field.
- Apply remaining N (urea) in 2 equal portions at 30 days and 50–60 days (panicle initiation) after emergence.
- In established crops, apply chemical fertilizer only in standing water and evenly across the whole field.
- Do not apply high rates of fertilizer for traditional varieties as they may have limited response and cause lodging.
- If application of pesticide is required, it has to be as per the recommendations of ICAR, SAU's and Agriculture Department.
- Do not use the hazardous substances which are prohibited for agricultural use.
- In case of rice produced for export, do not use pesticides which are prohibited by the importing country.

G. Pre-Harvest Quality Management

- Crop Uniformity and purity has to be maintained at all intervals. The off types need to be removed immediately. The quantity of off type rice plants shall not exceed 3%.
- Crop has to be regularly observed for diseases and pests and suitable control measures to be adopted.
- Produce shall not have any plant disease and damages by insect greater than 10%.

H. Harvesting and Post Harvest handling

- If threshing and/or harvesting is applied, equipments shall be properly cleaned and threshing process shall be handled carefully in such a way that it does not introduce any admixing grain to the produce. If the machine has been previously used to harvest or thresh other rice variety, it shall be cleaned to get rid of all remaining grains.
- If produce is not sold as wet paddy, it must be dried to reduce moisture within 24 hours after harvested.
- Drying practice shall not introduce any breakage to grain in such a way that the result of milling quality test of paddy is lower than the criteria required in Thai Agricultural Standard for each type of rice.
- The moisture of paddy shall not exceed 15% for traded rice and 14% for stored rice.

I. Transport, Holding and Storage

- Equipments, containers and carriages for transportation and storage shall be cleaned and able to prevent quality deterioration of produce and prevent from contamination of substances that may be harmful to consumers, and also from inducing of admixing grain.
- Grain collecting and storage room shall be managed in such a way that the condition is hygienical clean and well ventilated. They are able to prevent contaminations and inducing of admixing grain.
- If more than one variety of rice is handled or stored, management in storage shall be in such a way that admixture of grain between different varieties is prevented.
- The dried seeds to store in jute bags, properly stitched, labelled and stacked uniformly in manner with walking space on all four sides.
- The stacks to be away from walls and moisture and properly covered.
- Regular fumigation and rodent control methods to be adopted to prevent damage and losses.

J. Personal Hygiene

- The pesticides and herbicides are poisonous; if they are not used properly, children's reach. they can cause health and environment problems. Label them clearly and keep them out of
- Always use protective clothing when spraying. Do not wear raincoats when spraying as this increases sweating. Personnel and labour shall wear clean clothing suitable for the operations with which they are involved.
- Personnel working in the production area should refrain from non-hygienic behaviour, such as smoking, spitting, and chewing during the operations.
- Visitors who are allowed to enter the production area shall receive prior permission and adhere strictly to the personal hygiene instruction

K. Record Keeping and Traceability

A record in proper format to be maintained for seed sources and time of sowing/ planting, water source, soil preparation, the off-type rice removal, the survey of plant damages by pest and pest management, Date and time of application of pesticides, Time of harvesting and threshing, drying for moisture reduction and paddy packaging and storage.

Berseem Mescavi, Wardan, BL-1, BL-10, BL-2, BL-22, JB-1, JB-2, JB-3, BB-2, BB-3, BL-43, BB-5, BB-6, BB-7	Mid October (Punjab, Haryana and U.P) November (Gujarat and W.B)	Optimum time sowing: 25 kg/ha. Early sowing: 15-20% extra seed rate. Lowland Rice: 35 kg/ha	20 kg N/ha, 80-90 kg P2O5/ha and 30-40 kg K2O/ha at sowing,	16-18 irrigations in 10-12 days interval	Field infestation of chicory minimized by treating with 10% common salt and deep summer ploughing. Imazethapyr @ 0.10 kg a.i/ha as PPI was effective	First cut at 55 DAS and subsequent cuts at 25-30 days after previous cut.
Lucerne Sirsa-8, Anand-2, Anand-3, RL-88, Co-1, T-9	End of September to early December but best time is Middle of October.	Broadcasting: 20-25 kg/ha, line sowing 12-15 kg/ha. It can also be sown with seed drill at 25-30 spacing	20 t FYM/ha every year for perennial crop. Seed inoculation with <i>Rhizobium meliloti</i> . 20 kg N, 60-75 kg P2O5 and 40 kg K2O/ha	Crop requires 15-20 irrigations in a year along with pre sowing irrigation. Early stage frequent irrigation required	First weeding at 20-25 DAS. pendimethalin @ 1.2 kg a.i/ha A Pre-emergence or spot application of diquat @6-10 kg/ha as Post emergence (5-10 DAS) controls <i>Casputa</i>	First cut at 50-55 DAS and subsequent cuts at an interval of 25-30 days. In a year 8-10 cuts.
Napier Bajra Hybrid IGFRI-3, IGFRI-6, IGFRI-7 and IGFRI-	Planting through rooted slips at any time of the year except in	Sole crop: 35000 rooted slips or stem cuttings/ha at 75×50 cm spacing.	20-25 t FYM/ha. At sowing basal dose of 60 kg N, 50 kg P2O5 and 40 kg	During monsoon no need of irrigation except long dry spell.	Regular hand weeding/hoeing ensures good aeration	First cut at 60-65 DAS and subsequent cuts at an interval